A Lunar Robotic Construction System using Raw Regolith

by

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University of Toronto Institute for Aerospace Studies (UTIAS)
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Abstract

The renewed global interest in furthering human's presence on the Moon has catalyzed efforts to establish a sustainable lunar base. The incentive is not only for scientific opportunities and prospects of deep-space exploration, but also for demonstrating technologies that will extend our reach throughout the Solar System. Central to such efforts is the development of robust and scalable lunar construction technologies. This thesis investigates a robotic construction system that leverages the raw lunar regolith, the most abundant and readily-available construction material on the Moon, for the creation of lunar surface infrastructure. The lunar regolith is utilized through Regolith Containment Units (RCUs), analogous to terrestrial sandbags, to provide physical, radiation, and thermal protection for lunar surface structures. A robotic system for bagging and manipulating these RCUs is designed to a Pre-Phase A level.

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Chapter 1

Introduction

1.1 Motivation

The recent renewal of focus on the Moon by space agencies and private sector is motivated by not only its scientific opportunities and prospects of deep-space exploration, but also demonstrating technologies that will extend sustained human presence throughout the Solar System. The International Space Exploration Coordination Group (ISECG), a consortium consisting of 27 space agencies, revised its Global Exploration Roadmap (GER) in 2024 with a supplement updating mission scenarios and architecture for both human and robotic lunar surface missions [1]. The goal is to ensure a sustainable return to the Moon by establishing the required infrastructure, and to highlight the increasing interest in certain technologies such as lunar in-situ resource utilization (ISRU), while generating socio-economic benefits on Earth. Some notable efforts to pave the roadmap include the NASA-led Artemis Program, whose one of the key goals is to establish Sustained Artemis Base Camp Infrastructure and Commercial Operations at the lunar South Pole by the late 2020s, as a starting point for the future Mars-based infrastructure [2]. In a similar concept, the European Space Agency (ESA) has proposed the establishment of the Moon Village habitat in conjunction with the Massachusetts Institute of Technology (MIT), though no specific target date has been yet released [3, 4, 5, 6]. The Chinese National Space Administration (CNSA) and Russia's Roscosmos [7], in collaboration with 11 other nations to date [8], have also expressed their goal to send humans to the Moon before 2030, and to establish an International Lunar Research Station on the lunar surface. India, a relatively new player in lunar exploration, has also cited the long-term goal of establishing a lunar base in the Indian Space Research Organization (ISRO) Vision 2047 [9]. The Canadian Space Agency (CSA) has made its commitment to the development of lunar infrastructure through the Lunar Surface Exploration Initiative (LSEI), where Robotics is presented as one of the key subjects of study [10].

A crucial prerequisite to the lunar bases is the establishment of supporting infrastructure such as berms, shelters and other protective means for the equipment, machines and structures. A major challenge in establishing the lunar infrastructure is the scale of the work required for lunar construction. Planetary missions for years has been focused on exploration, where the primary goal is merely scientific, and machines and equipment need to minimally manipulate and interact with local resources. The construction of lunar infrastructure, on the other hand, will require extensive

in-situ resource utilization (ISRU) reliably and at scale. This necessity brings to light our limited knowledge of and experience with the lunar environment, construction, and robotics.

A variety of ISRU-based technologies have been investigated for the creation of such surface infrastructure. Confinement formation, or regolith bagging, is one technology of particular interest, due to the minimal amount of in-situ processing required. One study evaluates confinement formation as the best method for building lunar surface structures, with particular strengths in size, in-situ ratio, temperature, and production time [11]. Another work also proposes the construction of super-adobe structures using regolith bags [12]. Regolith bagging has been previously investigated, as well, by NASA[13, 14], including studies at Marshall Space Flight Center (MSFC) to produce garage-like structures [15, 16]. Regolith bagging can provide physical, radiation, and thermal protection to lunar surface structures. Therefore, this thesis investigates robotic systems for the implementation of confinement formation methods for the creation of lunar surface infrastructure, with a particular focus on lunar landing pads as an implementation scenario.

1.2 Goal and Objectives

The goal of this research is to perform the Pre-Phase A design of a robotic system that interfaces with a lunar regolith excavation (and processing) system, specifically the ISRU Pilot Excavator (IPEx) from NASA [17], to intake raw lunar regolith feedstock, fill Regolith Containment Units (RCUs), and manipulate them into desired configurations that can be used for creating structures, such as berms or shelters, for equipment in the lunar environment. The research objectives are broken down as follows.

1. Perform a thorough review of lunar construction to understand the available technologies and scope the design problem (condensed version in Chapter 2).
2. Formalize the design problem by quantifying the specifications, including objectives and constraints, and performing functional analysis (Chapter 3).
3. Survey the reference designs and propose possible solutions that satisfy the defined design specifications and functionalities (Chapter 3).
4. Perform a systematic assessment of the potential solutions to select the final solution (Chapter 3).
5. Design the selected solution at the Pre-Phase A level (Chapter 4).
6. Verify and validate the design using physics-based simulation environments (Chapter 4).

This thesis further aims to contribute to the academic community through journal publications. At the time of writing, a literature review on the state-of-the-art in lunar construction is in pre-publication with Progress Aerospace Sciences by Science Direct. Further, a second work on the system conceptualization is in pre-publication with Aerospace by MDPI as a part of a special edition on lunar construction. A third work on the system-level (Pre-Phase A) design of the robotic system is being completed for journal submission. Additionally, this thesis was conducted in collaboration with MDA Space, generating new collaborations with NASA centers, American academic institutions, and the European Space Agency. As part of this collaboration, some laboratory work using regolith simulant was conducted, which may lead to future publications.

Chapter 2

Background: Lunar Construction

There are several necessary components of lunar construction for establishing a lunar habitat, as depicted in Figure 2.1. This Chapter focuses on the components necessary for the development of a robotic system for implementing confinement formation methods on the lunar surface.

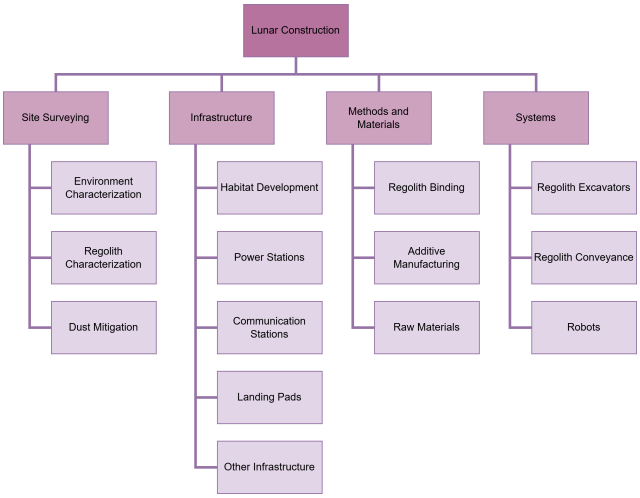


Figure 2.1: Lunar Construction Overview

2.1 Lunar Site Surveying

In general, most international efforts have focused on the lunar South Pole as the primary location for a sustainable lunar habitat. This is due to a variety of factors, including illumination and power profiles, the presence of volatiles (e.g., water ice) in Permanently Shadowed Regions (PSRs), and terrain [4, 18, 19, 20, 21, 22]. Some works have studied the illumination conditions at both polar regions and how these can impact power generation [21, 22]. The illumination conditions at the lunar poles are severely impacted by the local terrain, owing to the low inclination angle of the Moon's rotational axis at 1.54° [23]. This inclination creates two phenomena of interest at the lunar poles. The first phenomenon is *Peaks of Eternal Light* or *Peaks Of Light* (POL), where the terrain elevation is higher than the horizon, allowing for permanent sunlight and solar power generation.

The second phenomenon involves PSRs, where there are permanent shadows in the craters, which can trap essential resources and volatiles such as water ice. An ideal location would include a region of near permanent sunlight for power generation, next to a PSR for resource extraction, and relatively flat terrain for habitat construction and landing [24, 25]. Some works have also considered the implementation of solar towers, ranging from 2 m in height up to 2000 m, to induce permanent sunlight at the solar array locations, achieving constant or near-constant illumination in some cases [26, 27]. However, these strategies can also lead to additional challenges, such as casting large shadows by solar towers across the region of interest, due to the low incidence angle of the Sun on the lunar horizon [27]. Small scale solar towers are already being designed, such as the Lunar Array, Mast, and Power System (LAMPS), which can extend up to 15 m height [28].

2.1.1 Lunar Environment

Some overviews of the lunar environment are presented by [29] and [30]. The primary source of data stems from the "Lunar Sourcebook" [31], which curates and summarizes the findings of the Apollo missions and subsequent analysis of the lunar environment and regolith. The lunar gravity is approximately 1/6 g, or 1.63 m/s^2. The atmosphere is almost non-existent, and is largely composed of neon, hydrogen, helium, and argon, with a total mass of approximately 25000 kg and pressure of 3×10^{-15} atm.

The Moon has no electromagnetic field, so it is constantly bombarded by radiation from space. The radiation includes solar flares (called Solar Particle Events, SPEs), solar wind, galactic cosmic rays (GCRs), and radiation from the Moon [29, 30]. Galactic cosmic rays and solar wind are inversely proportional, since solar wind deflects GCRs. Solar wind and solar flares operate on an 11-year activity cycle. In general, humans have a maximum radiation allowance of 0.05 Sv/year, where 0.1 Sv results in a 0.5% increase in the chance of cancer later in life. The GCRs alone cause 0.3 Sv/year on the surface, and solar flares cause up to 0.6 Sv/h. This can be compared to the 0.001-0.002 Sv/year that the average human experiences on Earth. To mitigate the radiation effects, regolith shielding can be used. While the Earth's atmosphere provides 100 g/cm^2 of shielding, $2-3$ m of regolith provides 40 g/cm^2. It is commonly accepted that 1 m of regolith shielding would be sufficient for nominal conditions, but rare large flares would require 700 g/cm^2 of shielding [32].

Temperatures were found to vary by 282° C on the lunar surface at the Apollo landing sites, from −171° C to 111° [32], dropping below −213 °C in PSRs near the poles [33]. This variation is drastically reduced with depth, making the lunar regolith a good thermal insulator. At the Apollo landing sites, where the equilibrium temperature of lunar regolith is in the range of −15 °C to −20 °C, the variation is nearly 10 °C at 20 cm depth, and 1 °C at 40 cm depth [34]. A comprehensive mapping of lunar surface temperatures is compiled in [33], using 10 years of Diviner data from the Lunar Reconnaissance Orbiter, indicating that the temperature range and distribution remain similar between the North and South poles throughout seasons, with slight increases in the size of cold spots during the winter. This consistency can be explained by the Moon's lack of weather and low axial tilt. Temperatures vary from over 300 K in the hottest regions to below 60 K in the coldest regions, the latter typically coinciding with PSRs in the craters. The maximum bolometric temperatures for the North and South poles are shown in Figure 2.2, and the presence of lunar ice in Figure 2.3, indicating a large overlap between the coldest regions and water ice, particularly at the South Pole.

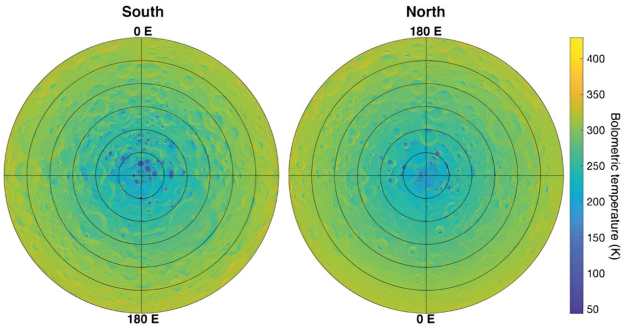


Figure 2.2: Maximum Bolometric Polar Temperatures [33]

Seismic activities on the Moon are limited, including deep and shallow "moonquakes" as well as meteorite impacts. The Apollo missions collected readings of approximately 12000 moonquakes, with most measuring 1-2 on the Richter scale, and a maximum of 3. Seismic activities are dominated by the meteorite impacts experienced on the surface [35]. Micrometeorites have an average mass of 10^{-10} to 10^{-8} kg, and an average speed of 72 km/s [29, 36]. In general, structures can be protected from micrometeorites by lunar regolith and structural elements, such as Whipple shields [37].

The lunar crust is composed of 60% oxygen by mass, or 40-50% SiO_2, and 10-20% FeO and CaO each. Lunar water ice is identified at the polar regions, particularly the South pole, making it an attractive target for settlement [30]. NASA's Moon Mineralogy Mapper instrument aboard Chandrayaan-1 was able to collect data confirming the presence of lunar ice at the polar regions, shown in Figure 2.3 [38].

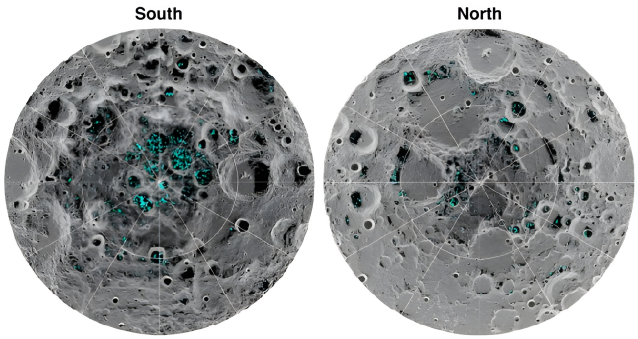


Figure 2.3: Distribution of Polar Surface Ice [38]

2.1.2 Lunar Regolith

The surface of the Moon has been pulverized by the collisions of both small and large asteroidal and cometary materials for billions of years, forming a fine-grained layer of "regolith" [31]. As the high-speed meteorites and micrometeorites impacted the surface, the soil particles broke up and began melting together, forming lithic fragments and clustering to form larger petrological elements, such as glass and agglutinates. The lunar regolith has a consistent particle and petrological distribution with depth, but is locally poorly sorted [39]. Overall, lunar regolith petrology can be broken down into a few groups: lithic fragments, breccia fragments, mineral fragments, glass fragments, and agglutinates. The lunar regolith itself is a generally fine, adhesive, sticky, grainy, and highly abrasive

material, which has properties unlike typical terrestrial soil. Several recent reviews of lunar regolith and its properties, specifically relating to engineering design and excavation, have been conducted by [39], [40], and [41].

With respect to the abundance of regolith on the surface of the Moon, one study estimates the regolith median depth to be 2-4 m in the maria and 6-8 m in the highlands [42]. A more recent study concludes that the lunar regolith may be thinner in the lunar maria than in the highlands, but estimates an average rock-free depth of 10 m in the maria and 12 m in the highlands [43]. Particle size properties remain consistent up to several meters of depth [41]. While some studies have attempted to fit distributions to the various properties of lunar regolith, such as size distribution, these models are heavily dependent on the samples that have been collected and analyzed [44, 45]. The work of [36] details many important properties of the lunar regolith and dust, such as its dynamics and mechanics, including dust charging, which can impact its movement. Generally, the particle size characterization of lunar regolith can be broken down into three categories: lunar dust (< 20 μm), lunar soil (< 1 cm), and mega-regolith (> 1 m) [46]. Lunar regolith particles are extremely fine, with an average size of 70 µm and average elongation of 1.35, making them ovular. The typical lunar regolith particle has nearly 8 times the surface area as an equivalent sphere, contributing to its high interlocking behaviour. Regolith particle interlocking results in packing that can make regolith properties anisotropic [40]. The regolith consists of a loosely packed upper layer, which rapidly compacts with depth past 15 cm. The bulk density stands at 1.3 g/cm^3 at the surface, 1.52 g/cm^3 at 10 cm depth, and 1.83 g/cm^3 at 100 cm. The bulk density asymptotically approaches 1.92 g/cm^3. Similarly, the intergranular porosity of lunar regolith was found to vary from 65% at the surface to less than 40% at depth (asymptotically) [41]. The bulk density and porosity are inversely related; as porosity increases, bulk density decreases [31].

The shear strength of lunar soil can be described by the Mohr-Coulomb Failure Criterion, which breaks down the soil strength into cohesive and frictional components. Similar to density, the cohesion and internal friction angle of the lunar regolith increase with depth. Consequently, the shear strength of lunar soil rapidly increases after 15 cm, and again after 30 cm [41]. This means that larger excavation forces are required to excavate lunar regolith at greater depth and higher compaction. Many sources agree that regolith excavation at depths greater than 15-30 cm would be challenging, especially due to the low traction caused by the low-gravity environment [39, 40, 41]. This means that most excavation concepts focus on the extraction of surface regolith. The range of various geotechnical properties of lunar regolith is presented in Table 2.1.

Table 2.1: Common Lunar Regolith Geotechnical Properties

Property	**Range**	**Units**	**Source**
Bulk Density (ρ)	1.3-1.92	g cm^{-3}	[31]
Porosity (n)	40-65	%	[41]
Specific Gravity (G)	3.1	-	[31]
Friction Angle (ϕ)	30-50	°	[31]
Cohesion (c)	0.1-3.4	kPa	[41]

Lunar regolith can also take on an electrostatic charge, and Van de Waals forces help dictate its movement [39]. The lunar regolith has low electrical conductivity and dielectric losses, resulting in electrostatic charging under ultraviolet radiation. This phenomenon is particularly challenging for surface operations, since lunar regolith adheres electrostatically to whatever material it comes into

contact with. The electrostatic charging effect is especially pronounced during the lunar night, when lunar regolith electrical conductivity decreases. The electrostatic charging effect can result in the coating of equipment, such as solar panels, potentially causing problems for lunar surface systems [31].

Lunar dust has been documented by Apollo astronauts as being highly abrasive, highly adhesive, and electrostatically charged [31]. This can pose challenges for humans, but also machinery, since it can embed and compact itself into exposed cracks, cover and scratch surfaces, cut through materials, and damage electrical components by electrostatic discharge. This motivates the development of dust mitigation technologies, which can be broadly sorted into passive, active, and implicit mitigation (or dust tolerant design), as shown in Figure 2.4. Some recent reviews on lunar dust mitigation methods include the works of [46] and [47], as well as the trade study by [48]. Further, NASA has released a guide compiling their experiences and suggestions regarding lunar dust mitigation, including specific studies on optical surfaces, radiator and thermal control surfaces, communications and electronics, as well as mechanisms and mechanical assemblies [49].

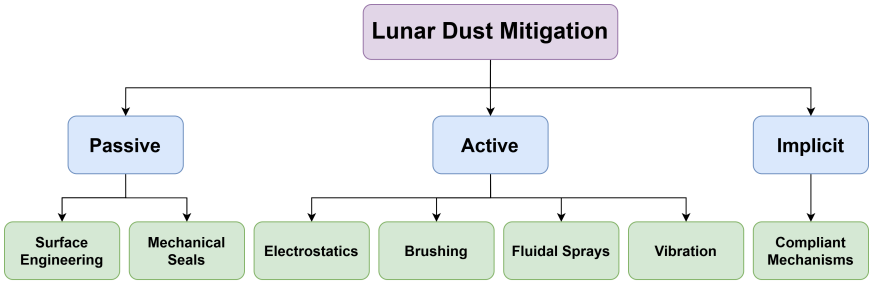


Figure 2.4: Lunar Dust Mitigation Method Classification

2.2 Lunar Infrastructure

The vast majority of proposed habitat locations on the lunar surface involve the lunar North and South poles. In 2022, NASA identified 13 candidate landing regions within 6 degrees of latitude of the lunar South Pole for the Artemis missions [18]. These missions would aim to establish a sustainable human presence on the Moon with the Artemis Base Camp [2]. NASA cites the presence of in-situ resources in PSRs, particularly water ice, as well as large periods of continuous sunlight, as some of the primary considerations for these selections [18]. Similarly, the ESA's Moon Village concept has identified areas around the Shackleton crater at the lunar South Pole as a target location [3, 4]. The CNSA has also cited the lunar South Pole as a target habitat location, with some researchers conducting a thorough site selection based on the trade-off between scientific and engineering objectives, with a particular attention to the Amundsen and Malapert craters [19]. Another study investigates the lunar North and South poles as possible sites for a liquid mirror observatory, finding the North pole to be optimal, due to the larger slopes at the Shackleton South Pole location, unless the habitat and observatory are decoupled [20]. Finally, a study for a *PneumoPlanet* inflatable habitat investigated possible habitat locations and landing sites within 1° angular distance from both poles, proposing ideal locations in both regions [21].

2.2.1 Landing Pads

A major challenge to the goal of repeatable lunar landings is the interaction of rocket engine exhaust plumes with the lunar regolith, which can create blast ejecta and cratering. The ejecta can reach velocities over 2000 m/s, and travel large distances in a vacuum, potentially entering orbit and cis-lunar space. It is estimated that the Apollo lunar lander displaced 2600 kg of regolith during landing, leading to erosion and cratering [50]. The ejecta creates high velocity particles that can damage surface structures, and remain suspended in orbit, and the cratering results in uneven terrain, both of which could hinder future lunar landings [51, 52]. While current rocket and localization technologies may require landing pads of up to 100 m diameter, the landing area is projected to be reduced to 25 m as landing accuracy improves [53, 54, 55]. One trade study investigates the cost of candidate landing pad construction methods, dividing the task into an inner zone (12 m radius) and outer zone (27 m radius). The inner zone must withstand the high temperature and pressure of the plume stagnation region, while the outer zone must withstand erosion. The study broadly considers grading, compacting, sintering, paving, gravel, and polymer as candidates for the inner and outer zones [56]. Another trade study focuses largely on AM methods for the construction of lunar landing pads, considering only a few non-AM implementations [57]. two recent reviews on lunar landing pads have been conducted by [51] and [52].

The Lunar Spaceport by the Open Lunar Foundation proposes four architectures for a potential lunar landing pad, culminating in a final robust landing pad with target construction in 2120 [58]. The first concept, The Skinny, represents the current state of technology, with an inner zone consisting of an octagonal platform made of metal panels, surrounded by an outer zone composed of a heat-resistant textile apron. Next, there is The Machine, which is a spacecraft-like system elevated off of the ground with legs, and uses a robotic arm for servicing the visiting vehicle. There is also The Brute, which consists of a 3D printed high-temperature polymer pad on top of an elevated mound of regolith. The fourth concept is The Works, which would constitute a fully operational spaceport. This final concept would have a multi-zone pad, an ejecta shield, and subterranean systems for refueling and capturing exhaust gases. The pad itself would be made of grouted, oven-sintered basalt setts connected by liquid geopolymer.

A NASA study investigates the regolith stabilization problem for a vertical take-off and landing pad using ISRU, and finds that a combination of techniques would be ideal, such as combining solar sintering for an inner zone and polymers for an outer zone [59]. The study also conducts field testing using the PISCES rover and robotic arm, successfully demonstrating the fabrication and deployment of 100 sintered basalt interlocking pavers over a 3m×3m inner zone. The rover further constructs a 20m-diameter outer zone, leveling and grading the zone, then covering it in gravel [60]. Another recent concept for a lunar landing pad considers the compaction of surface regolith and the usage of stacked rocks for stabilization. The study finds that although simple gravel landing pads would not be effective, a multi-zone approach could be effective and feasible, where the central zone is made by interlocking pavers and the outer zone is made of gravel and rocks [61]. Another work also investigats the possibility of instant landing pads created using the Masten in-Flight Alumina Spray Technique (FAST), which would spray the surface upon arrival. The technique aims to mitigate the landing plume effects and to allow greater flexibility in landing location, particularly for scientific missions [62]. A different approach to landing pad construction is being taken by Project Olympus, a collaboration between NASA and the large-scale 3D printing company ICON, which seeks to leverage

3D printing technology to build whole habitats, including landing pads [63]. The technology has been tested on a subscale level in practice, exploring various morphologies for the landing pad structure [64]. Alternatively, it has also been suggested that lunar regolith bags could be used as a base for a lunar landing pad [51]. Overall, a robotic construction system for landing pads will realistically need to be able to accommodate a variety of construction techniques, catering to both the inner and outer zones.

2.2.2 Other Infrastructure

Other infrastructure, such as blast berms and non-pressurized shelters, will also be necessary when establishing a base on the lunar surface [65]. With regard to using blast berms to protect surface structures during landing and take-off, simulations show that larger particles of the ejecta would be traveling with supersonic speeds at less than 3° from the touchdown, and smaller particles would be ejected with slower velocities at close to 17° [53, 66]. A few different methods have been proposed for the creation of such blast berms, including piling raw regolith [67], additive manufacturing [53, 58], and regolith sandbags [12, 68]. One concept proposes the stacking of irregular and unprocessed lunar stones for the creation of blast berms [69]. It is also suggested that lunar stones could be stacked in three layers, with average rock diameters of 144.7 mm, 72.4 mm and 14.5 mm for the top, middle, and bottom layers, respectively [61].

Non-pressurized shelters may be used to protect equipment or surface systems. These could be constructed through deployable components [70], additive construction [53], or regolith sandbags [71]. Some studies have been undertaken to analyze the stability of potential non-pressurized regolith bag structures, finding regolith bag arch structures to be stable and feasible [71, 16]. These structures could serve purposes such as robotic storage, server storage, or communications [68].

Although roads will also be a crucial piece of infrastructure for the expansion of future lunar bases, little work has been done to consider their construction specifically. A collaboration between NASA and The University of Texas at Austin investigates the construction of lunar roadways and relevant technologies, including graders, compactors, blades, augers, and various drive configurations [72]. Another more recent study considers how the in-situ regolith could be used for the creation of lunar roadways [73]. The lack of emphasis on the research on lunar roadways can likely be attributed to the fact that the technologies used for building lunar landing pads can be similarly utilized for roadways, such as sintering and paving technologies, amongst others [74]. Most AM, sintering or paving technology research declares the creation of lunar roadways as a general goal, such as Project Olympus by ICON and NASA [53]. One specific lunar roadway building project is PAVER by ESA [75, 76], which seeks to raise the TRL of casting and 3D printing technology for paving from 3 to 5. The project leverages a 12kW CO_2 laser with spot beam up to 100 mm to achieve the in-situ melting or sintering of the lunar regolith for the creation of roadways and other paving elements.

2.3 Lunar Construction Using Raw Materials

Almost all habitation concepts for a lunar base involve the utilization of regolith for shielding from the harsh lunar environment [77]. As such, it becomes necessary to investigate the best way to move and shape the regolith. Beyond movement of the raw regolith, some works have investigated the possibility of containment units, typically of "sandbag" form, as a construction unit for various

structures. It has been argued that such Regolith Containment Units (RCUs) benefit from a high in-situ material ratio, short production time, and less stringent temperature requirements, suffering only from slightly lower compressive strengths as compared to competing methods [11]. An early study on such lunar regolith containers is conducted by NASA, investigating potential materials and filling mechanisms. A container filling mechanism is designed, but not tested, using a conveyor and funnel system with a theoretical capacity of filling 120 container per hour for 8 hours a day. The containers are selected to be 0.0283 m^3, and made of a fiberglass composite weave [13]. A similar study suggests a scoop mechanism for filling 0.487 m^3 containers made of Kevlar 149. With such a mechanism, it is estimated that the system could fill 10 containers per hour for 8 hours a day [14].

One particular NASA project involves the creation of a lunar RCU to build a garage-type structure [15]. A multitude of materials are considered and tested, including Vectran, Zylon, Twaron, Nextel, Nomex, and Gore PTFE. After testing, Vectran is recommended for further study, followed by Kevlar and Twaron. The project also outlines various approaches to filling the containers, including shovels, funnels, tubes, sandblasters, leafblowers, etc. In the end, the best approach is determined to be a hopper attached to a helical flexible screw conveyor system built by Hapman Inc. The final structures are erected at Marshall Spaceflight Center (MSFC). The work is continued by investigating the stability of these structures [16]. It is found that structures made from top connected containers are more stable as compared to center connected containers (examples shown in Figure 2.5). It is also found that a screw conveyor system is the most reliable method for filling the containers. Another study also considers potential container filling mechanisms, focusing on use cases for the pneumatic lunar regolith excavator from Honeybee Robotics [78]. The work proposes a few conceptual operational configurations for the excavator, including filling containers directly at the mining site before manipulation, centralized filling at a lunar regolith silo, and filling the containers at their final location. However, these concepts are not demonstrated beyond the proof of concept only for the pneumatic excavator. Another study investigates the design of an RCU filling and stacking system for the burial of a lunar outpost [79]. The design involves a wall-scaling system whereby the regolith is taken in, the RCU is formed and ejected directly on top of the stack. The final concept is tested using sand on a one-tenth scale.

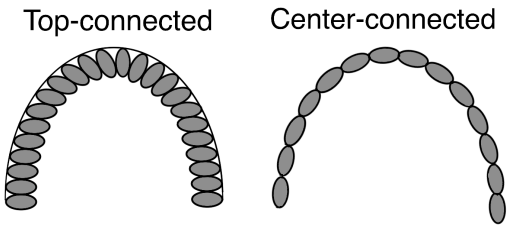


Figure 2.5: Top-connected versus Center-connected RCUs

A recent concept proposes containers equipped with microprocessors, piezoelectric power generators, RF transponders, and RFID tags for potential implementation in a lunar environment [12]. The work argues that RCUs present the lowest entry cost barrier to building lunar infrastructure, since they do not require the extraction of in-situ resources or the modification of regolith as much as some other technologies do, e.g., AM. The concept involves a collection of such containers to form a structure, which are able to actively communicate health and placement data to the assembly

and monitoring systems. The work continues to consider the high-level design of the surrounding architecture, such as the lunar regolith excavation, the RCU manipulation and filling subsystems [68]. The study focuses on the creation of berms and superadobe structures for equipment protection and storage, as well as protection of greenhouses and habitats. Another recent study also proposes aramid fabric RCUs as key components in habitat construction, testing their capabilities as load bearing columns, self-supporting arches, and external protection for habitat structures. It finds that RCUs provide a viable construction material when used in tandem with pre-fabricated and inflatable structures, with a column compressive strength of 6.9 MPa and 90% strength retention after vacuum aging [71].

The use of raw materials in the form of lunar rocks and gravel is also investigated, particularly for the construction of landing pads [61, 80]. These studies find that, although the central area of a lunar landing pad would need to be solid, the outer blast shield could be composed of loose gravel and rocks of varying sizes organized into three size-distributed layers. The required rock sizes are found to range from 14.5 mm, for the bottom layer, to 144.7 mm, for the top layer. It is calculated that the volume of rock required for the landing pad increases with the squared radius, resulting in 3140 m^3 of rock needed for a landing pad of 100 m in diameter.

2.4 Lunar Construction Systems

2.4.1 Lunar Regolith Excavators

The regolith excavation task has been under development since the early 2000s, with the most promising concepts reaching flight-readiness. Lunar regolith excavation systems are broadly classified as continuous or discrete, based on whether or not their cutting surfaces remain in constant contact with the soil [81]. Discrete systems, such as front-end loaders and dozers, interrupt contact to clear or dump material. Continuous systems, such as bucket wheels and drums, maintain constant engagement, and are often more efficient in low-gravity conditions. These systems are further distinguished as complete (with integrated mobility) or partial (stationary) platforms. NASA's Kennedy Space Center (KSC) has led recent developments in continuous-complete systems with the Regolith Advanced Surface Systems Operations Robot (RASSOR) series [82]. These dual bucket-drum excavators counteract lunar gravity by using counter-rotating drums, and they have progressed from the teleoperated RASSOR [83] to RASSOR 2.0 [84] that features semi-autonomous control and improved mass-efficiency, excavating up to 2.7 metric tons of regolith per day with just 4 W/kg of power consumption. The latest iteration, ISRU Pilot Excavator (IPEx) [17], reduces system size by 30%, and continues development toward TRL 5 with enhanced bucket drum [85] and wheel designs [86].

Some other excavation concepts have also been investigated. The Lunar Excavation and Size Separation System (LES^3), integrated with the 40–60 kg LUVMI-X rover [87], is a compact regolith excavation prototype weighing under 2 kg. It excavates 100 g per cycle using a low-force cylindrical scoop, optimized for shallow surface layers to avoid compaction effects. Regolith is classified into two size fractions via dry sieving and vibration, leveraging vibration to also reduce excavation forces. A larger-scale concept combines a bucket chain excavator with ballistic conveyance, launching sealed regolith containers to remote sites [88]. While Discrete Element Modeling (DEM) of the concept

supports feasibility in lunar conditions, the absence of supporting infrastructure raises deployment concerns. A bucket-ladder system has also been developed which aims to reach TRL 5–6, building on prior results showing >1000 kg/h excavation rates at 0.01 W/kg/h in Earth conditions [89]. However, the mechanical complexity of bucket-ladder systems and the lack of lunar-environment testing remain as major challenges to practical implementation. A simpler system, the CASPER rover, employs an Archimedes screw-based propulsion and excavation system [90], achieving a 30 kg/h excavation rate with only 3.4 kg mass and 30 W power consumption. Testing has been limited to silica sand, which differs significantly from lunar regolith, highlighting the need for higher-fidelity validations. Similar works include auger and flexible screw systems for regolith collection, conveyance, and drilling tasks [91, 92], sharing core principles with CASPER but varying in mechanical design and intended function.

2.4.2 Lunar Regolith Conveyance

Similarly to the excavation techniques, lunar regolith conveyance methods can also be broken down into three categories, namely, batch, continuous, and transfer. Batch methods involve the discrete conveyance of finite amounts of lunar regolith using systems such as haulers, aerial cableways, railcars, ballistics, sleds, and robotic arms. Continuous methods keep the lunar regolith in constant motion using systems such as drag chains, conveyor belts, vibrating surfaces, augers, bucket-ladders, and pneumatic conveyors. Transfer mechanisms allow lunar regolith to be transferred between different points in the system, including at intake. Some examples are auger-and-hopper mechanisms, hoppers, and bowl feeders. A review and assessment of lunar regolith conveyance methods is conducted by [48], examining long-distance conveyance, short near-horizontal conveyance, short near-vertical conveyance, and transfer points. Wheeled haulers are recommended for long distances, while conveyor belts, vibratory systems, and augers all perform well for shorter distances, and robotic arms are also viable options if regolith is pelletized. Auger-and-hopper systems with gravity discharge are found to be the best for transfer points along a conveyance route, with robotic arms also performing well for pelletized regolith.

Despite the many student prototypes related to the Lunabotics excavation competition, little work has been done to formally raise the TRL of many conveyance concepts. Some recent works have investigated vibratory conveyance, with a few studies testing the concept in both vacuum and atmospheric conditions. One study uses two dielectric elastomer actuators to vibrate an acrylic plate with lunar regolith simulant, enabling a maximum mass flow rate of 1.95 g/s in Earth gravity. It is found that 30 Hz vibration frequency at 60° inclination angle would result in optimal conveyance [93]. Another study investigates the conveyance of lunar regolith simulant by the vibration of a small (26-mm diameter) tube using a piezoelectric actuator, instead of vibrating a flat surface, achieving a mass flow rate of 7 g/s. The study also finds that regolith can be vibrated upwards at an angle of up to 24° in Earth conditions [94]. A follow-up study uses a similar actuator setup, but incorporates some numerical calculations, through DEM and single-particle modelling, to predict the conveyance performance in lunar gravity. Considering gravity and air drag, calculations predict that the conveyance system would be 1.4 times more effective in lunar conditions, achieving conveyance speeds of nearly 25 mm/s [95]. A more recent work tests an eccentric mass vibratory conveyor, where a motor spins two eccentric masses while connected to a pulley. The system achieves an average lunar regolith simulant mass flow rate of 209.86 g/s at a frequency of 3.46 Hz in Earth gravity and

in vacuum [96].

A number of experiments have been conducted by NASA Kennedy Space Center (KSC) to develop and test a pneumatic conveyor for feeding an ISRU oxygen production reactor, in both 1g and $\frac{1}{6}$g gravity [97, 98]. Both air and helium are used for the transport of various lunar regolith simulants, achieving mass flow rates of up to 6.4 kg/min [99]. As a part of the same series of experiments, augers are also field-tested at NASA's Mauna Kea Analog Site for feeding lunar regolith to the oxygen production reactor [100, 101, 102]. Flexible screw conveyors have also been conceptualized and tested, enabling the curved transport of lunar regolith [91]. One particular auger system is found to be capable of conveying an average of 298.35 g/s of lunar regolith simulant in atmospheric conditions, and an average of 215.71 g/s in vacuum [96].

Hoppers are another promising conveyance technology, with much work being done to investigate the optimal outlet widths and inclination angles in both 1g and $\frac{1}{6}$g conditions. One experiment achieves a maximum mass flow rate of 22 g/s in $\frac{1}{6}$g using a 75° inclination and 18mm outlet width [103]. Another experiment achieves a mass flow rate of over 15000 g/min with a 5cm funnel opening in Earth conditions [104]. However, such experiments further conclude that it remains difficult to reliably predict the flow characteristics of lunar regolith.

Several experiments have also been conducted on other conveyance methods. A concept leverages the electrostatic nature of the lunar regolith, using electrostatic travelling waves, to transport regolith at a rate of 13.5 g/min [105]. Another conveyance technology also involves the electrostatic or magnetic transport of lunar regolith particles. One experiment successfully demonstrates the conveyance of lunar regolith using one or multiple solenoid coils along a tube with 200 mm length and 16 mm diameter at a 45° upwards angle [106]. The work then extrapolates the results, and concludes that this method could propagate 1.54 g/pulse of lunar regolith over a 1m long tube. A more recent conveyor belt concept is the Tandem Of Lunar REgolith CONveyors (TOLRECON), a rod scraper conveyor that horizontally transports lunar regolith to a bucket elevator system. The concept has been simulated and field-tested as a part of the Over the Dust Moon Challenge by the Colorado School of Mines [107].

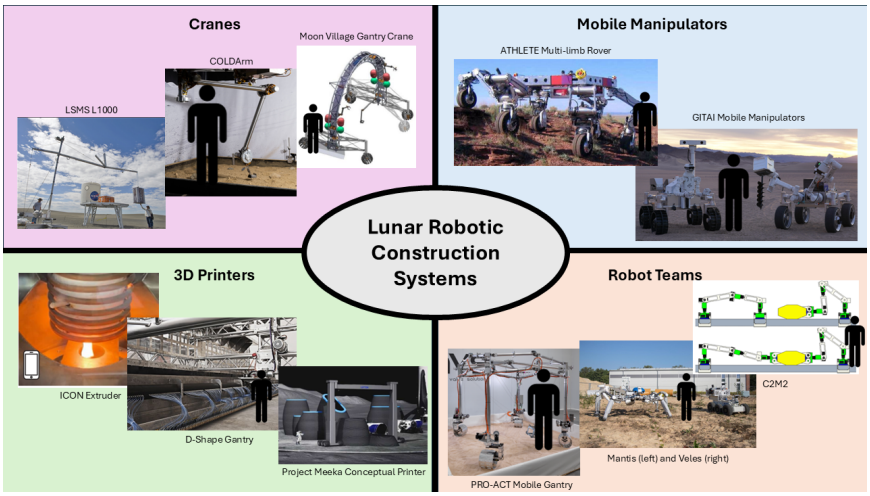


Figure 2.6: Overview of Lunar Robotic Construction Systems

2.4.3 Lunar Robotic Construction Systems

The existing concepts for lunar robotic construction systems and technologies are primarily divided into four groups: cranes, mobile manipulators, 3D printers, and robot teams (see Figure 2.6). This section focuses primarily on Cranes and Mobile Manipulators as the most relevant reference designs for the confinement formation task.

Cranes

A number of generations of lunar cranes have been conceptualized and demonstrated (see [108] for a survey.) Early lunar crane concepts include the Lunar Excursion Vehicle Payload Unloader (LEVPU) by NASA [109] and the Stewart Platform Instrumented Drive Environmental Robot (SPIDER) by the National Institute of Standards and Technology [110]. The most recent and advanced examples of lunar cranes are the multiple generations of NASA's Lunar Surface Manipulation System (LSMS), with the first generation being proposed and field-tested in 2008 [111, 112, 113]. The cable-driven system is proposed for payloads too heavy for astronauts to handle, ranging from 100 kg to 3000 kg. The mass-to-payload ratio for the first generation is 3%, emphasizing its ability to manipulate large loads at low mass. The first generation is later enhanced with automated control [114] and hardware improvements [115, 116]. The second LSMS generation has also been designed in parallel, with a focus on reducing the weight of the system and implementing self-offloading capabilities. The second generation also includes a single-link variation that would largely act as a crew elevator to and from the lander, including the lifting of incapacitated astronauts, along with payload offloading. The work has been continued through the design of a heavy-lift variant, the LSMS-H [117], which focuses on the offloading of more massive cargo, such as a habitat module. The LSMS-H also explores the possibility of mounting the LSMS onboard the Chariot mobility base to enable operations with a third-party ESA lander, with a focus on determining the tip-over boundary. The development of the LSMS has been continued, with the newest generations reaching the protoflight stage, including the L1000 [118], L500, and L35 [119], with 1000 kg, 500-600 kg, and 35 kg load capacity, respectively. Notably, the LSMS L35 has been selected for a technology demonstration as a part of the LSMS AutoNomy capabilities Development for surface Operations and construction (LANDO) project, being paired with Astrobotic's Peregrine lander for payload offloading. Further developments on the LSMS continues, with recent studies exploring the reproducibility of the LSMS [120], safe operational envelope [121], and trajectory tracking [122]. A summary of the various LSMS generations is presented in Table 2.2.

Table 2.2: LSMS Variations

LSMS Generation	Lander	Height (m)	Reach (m)	Lunar Wrist Capacity (kg)	Mass (kg)
Generation I	Altair	3.75	7.5	1000	309
Generation II	Altair	4.25	8.5	3000	190
Single-Link LSMS	Altair	3.75	7.5	4879	108
LSMS-H (Heavy)	Altair	4.17	8.34	6907	169
L1000	Blue Moon	4.0	8	1000	100-150
L500	Griffin	-	5.0	500-600	-
L35	Peregrine	0.9	2.25	35	7

As part of the ESA's Moon Village project, two mobile crane configurations have been conceptualized for transporting the habitat from the landing site, a gantry type crane and a pick-and-carry

type crane. The gantry crane concept is based on the mobile gantry system for the RSLO2 project [123], which builds off of the ATHLETE mobility platform [124, 125, 126], achieving a 15000kg carrying capacity with only a 6600kg mass. Each gantry can be equipped with up to four robotic arms and a variety of gripper, maintenance, and excavation tools. It has been proposed that one lander can deliver up to two gantry systems, each deploying their mobility system upon landing [123].

The Cold Operable Lunar Deployable Arm (COLDArm) is a manipulator that can function as a crane, designed by NASA and intended to be mounted on the deck of a lunar lander [127]. The arm has 4 DoFs, and is approximately two meters long, enabling a tip payload of ∼65 kg on the surface of Moon. The arm has been designed to work in extremely cold cryogenic temperatures without the use of heaters, enabling its operation in extreme environments, such as the lunar night. Eliminating heaters helps reduce the input energy required and heat exposure during volatile collection. The COLDArm has been tested with a variety of lunar regolith simulants, demonstrating its ability to operate in the lunar environment.

RoboCrane [128] is an alternative usage of lunar crane technology, being used to support power and communication architectures. Specifically, RoboCrane aims to assist with the exploration of lunar lava tubes and caves. The system would lower a rover of approximately 400 kg into these tubes or caves and support the subsequent exploration (or other tasks) by providing a power and communication interface with the surface.

Mobile Manipulators

The development of mobile manipulators for lunar construction has recently entered the prototyping phase. The Pacific International Space Center for Exploration Systems (PISCES) has created a mobile manipulator named Helelani [60]. The four-wheeled rover has been field-tested in lunar analogue conditions in Hawaii, as a part of the vertical takeoff vertical landing (VTVL) pad experiment. The rover was used as a robotic manipulator for paving the landing site, as well as a leveler equipped with a leveling back-blade. Another wheeled manipulator concept is the Chinese Super Mason (CSM), with a 6-Degree-of-Freedom (DoF) arm for construction on the Moon [129]. The manipulation concept is implemented using the ABB IRB 6700 6-DoF manipulator. The robot arm is demonstrated in a controlled Earth environment, building a 2.8m × 0.8m × 1.1m arched-framework structure in 8.5 h. However, a fully integrated concept including the mobility platform for the arm has yet to be demonstrated. The Lunar Exploration and Robotic System (LERS) is a similar robotic platform [130], featuring a 6-DoF manipulator mounted on a 4-wheeled base. It is envisioned that the system could support a variety of lunar surface activities, including construction and infrastructure development. The design considers lunar environmental factors, such as terrain, and a preliminary prototype has been assembled, but testing has yet to be performed in either lunar analogue or terrestrial environments.

The ATHLETE wheel-on-limb mobility platform has also been demonstrated and conceptualized with various interchangeable manipulators and tools to support lunar construction [124, 125, 126, 131]. In its typical configuration, the ATHLETE mobility platform has 6 multi-link robotic limbs attached to a flat (interchangeable) base. The minimal configuration, called the Tri-ATHLETE, is composed of 3 limbs. Each limb is capable of functioning independently as a 6- or 7-DoF manipulator with interchangeable end-effectors. These end-effectors are often wheels to enable mobility configurations, particularly for high-capacity transport, but can also include other implements for tasks

such as grappling, drilling, construction, additive manufacturing, and many more. The ATHLETE limbs can also operate independently, with the ability to function as independent manipulators or in unique configurations, such as a high-capacity crane [125]. The ATHLETE individual limbs have lunar payload capacities ranging from 500 kg to 11456 kg, while the complete mobility system capacity ranges from 1000 kg to 22912 kg. Overall, the design of the ATHLETE system has focused on prioritizing interchangeability in operation, allowing it to be modified to tackle a diverse range of tasks.

More recently, a four-wheeled rover base capable of mounting various manipulators is being developed by GITAI [132]. Two of these manipulators are the S1 and Inchworm robot manipulators. The S1 manipulator has 9-DoFs (including the hand), a maximum reach of 1 m, and it is capable of enacting 100 Nm of torque [133]. The Inchworm manipulator emphasizes the development of multi-purpose tools, with the ability to be transferred between the rover, lander, habitat, or other modules. It is a 7-DoF manipulator, plus two DoFs for the end-effector, with a 2m reach, capable of enacting 368 Nm of torque [134]. The GITAI mobile manipulator has demonstrated its ability to perform a variety of tasks, such as resource excavation, welding, and solar panel construction. The system has also been field-tested on the ISS and in lunar analogue settings, and has demonstrated features mitigating lunar environmental effects such as lunar dust [135].

Chapter 3

Design Conceptualization

This thesis focuses on the usage of Regolith Containment Units (RCUs), analogous to terrestrial "sandbags" and inspired by the aforementioned confinement formation methods. These RCUs will be filled and manipulated by a Lunar Robotic Construction System (LRCS) to create lunar surface structures, such as berms and shelters for surface equipment and machines. In this chapter, the conceptualization of the LRCS is presented (Section 4.2), outlining the environmental considerations (Section 2.1.1), the operational scenario (Section 3.1.1), a functional analysis using the glass-box method (Section 3.1.2), and a set of high-level system requirements (Section 3.1.3). Once this framework is established, an assessment process is undertaken for the four primary LRCS functional modules using the Analytic Hierarchy Process (AHP, Section 3.2). This includes an expert survey of design assessment criteria priorities (Section 3.2.1), using the ZM_{II} ranking aggregation method to integrate the results into the AHP process in a novel way (Section 3.2.2). The assessment is conducted (Section 3.3) for each of the four primary LRCS functional modules (Sections 3.3.1 – 3.3.4), using specific metrics to assess the candidate solutions with respect to the design criteria. Once the final morphology is evaluated (Section 3.3.5), the surrounding architecture for the LRCS is also preliminarily selected (Section 3.4).

3.1 System Conceptualization

The conceptualization of the LRCS can be broken down into three stages (Figure 3.1), namely, Task Definition, System Conceptualization, and Design Selection. The Task Definition stage includes characterization of the lunar environment (previously done in Section 2.1.1) as well as a breakdown of the expected system operation (Section 3.1.1), including any operational trade-offs. The information is then used in the System Conceptualization stage to conduct a functional analysis (Section 3.1.2), using the glass box method [136], and define a set of system-level requirements iteratively (Table 3.2). The Design Selection stage then uses the functional analysis and system requirements to make a systematic assessment of various design alternatives, using the Analytic Hierarchy Process (AHP) [137], to select the best candidates according to a set of criteria. The criteria are prioritized through a survey from the field experts, using the ZM_{II} method for ranking aggregation [138, 139, 140]. In this thesis, four key functional modules are highlighted in the assessment process. Once the morphologies for these functional modules are determined, the morphologies for the remaining

(standard) functional modules morphologies will be selected, e.g., power generation.

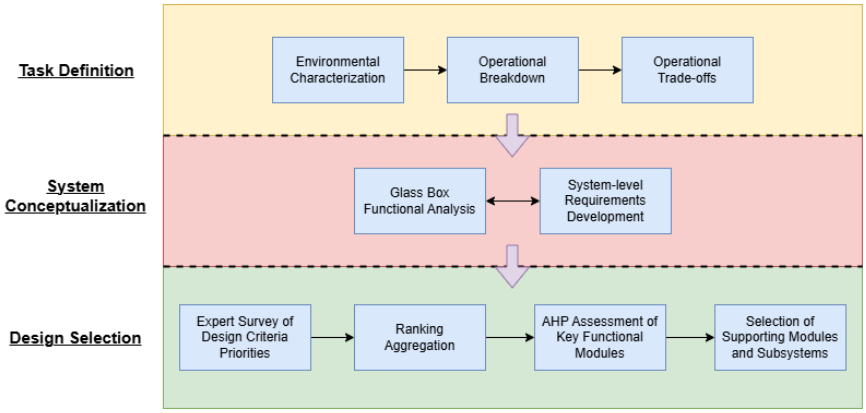


Figure 3.1: Conceptualization Process for the LRCS

3.1.1 Operation

The LRCS will need to perform a series of tasks to accomplish the overall goal, which can be broken down as shown in Figure 3.2. Of particular importance, the LRCS will need to be capable of interfacing with NASA's ISRU Pilot Excavator (IPEx) to intake lunar regolith. Without loss of generality and to be specific with the conceptualization process, this thesis focuses on the operation for constructing a blast berm for landing pads. A major challenge to the goal of repeatable lunar landings is the interaction of rocket engine exhaust plumes with the lunar regolith, which can create blast ejecta and cratering. The ejecta can reach velocities over 2000 m/s, and travel large distances in a vacuum, potentially entering orbit and cis-lunar space. It is estimated that the Apollo lunar lander displaced 2600 kg of regolith during landing, leading to erosion and cratering [50]. The ejecta creates high velocity particles that can damage surface structures, and remain suspended in orbit, and the cratering results in uneven terrain, both of which could hinder future lunar landings [51, 52]. Simulations show that larger particles of the ejecta would be traveling with hypersonic speeds at less than 3° from the touchdown, and smaller particles would be ejected with slower velocities at close to 17° [53, 66]. While current rocket and localization technologies may require landing pads of up to 100 m diameter, the landing area is projected to be reduced to 25 m as landing accuracy improves [54, 55, 53]. As such, protecting surface structures against the most harmful hypersonic particles ejected at 3° from a 100m diameter landing pad, designed for current rocket technologies, will be considered as the operational task. To achieve this, the berm will need to be $50 \times tan(3°) \approx 2.6$ m tall. The RCUs may be stacked using either supported or unsupported architectures to achieve the desired height, as shown in Figure 3.3. This thesis focuses on unsupported RCU structures, but supported structures, using frames built by other systems, would be a practical implementation that could drastically reduce the required number of RCUs while remaining structurally sound. As shown in Figure 3.4, multiple stacking of RCUs, e.g., double or triple stacking, can be considered as a modification to simple pyramidal stacking to reduce the amount of RCUs required. The limit for multiple stacking is dependent on the specific structural properties of the RCUs, which should be further investigated. It is anticipated that, due to low gravity as well as the lack of environmental effects (e.g., wind), the RCUs could be stably stacked high with a relatively small base. Other stacking concepts, including

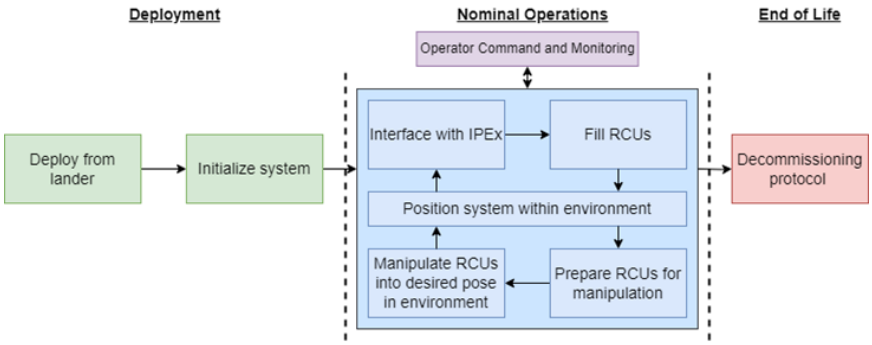


Figure 3.2: Operational Stateflow Flow Diagram for the LRCS

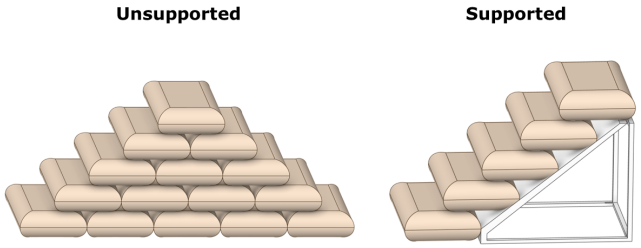


Figure 3.3: Unsupported and Supported RCU Structures

asymmetrical stacking or non-pyramidal stacking, would require a more thorough stability analysis considering the non-rigidity of the RCUs and the inter-RCU friction. The number of RCUs for each stacking concept can be determined, starting with simple symmetrical (pyramidal) stacking as a base case, by considering the Gauss series

$$N_{pyramid} = 1 + 2 + \ldots + N_{\text{base}} = \frac{N_{base}(N_{base} + 1)}{2}, \tag{3.1}$$

where N_{base} is the total number of RCUs in the base row of a single pyramid. Multiple RCU stacking per row can be incorporated such that

$$N_{\text{base}} = \underbrace{\left\lceil \frac{2.6\,\text{m}}{N_{\text{stack}} \times 0.1\,\text{m}} \right\rceil}_{\text{base depth based on desired height}} \tag{3.2}$$

where N_{stack} is the number of rows with the same amount of RCUs, e.g., single, double or triple stacked. By substituting equation 3.2 into equation 3.1 and multiplying by N_{stack}, the total number of RCUs required for a particular configuration would be

$$N_{\text{pyramid}} = N_{\text{stack}} \times \frac{\left\lceil \frac{2.6\,\text{m}}{N_{\text{stack}} \times 0.1\,\text{m}} \right\rceil \left(\left\lceil \frac{2.6\,\text{m}}{N_{\text{stack}} \times 0.1\,\text{m}} \right\rceil + 1 \right)}{2}. \tag{3.3}$$

Therefore, the total number of RCUs required for the construction of the berm in the various configurations is

$$N_{\text{total}} = \underbrace{\left\lfloor \frac{\pi \times 100\text{m}}{0.43\text{m}} \right\rfloor}_{\#\text{ pyramids}} \times N_{\text{pyramid}} \,. \tag{3.4}$$

Figure 3.5 shows the number of required RCUs for the berm as a function of N_{stack}. The greatest challenge with regard to pyramid stability would be disturbances due to the manipulator operations. Therefore, the ability to stack these RCUs repeatably is taken as a requirement for the LRCS manipulator. In this thesis, 4 RCUs is taken as a reasonable and achievable stack height, beyond which the reduction in the total number of RCUs is less considerable. Consequently, $N_{\text{total}} = 81760$ RCUs are deemed to be required for the berm construction.

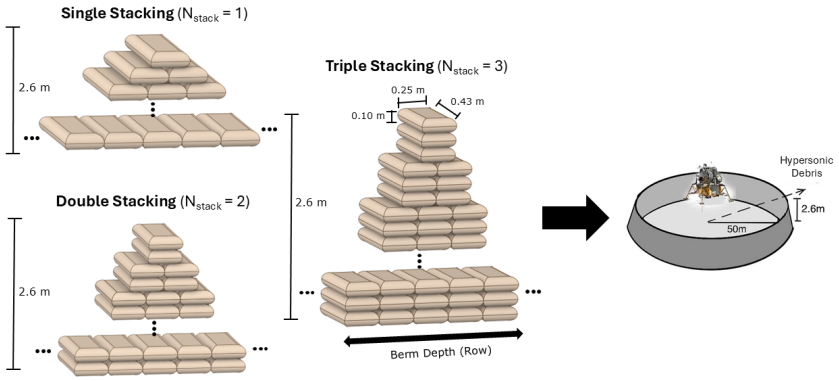


Figure 3.4: RCU Stacking Methods

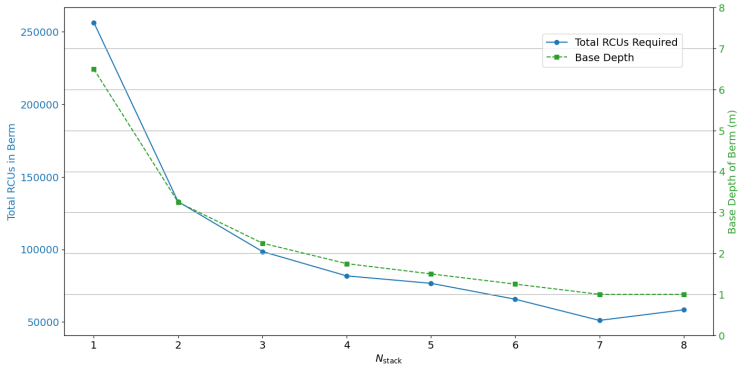


Figure 3.5: Total Number of RCUs and Berm Base Depth versus Number of Stacked RCUs per Row

The total amount of regolith required for this process will be $M_{\text{total}} = 22$ kg $\times$ 81760 RCUs $=$ 1798720 kg. If the regolith were to be removed from the landing pad area directly, for a highest

regolith density of $\rho = 1.92$ g/cm^3[31], the necessary excavation depth (D) would be

$$D = \frac{M_{\text{total}}}{\rho \times A_{\text{landing pad}}} = \frac{1798720kg}{1920\text{kg/m}^3 \times \pi(50\text{m})^2} = 0.12\,\text{m}\,. \tag{3.5}$$

Operational Trade-offs

One key operational consideration is the required frequency or speed of operation. This is highly dependent on the rate at which lunar regolith will be brought to the system by IPEx. Some properties of IPEx are summarized in Table 3.1. Of particular importance is the load-dump cycle time, which indicates that 30 kg of lunar regolith can be brought to the LRCS every 13.5 minutes. Assuming that a single RCU can be filled using one IPEx load, and that just one IPEx system is deployed, the LRCS needs to intake the lunar regolith and fill one RCU in under 13.5 minutes.

Table 3.1: IPEx Properties (Courtesy: NASA KSC)

Excavation System	Counter-rotating bucket drums
Maximum Traversable Slope	30°
Maximum Obstacle Height	7 cm
Excavator Dry Mass	35 kg
Regolith Capacity	30 kg
Load-dump Cycle Time	13.5 min
Length × Width × Height*	76 cm x 62.5 cm x 68.2 cm

* Stowed configuration - bucket drums up (see Figure 3.6)

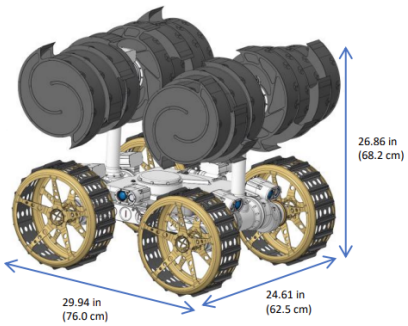


Figure 3.6: IPEx CAD Model (Courtesy: NASA KSC)

It is also important to consider the possibility of decoupling the RCU filling and manipulation tasks. In other words, the LRCS could be one stand-alone system (mobile or otherwise), or it could be configured as two separate subsystems, one completing the RCU filling task and the other subsystem being a mobile manipulator that places the RCU in the environment. A key factor for making such a decision is the environment where the LRCS is likely to be deployed. The vast majority of proposed habitat locations on the lunar surface involve the lunar North and South poles. In 2022, NASA identified 13 candidate landing regions within 6 degrees of latitude of the lunar South Pole for the Artemis missions [18]. The missions will aim to establish a sustainable human presence on the Moon with the Artemis Base Camp [2]. NASA cites the presence of in-situ resources in PSRs, particularly water ice, as well as large periods of continuous sunlight, as some of the primary considerations for these candidates [18]. The broader international community also considers establishing a presence primarily at the lunar South Pole, due to largely overlapping factors such as illumination and power

profiles, presence of volatiles in PSRs (e.g., water ice), and terrain [20, 4, 22, 21, 19].

The illumination conditions at the lunar poles are highly impacted by the local terrain, owing to the low inclination angle of the Moon's rotational axis at 1.54° [23]. This inclination creates two phenomena of interest at the lunar poles. The first phenomenon is *Peaks of Eternal Light* or *Peaks Of Light* (POL), where the terrain elevation is higher than the horizon, allowing for permanent sunlight and solar power generation. The second phenomenon involves PSRs, where there are permanent shadows in the craters, which can trap essential resources and volatiles such as water ice. An ideal location would include a region of near permanent sunlight for power generation, next to a PSR for resource extraction, and relatively flat terrain for habitat construction and landing [24, 25]. Some works have also considered the implementation of solar towers, ranging from 2 m in height up to 2000 m, to induce permanent sunlight at the solar array locations, achieving constant or near-constant illumination in some cases [26, 27]. Small scale solar towers are already being designed, such as the Lunar Array, Mast, and Power System (LAMPS), which can extend up to 15 m height [28]. Considering these factors, as well as the size of the anticipated blast berm, a decoupled system would be ideal to leverage the possibility of near-constant solar power at the South Pole, while still being able to manipulate the RCUs into the environment. This means that the lunar regolith intake and the RCU filling, along with the core functionalities of the LRCS, e.g., power generation (solar cells), communications, etc., would be undertaken by a collective assembly with limited or no mobility. Meanwhile, the RCU manipulation task would be undertaken by a mobile manipulator, with its power obtained from either an Energy Storage System (ESS) or a Radioisotope Thermoelectric Generator (RTG). Another advantage of the decoupled system is that the most massive elements of the system would not need to be mobile, which helps to reduce the overall power consumption and the risk of damage to these components. Instead, only the components necessary to transport and manipulate the RCUs would require a high degree of mobility.

The key functions that a system is required to perform to accomplish the design objectives can be identified through the functional analysis, which consists of two primary processes, namely, functional decomposition and functional allocation [141]. Functional decomposition involves hierarchically breaking down the top-level function of the system (or goal of the design) into unique sub-functions. Functional allocation will then involve the mapping of sub-functions to subsystems (or modules). The two processes can continue recursively, as depicted in Figure 3.7 (for two levels). Based on the principles of taxonomy design [142], the ending condition for each level would be to not have any further addition, merging or splitting of functions and subsystems (or modules). Further, subsystems and those functions within each subsystem should be uniquely identified.

3.1.2 Functional Analysis

In this thesis, both functional decomposition and allocation processes are performed through the glass-box method [136], which treats the system initially as a "black box" with certain defined inputs and outputs, and then incrementally reveals the box by following the flow of inputs throughout the system.

Figure 3.8 shows the first level of functional analysis for the LRCS . The primary subsystems include: RCU Filling, RCU Transfer, RCU Manipulation, Thermal Management, Power Management, Communications, Onboard Computing, and Mobility. In addition, a subsystem is included for a third-party regolith analysis and processing tool to support potential science objectives, such

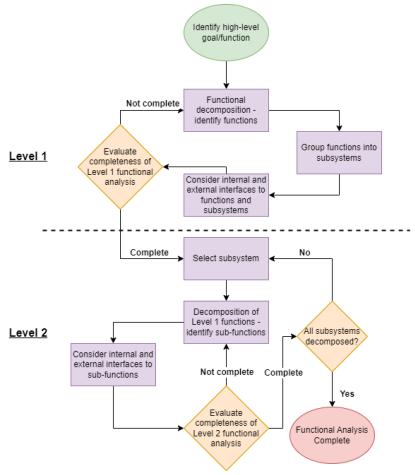


Figure 3.7: Functional Analysis Flow Chart

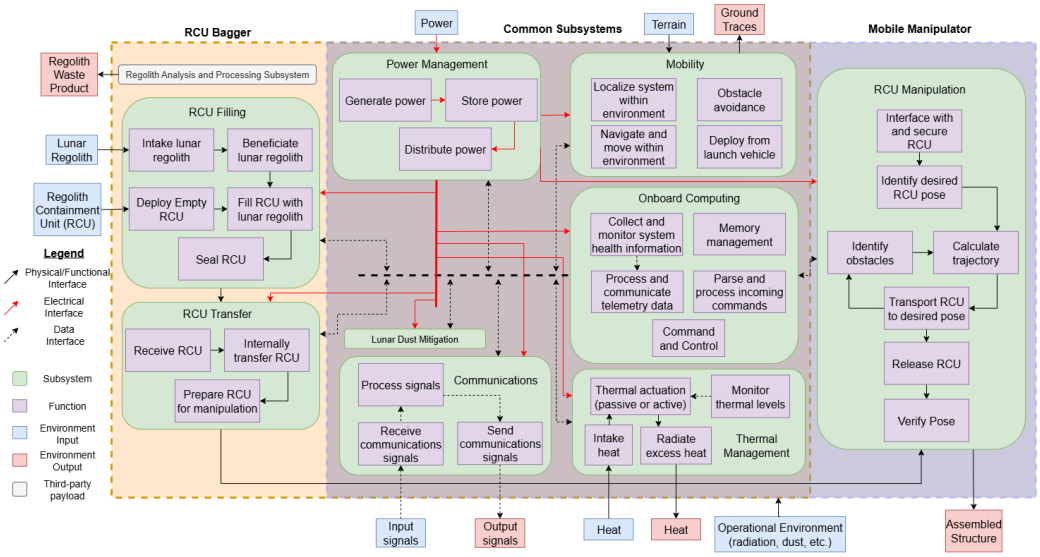


Figure 3.8: Level 1 LRCS Functional Analysis

as those in the Artemis Program plan [2]. The beneficiation of lunar regolith is a necessary function during regolith intake, involving the filtering of large rocks and particles that would constitute undesirable feedstock for the filling of the RCUs. The filtered materials will be ejected from the system as the primary form of waste, along with the heat. The Mobility subsystem will ensure a large workspace, as well as proper deployment of the system from its launch vehicle.

A number of functional modules from Figure 3.8 are selected as the primary modules for design selection: intake lunar regolith, deploy empty RCU, interface with and secure RCU, and transport RCU to desired pose. The other functional modules will be selected to support the morphology of these primary modules.

3.1.3 System Requirements

Table 3.2 displays some of the key requirements for the LRCS. The functional requirements are directly derived from the functional analysis presented in Figure 3.8. The constraints on the stowed volume are based on the properties of the Apollo Lunar Roving Vehicle (LRV) as an upper bound [143]. The LRV remains one of the largest robotic systems sent to the Moon, and is a good estimate for what we know we should be able to deliver to the lunar surface. An estimate is necessary, as most landers currently being developed for the lunar surface do not yet specify volumetric constraints.

The average daily radiation experienced on the lunar surface has been measured at 1369 μSv/day, including both solar wind and galactic cosmic rays [144]. Assuming an expected lifetime of one year, this comes to 1369 μSv/day $\times$ 365 days = 499685 μSv or 0.5 Sv. The SLS-SPEC-159 Cross-Program Design Specification for Natural Environments by MSFC [145] specifies the expected temperature ranges at an 85 degree lunar latitude. These values are rounded up for the maximum and down for the minimum to find the operable temperature range of 60 K to 190 K (based on mean temperature) and survivable temperature range of 40 K to 230 K (based on 1 Sigma temperature). The LRCS will also need to withstand an ambient pressure of 3×10^{-15} atm [29], a maximum solar irradiance of 1425.7 W/m^2 [146], and exposure to lunar regolith dust [147].

According to the U.S. Army Corps of Engineers [148], the most commonly used sandbags for terrestrial missions measure 35.56 cm $\times$ 60.96 cm on a flat surface before filling. The corresponding sandbag unit filled to two-thirds full capacity (for desired engineering properties) measures 10.16 cm $\times$ 25.4 cm $\times$ 43.18 cm, or a volume V = 11143.20 cm^3. Considering the highest possible density for lunar regolith (worst case) of ρ = 1.92 g/cm^3 [41], the estimated mass M for the RCU can be obtained as:

$$M = \rho \times V \approx 22\ kg \tag{3.6}$$

Considering that IPEx holds a maximum load of 30 kg (see Table 3.1), this represents approximately a 26.7% margin between a single anticipated IPEx load and the required regolith to form one RCU.

3.2 Design Assessment: Methodology

The potential morphologies for major functional modules are assessed using the Analytic Hierarchy Process (AHP) method [137]. The method is a value-based decision-making process, where the value of a candidate solution with respect to each criterion and the weight of the criteria are obtained based on pair-wise comparisons. Specifically, it involves quantitative representation and evaluation of the candidates for each module based on their relative preference (RP) with respect to each design criterion, as well as the relative importance (RI) of the criteria. The first step is to quantify the design criteria that are to be used for the evaluation, which are identified by the proposed design requirements. The pair-wise RP evaluation of morphologies is then performed with respect to each criterion, as well as the pair-wise RI evaluation of the criteria. All comparisons are conducted on an ordinal 1-to-9 scale (see Table 3.3.)

The pair-wise RI of P criteria is entered into a $P \times P$ matrix S, and each element s_{ij} is then column-normalized:

$$m_{ij} = \frac{s_{ij}}{\sum_{i=1}^{P} s_{ij}}, \tag{3.7}$$

Table 3.2: System Requirements (with some modifications)

ID	**Requirement:** The system shall...
	Functional Requirements
SYS-F-01	intake lunar regolith from an external lunar regolith excavation and processing system.
SYS-F-02	beneficiate lunar regolith.
SYS-F-03	transfer excess regolith to a third-party Regolith and Analysis Processing Subsystem.
SYS-F-04	fill the RCUs with lunar regolith.
SYS-F-05	seal the RCUs.
SYS-F-06	internally transfer the RCUs from the filling configuration to the manipulation configuration.
SYS-F-07	be decoupled into a mobile manipulation system and a centralized RCU filling system.
SYS-F-08	interface with and secure the RCUs for manipulation.
SYS-F-09	manipulate the RCUs into a desired pose within the environment.
SYS-F-10	be able to move and localize itself in the lunar environment.
SYS-F-11	have onboard computing capabilities.
SYS-F-12	collect and store system health and history data.
SYS-F-13	be able to communicate with lunar surface and orbital communication systems.
SYS-F-14	generate, store, and manage sufficient power for operation during the operational lifetime.
SYS-F-15	be able to maintain its thermal state within operable ranges.
SYS-F-16	be capable of operating in the lunar environment.
SYS-F-17	be deployable from a launch vehicle to the lunar surface.
	Performance Requirements
SYS-P-01	support a sustained operational (average) power draw of at least 1100 W (see Section 4.6.1).
SYS-P-02	support a peak operational power draw of at least 2600 W for 1 min (see Section 4.6.1).
SYS-P-03	be capable of filling and sealing 1 RCU in under 2.7 minutes (see Section 4.1).
SYS-P-04	be capable of manipulating an average of 1 RCU into the berm in under 163 seconds (see Section 4.1).
SYS-P-05	have an operational lifetime of 6 months (see Section 4.1).
SYS-P-06	have a total battery charge of at least 110 Wh (see Section 4.6.1).
SYS-P-07	be capable of reaching a manipulation workspace volume of 2m $\times$ 2.6m $\times$ 2m (depth, height, width).
SYS-P-08	be capable of reaching a mobility workspace area of $\pi \times (50\text{m})^2 = 2500$ m^2.
SYS-P-09	be capable intaking at least 60 kg of lunar regolith (double IPEx capacity).
SYS-P-10	be capable of bagging and stacking 81760 RCUs (see Section 3.1.1).
SYS-P-11	be capable of repeatably stacking 4 RCUs vertically 20440 times (see Section 3.1.1).
	Environmental Requirements
SYS-E-01	have an operational temperature range of 60 K to 190 K [145].
SYS-E-02	have a survivable temperature range of 40 K to 230 K [145].
SYS-E-03	maintain the temperature of electronics between 233 K and 358 K.
SYS-E-04	be capable of withstanding a total radiation dose of at least 0.5 Sv.
SYS-E-05	operate while exposed to a maximum solar irradiance of 1425.7 W/m^2 [146].
SYS-E-06	operate while exposed to lunar regolith and lunar regolith dust [147].
SYS-E-07	operate in an ambient pressure of 3×10^{-15} atm [29].
	Constraints
SYS-C-01	have a maximum mass of 810 kg (for medium class lander).
SYS-C-02	have a maximum stowed volume of 6 m^3.
SYS-C-03	use non-rigid RCUs with dimensions of 0.43m $\times$ 0.25m $\times$ 0.10m.
SYS-C-04	use RCUs with a mass of 22 kg.
SYS-C-05	have a regolith intake interface of dimension 0.625m $\times$ 0.312m to accommodate IPEx.

Table 3.3: AHP Ordinal Scale

Value	Description of RP/RI
1	Equally preferred/important
3	Moderately preferred/important
5	Strongly preferred/important
7	Very strongly preferred/important
9	Extremely preferred/important
2, 4, 6, 8	Intermediate values

and each row is averaged:

$$t_i = \frac{1}{P}\sum_{j=1}^{P} m_{ij}. \tag{3.8}$$

The value t_i $(i = 1, 2, \ldots, P)$ is the RI value for criterion i. Similarly, for each criterion, the pair-wise RP of N morphologies is entered into an $N \times N$ matrix R, and each element r_{kl} is then column-normalized:

$$n_{kl}^{i} = \frac{r_{kl}^{i}}{\sum_{k=1}^{N} r_{kl}}, \tag{3.9}$$

and each row is averaged:

$$f_k^i = \frac{1}{N}\sum_{l=1}^{N} n_{kl}. \tag{3.10}$$

The value f_k^i $(i = 1, 2, \ldots, P\ ;\ k = 1, 2, \ldots, N\ )$ is the RP value for morphology k with respect to criterion i.

The RP and RI values are then used to produce the overall decision value D_k of each candidate morphology k:

$$D_k = t_1 \times f_k^1 + t_2 \times f_k^2 + \cdots + t_P \times f_k^P, \quad (k = 1, 2, \ldots, N). \tag{3.11}$$

To ensure coherent and rational decision-making throughout the process, the logical consistency of the pair-wise rankings made at each step is checked by a quantitative consistency index. An ideally consistent pair-wise ranking of three elements a, b and c should follow the transitivity relationship, given that an element is equally preferred/important to itself:

$$h_{ac} = h_{ab} \times h_{bc}, \tag{3.12}$$

where h_{ac} is the relative ranking of element a to c. The consistency of a ranking matrix can then be measured by a *Consistency Index* (CI) computed as follows:

$$CI = \frac{\lambda_{max} - q}{q - 1}, \tag{3.13}$$

where λ_{max} is the largest eigenvalue of the ranking matrix, and q is the number of morphologies N or criteria P. It can be shown that for an ideally consistent ranking matrix the consistency index CI is equal to zero. However, in practical cases a ranking matrix can hardly end up ideally consistent, and the ranking can still be acceptable if CI is reasonably smaller that that of a matrix of the same dimension with all entries randomly selected from the 1-to-9 range. Such a random matrix can numerically be constructed repeatedly, and it can be shown that for each dimension q the consistency index $(CI)_R$ of the matrix converges to a value shown in Table 3.4. Heuristically, a

ranking matrix can be considered acceptable if

$$CR = \frac{CI}{(CI)_R} \leq 0.10, \tag{3.14}$$

where CR is called the *Consistency Ratio* (CR) [149].

Table 3.4: $(CI)_R$ Values for Consistency Ratio [137, 150]

q	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
$(CI)_R$	0	0.58	0.9	1.12	1.24	1.32	1.41	1.45	1.49	1.51	1.48	1.56	1.57	1.59	1.605	1.61	1.615	1.62	1.625

Examples of the AHP process can be found in [151] and [152]. In this thesis, the AHP process is conducted in a sequential fashion, following the flow of regolith through the system and evaluating at each stage.

3.2.1 Expert Criteria Prioritization and Ranking Aggregation

To help establish the RI of the criteria, a survey of experts is first conducted to obtain their ranking of the criteria. Then, the experts rankings are aggregated to achieve a set of output importance grades to inform the assignment of criteria priorities [153]. Consider M experts who express their judgments on the ranking of P criteria. These judgments are to be expressed as a set of *strict* and *complete* subjective rankings. Strict ranking does not permit ties or indifference relationships, and complete ranking implies that experts have ranked every criterion.

this thesis utilizes a ranking aggregation technique, called ZM_{II}, to produce the set of aggregated criteria rankings [138, 139]. The ZM_{II} technique is selected since it achieves relatively high aggregated consensus (equations (3.29) and (3.30)), and it outputs a ratio scaling of the criteria, which can be used to better inform the RI values than simply ordering the criteria, as well as producing an ideally consistent RI matrix (double check) [140, 154]. Further, the ZM_{II} method does not require pairwise comparisons, reducing the degree of subjectivity of the process and the required cognitive load on experts to collect data. The technique is an improvement to the Thurstone's Law of Comparative Judgment (LCJ) [155, 156] by introducing anchor elements C_t (M-element) and C_b (Z-element), which represent the most and least important (dummy) criteria, respectively. This means that no criterion should be positioned higher than C_t or lower than C_b. Consequently, the total number of criteria in the ZM_{II} ranking process is considered as $(P + 2)$.

The ZM_{II} method is a random-distribution-based technique, which is formulated by considering a random value c_i for the ranking position of each criterion i. The position of each criterion is postulated to be distributed normally, such that $c_i \tilde{} \mathcal{N}(\mu_i, \sigma_i^2)$, where μ_i and σ_i^2 are the (unknown) mean value and variance, respectively, in terms of the ranking of the criterion. Then, assuming that the variance (and covariance) being the same for all c_i's (and c_j's), it can be asserted that [140]:

$$p_{ij} = \mathcal{P}[(c_i - c_j) > 0] = 1 - \Phi[-(\mu_i - \mu_j)], \tag{3.15}$$

where p_{ij} is the probability that the value of c_i is higher than c_j, with Φ being the cumulative distribution function (CDF) of the right-tailed standard normal distribution $\mathcal{N}(\mu_i, \sigma_i^2)$, where $\mu_i = 0$ and $\sigma_i^2 = 1$. The value of p_{ij} can be estimated by its mean $\bar{p}_{ij} = \frac{k_{ij}}{m_{ij}}$, where k_{ij} is the total comparison score across all experts between C_i and C_j, and m_{ij} is the number of experts who

ranked criteria i and j. For a given comparison by an expert, a score of 1 is assigned for strict preference ($c_i > c_j$) and a score of 0.5 for indifference ($c_i = c_j$). Further, in the case of *complete* ranking, m_{ij} equals the total number of experts M.

By re-arranging equation (3.15), we express each pairwise comparison as a linear constraint between the unknown mean rankings μ_i and μ_j, such that

$$\mu_i - \mu_j = -z_{ij}, \quad \text{where } z_{ij} = \Phi^{-1}[1 - \bar{p}_{ij}], \tag{3.16}$$

forms the skew-symmetric matrix $\mathbf{Z} \in \mathbb{R}^{(P+2)\times(P+2)}$. The model breaks down if all experts express the same judgments (CDF approaches $\pm\infty$), which is avoided by the simple approximation $\Phi^{-1}[1 - \bar{p}_{ij}] = -1.995$ for $\bar{p}_{ij} \geq 0.977$, or $\Phi^{-1}[1 - \bar{p}_{ij}] = 1.995$ for $\bar{p}_{ij} \leq 0.023$. For each pair of criteria, equation (3.16) corresponds to a single expert-derived constraint between them, and the complete system is constructed from all comparisons:

$$\mathcal{C} = \{(i, j) \mid 1 \leq i < j \leq (P + 2)\},$$

where $(P + 2)$ is the total number of criteria, including the two dummy criteria that act as anchor elements. The total number of comparisons is $|\mathcal{C}| = Q$, and each comparison yields one row of an overdetermined set of linear equations:

$$\tilde{\mathbf{A}}\mathbf{X} = \tilde{\mathbf{B}}, \quad \tilde{\mathbf{A}} \in \mathbb{R}^{Q\times(P+2)}, \quad \tilde{\mathbf{B}} \in \mathbb{R}^{Q\times 1}, \tag{3.17}$$

where $\mathbf{X} = [\mu_b, \mu_t, \mu_1, \mu_2, \ldots, \mu_P]^T \in \mathbb{R}^{(P+2)\times 1}$ is the column vector of unknown mean rankings, and μ_t and μ_b are the obtained mean values for the dummy criteria C_t and C_b, respectively. Each row of $\tilde{\mathbf{A}}$ encodes a constraint for some pair $(i, j) \in \mathcal{C}$, with $+1$ at position i, -1 at position j, and all other entries zero. From equation (3.16), the corresponding entry in $\tilde{\mathbf{B}}$ is $-z_{ij}$. To produce a final square system of the form

$$\mathbf{A}\mathbf{X} = \mathbf{B}, \tag{3.18}$$

the constraint pairs in $\tilde{\mathbf{A}}$ are collapsed into a Laplacian matrix $\mathbf{A} \in \mathbb{R}^{(P+2)\times(P+2)}$, such that each row i of $\mathbf{A}$ aggregates all the constraints for criterion i. In the case of *strict* and *complete* rankings, this results in a symmetric matrix $\mathbf{A}$ with -1 in all off-diagonal entries, and $(P + 1)$ in all diagonal entries. The column matrix $\mathbf{B} \in \mathbb{R}^{(P+2)\times 1}$ is the resulting sum of the constraints, effectively meaning that $b_j = \sum_{i=1}^{(P+2)} z_{ij}$. The system (3.18) can be solved in a closed form to obtain an estimate of the mean of the criteria rankings [140]:

$$\mathbf{X} = (\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T\mathbf{B}. \tag{3.19}$$

The solution (3.19) assumes that the uncertainties for elements of $\mathbf{X}$ are identical and they do not affect each other, which is not true in real-life rankings. To address these shortcomings, the solution can be modified by applying the Generalized Least Squares (GLS) method [157]:

$$\mathbf{X} = (\mathbf{A}^T\mathbf{W}\mathbf{A})^{-1}\mathbf{A}^T\mathbf{W}\mathbf{B}. \tag{3.20}$$

The matrix $\mathbf{W}$ is constructed as follows [138]:

$$\mathbf{W} = [\mathbf{J_p}\mathbf{\Sigma_p}\mathbf{J_p}^T]^{-1}\,, \tag{3.21}$$

where the Jacobian matrix is approximated by

$$\mathbf{J_p} = \frac{\partial[\Phi^{-1}(1-\bar{p}_{ij})]}{\partial \bar{p}_{ij}} \approx \begin{cases} \left| \dfrac{2\sqrt{2}\bar{p}_{ij} - \sqrt{2}}{\bar{p}_{ij}(1-\bar{p}_{ij})\sqrt{-2\sqrt{2\pi}[\ln(4) + \ln(\bar{p}_{ij} - \bar{p}_{ij}^2)]}} \right|, & \text{if } \bar{p}_{ij} \neq 0.5 \\ 2.506628, & \text{if } \bar{p}_{ij} = 0.5 \end{cases} \tag{3.22}$$

and the covariance matrix $\mathbf{\Sigma_p}$ is a diagonal matrix (assuming that all p_{ij} estimates are statistically independent), whose diagonal elements are the variance of p_{ij}:

$$\sigma^2_{p_{ij}} = \frac{\bar{p}_{ij}(1-\bar{p}_{ij})}{m_{ij}}. \tag{3.23}$$

.

The system of equations (3.20) can now be solved for $\mathbf{X}$. To scale the solutions into a $[0, K]$ range, the following transformation is applied:

$$\mathbf{X}' = \frac{K}{\mu_t - \mu_b}[0, (\mu_t - \mu_b), (\mu_1 - \mu_b), (\mu_2 - \mu_b), \ldots, (\mu_P - \mu_b)]^T. \tag{3.24}$$

The calculated mean values ($\mathbf{X}$ or $\mathbf{X}'$) for the criteria can be used to recover an aggregated ranking by simply ordering them from the highest mean value (top rank) to the lowest (bottom rank).

The uncertainty in the estimation of mean values of the criteria ranking can be quantified by using the Multivariate Law of Propagation of Uncertainty (MLPU) [157]:

$$\Sigma_{\mathbf{X}} = (\mathbf{A}^T\mathbf{W}\mathbf{A})^{-1}, \tag{3.25}$$

and using the transformation relation yields:

$$\Sigma_{\mathbf{X}'} = \mathbf{J}_{\mathbf{X}'}\Sigma_{\mathbf{X}}\mathbf{J}^T_{\mathbf{X}'}\,. \tag{3.26}$$

The Jacobian in equation (3.26) can be constructed as [138]:

$$\mathbf{J}_{\mathbf{X}'} = \begin{bmatrix} \frac{\partial x'_b}{\partial x_b} & \frac{\partial x'_b}{\partial x_t} & \frac{\partial x'_b}{\partial x_1} & \cdots & \frac{\partial x'_b}{\partial x_P} \\ \frac{\partial x'_t}{\partial x_b} & \frac{\partial x'_t}{\partial x'_t} & \frac{\partial x'_t}{\partial x_1} & \cdots & \frac{\partial x'_t}{\partial x_P} \\ \frac{\partial x'_1}{\partial x_b} & \frac{\partial x'_1}{\partial x_t} & \frac{\partial x'_1}{\partial x_1} & \cdots & \frac{\partial x'_1}{\partial x_P} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \frac{\partial x'_P}{\partial x_b} & \frac{\partial x'_P}{\partial x_t} & \frac{\partial x'_P}{\partial x_1} & \cdots & \frac{\partial x'_P}{\partial x_P} \end{bmatrix} = \frac{K}{\mu_t - \mu_b} \begin{bmatrix} 0 & 0 & 0 & \cdots & 0 \\ 0 & 0 & 0 & \cdots & 0 \\ -\frac{\mu_t - \mu_1}{\mu_t - \mu_b} & -\frac{\mu_1 - \mu_b}{\mu_t - \mu_b} & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ -\frac{\mu_t - \mu_P}{\mu_t - \mu_b} & -\frac{\mu_P - \mu_b}{\mu_t - \mu_b} & 0 & \cdots & 1 \end{bmatrix} \tag{3.27}$$

A 95% confidence interval can then be calculated assuming a normal distribution for p_{ij} and x'_i:

$$\text{confidence interval} = x'_i \pm 2\sigma'_i, \quad i = 1, 2, \ldots, P \tag{3.28}$$

where $\sigma'_i = \sqrt{\Sigma_{\mathbf{X}'(i,i)}}$.

A variety of methods can be employed to examine the *degree of consensus*, i.e., the agreement between aggregated ranking and those provided by the experts. this thesis uses a global p-indicator that quantifies the degree of consensus by calculating a ratio of the number of comparison agreements between individual expert rankings and the aggregated ranking over the total number of pairwise comparisons across all experts [154]:

$$p = \frac{\sum_{k=1}^{M} a_k}{M \times \binom{P}{2}} = \frac{1}{M} \sum_{k=1}^{M} \left[\frac{a_k}{\binom{P}{2}} \right], \tag{3.29}$$

where a_k is the sum of the agreements for $\binom{P}{2} = \frac{P(P-1)}{2}$ pairwise comparisons between the k-th expert's ranking and the aggregated ranking. Consequently, this indicator varies between 0 for full disagreement and 1 for full agreement between the expert rankings and the aggregated ranking.

Further, it is possible to quantify the degree of inter-expert agreement using Kendall's *coefficient of concordance* for strict and complete rankings [139, 158]:

$$w = \frac{12(\sum_{i=1}^{P} R_i^2) - 3M^2P(P+1)^2}{M^2P(P^2-1)}, \tag{3.30}$$

where R_i is the sum of the rank positions (e.g., 1, 2, ..., or P) of the i-th criterion across all experts. This indicator also varies between 0 for full disagreement and 1 for full agreement between the expert rankings.

3.2.2 Criteria Prioritization and Evaluation

Four core functional modules from the functional analysis in Figure 3.8 are selected for expert evaluation and further analysis, namely, intake lunar regolith (FM-1), deploy empty RCU (FM-2), interface with and secure RCU (FM-3), and transport RCU to desired pose (FM-4). A total of 47 engineers working in aerospace were surveyed on criteria for each of the four selected functional modules. The pool consists primarily of engineers from MDA Space, NASA, and University of Toronto affiliates. The expert pool breakdown by specialization can be seen in Figure 3.9, and by years of experience in Figure 3.10. There is a fair distribution in specializations, with most respondents specializing in GNC (Guidance, Navigation and Control) or Mechanical engineering. Meanwhile, the number of respondents has a reverse relationship with the experience duration of experts, which is to be expected as it is more difficult to find and get a hold of the most experienced engineers, i.e., over 10-15 years experience, who may be in more prominent positions.]

The collective rankings of the selected criteria for each of the four selected modules are shown in Table 3.5. It can be observed that criteria such as reliability, longevity, exposure to dust, and complexity are consistently ranked in positions 1-4. This can be attributed to the fact that producing fail-safe systems is a priority in the aerospace industry (from which experts were pooled), and that these criteria appear relatively similar without additional information being presented to the survey respondents. The remaining criteria experience a fair amount of variability in their ranking positions, although it can be noted that module-specific criteria, e.g., "Delivery flow rate" for FM-1, are generally ranked near the bottom.

Although the w and p indicators cannot be directly compared, one can generally observe that there was a low inter-expert concordance (below 35% for w) before applying the aggregation tech-

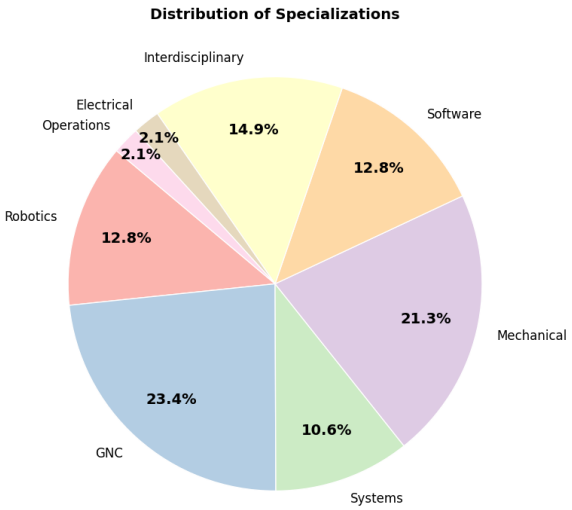


Figure 3.9: Expert Pool Specialization Breakdown

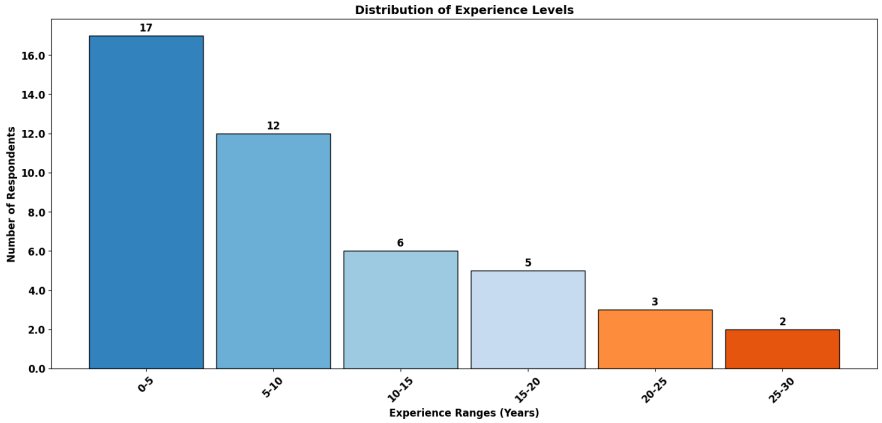


Figure 3.10: Expert Pool Experience Breakdown

nique. After applying the ZM_{II} aggregation technique, the p-indicator lies in the range from 65% to 75%, indicating a relatively good performance for the aggregation technique. The variance in the initial expert responses (represented by w) inherently limits how much the p-indicator can tend towards 100% agreement with the collective ranking.

The aggregated rankings are used to determine the RI of the criteria for the AHP, as explained in the subsequent sections. Since the scaled aggregated mean values of expert rankings $\mathbf{X}'$, derived from the ZM_{II} method, are ratio scale representations of the expert preferences, by normalizing them they can effectively replace the normalized RI values t_i in the AHP (see equation (3.8)):

$$\mathbf{T} = [\frac{x'_1}{\sum_{i=1}^{P} x'_i}, \ldots, \frac{x'_P}{\sum_{i=1}^{P} x'_i}]^T = \frac{K}{(\mu_t - \mu_b)\sum_{i=1}^{P} x'_i}[(\mu_1 - \mu_b), \ldots, (\mu_P - \mu_b)]^T \tag{3.31}$$

where $\mathbf{T} \in \mathbb{R}^{P\times 1}$ is a column vector containing all the normalized RI values for the P criteria (ignoring dummy criteria). The normalization maintains the relative ratios between the criteria, preserving the ratio scale. The new RI values are used directly when calculating the decision values

Table 3.5: Collective Rankings Produced by ZM_{II} Aggregation Method

Rank	FM-1	FM-2	FM-3	FM-4
1	**Reliability**: 66.8 ± 34.3	**Reliability**: 67.0 ± 36.1	**Reliability**: 70.1 ± 32.5	**Reliability**: 67.6 ± 30.7
2	**Longevity**: 59.9 ± 31.6	**Longevity**: 63.5 ± 34.6	**Longevity**: 64.2 ± 30.4	**Longevity**: 62.8 ± 29.0
3	**Exposure to dust**: 54.4 ± 29.3	**Complexity**: 50.9 ± 29.2	**Exposure to dust**: 57.4 ± 27.9	**Exposure to dust**: 58.9 ± 27.5
4	**Complexity**: 52.6 ± 28.5	**Exposure to dust**: 50.1 ± 28.9	**Complexity**: 55.0 ± 26.9	**Complexity**: 51.5 ± 24.6
5	**Mass**: 50.4 ± 27.5	**Power**: 48.1 ± 27.7	**Power**: 50.5 ± 24.8	**Workspace - module volume ratio**: 49.3 ± 23.6
6	**Dust production**: 46.8 ± 25.9	**Mass**: 47.3 ± 27.5	**Dust production**: 49.2 ± 24.5	**Power**: 48.9 ± 23.4
7	**Power**: 46.3 ± 25.5	**Storage volume**: 40.4 ± 24.2	**Mass**: 48.0 ± 24.0	**Manipulation accuracy**: 47.1 ± 22.7
8	**Regolith compaction**: 36.4 ± 21.0	**Transport speed**: 32.7 ± 20.2	**Attachment force**: 47.0 ± 23.4	**Dust production**: 46.5 ± 22.5
9	**Delivery flow rate**: 36.2 ± 20.7	-	**Rigid attachment**: 46.7 ± 23.2	**Mass**: 45.5 ± 22.0
10	-	-	**Attachment points**: 32.6 ± 17.0	**Structural rigidity**: 41.4 ± 20.3
11	-	-	**Attachment speed**: 29.3 ± 15.5	**Dexterity**: 40.8 ± 20.1
12	-	-	-	**Manipulation resolution**: 39.6 ± 19.5
w	0.257	0.311	0.332	0.204
p	0.689	0.713	0.730	0.661

in equation (3.11).

It is worth noting that the ZM_{II} method offers a way to reconcile non-strict (tie-permitting) and incomplete rankings with the AHP, allowing for some information loss due to the lack of pairwise comparisons. Typically, it would not be possible to have incomplete pairwise comparisons for the AHP, but the ZM_{II} method allows for a ratio scale output using a variety of input formats. This can make the data collection process easier, since it avoids the need for experts to conduct the tedious number of pairwise comparisons conventionally required for the AHP.

To produce the RP matrices of solutions with regard to the criteria, each criterion is quantified by a metric obtained from physical characteristics of the design problem. Given some metric evaluation $\bar{r}_k$ for a candidate solution k ($k = 1, \ldots, N$), the metric evaluations are adjusted to a 1-to-9 scale:

$$Q(\bar{r}_k) = 1 + \frac{\bar{r}_k - \bar{r}_{min}}{\bar{r}_{max} - \bar{r}_{min}} \times (9 - 1)\,. \tag{3.32}$$

Then, the AHP ratio for an RP matrix $\mathbf{R}$ is formulated as

$$r_{kl} \approx \frac{Q(\bar{r}_k)}{Q(\bar{r}_l)}, \quad k, l = 1, \ldots, N \tag{3.33}$$

The entries r_{kl} are rounded to the nearest AHP 1-9 scale value (see Table 3.3). This would form the RP matrix for a given criterion, adjusting all metric comparisons to the AHP 1-9 scale.

3.3 Design Assessment: Application

3.3.1 Intake Lunar Regolith

The RCU Filling subsystem consists of five functional modules, namely, regolith intake, beneficiation, filling, sealing, and deploying the empty RCU (see Figure 3.8). A primary morphological assessment of the RCU Filling subsystem is with regard to the intake mechanism, which directly manipulates the lunar regolith toward the filling module. The criteria for evaluation of the intake module, along with the normalized RI of the criteria (calculated as described in Section 3.2.2), are shown in Table 3.6.

The candidate solutions for the intake module are depicted and described in Table 3.7. These candidates focus on keeping the regolith consistently flowing to prevent the compaction and high-interlocking behaviour of the lunar regolith. A number of candidates have been inspired from the study in [48] regarding lunar dust mitigation and lunar regolith conveyance (and intake) technologies. Of particular interest is the success of hopper and auger systems as well as vibrating surfaces and bowl feeders. Another study specifically designed an RCU creation and stacking system, using a funnel and propeller shaft for sand (regolith analogue) intake [79]. Other studies have successfully implemented auger [159, 160] and vibration [161, 162, 160] conveyance of lunar regolith simulants. Furthermore, terrestrial sandbaggers are also considered, which largely follow a hopper and auger or funnel morphology. Examples include the Sandbagger [163] and the PanPac Engineering end-to-end palletization system [164]. Some works have also investigated funnel systems for the conveyance of lunar regolith simulant, successfully demonstrating their feasibility [165, 166]. Finally, front loaders are commonly used mechanisms for moving soil in terrestrial applications, and have been tested for lunar excavation purposes [81].

The calculations and formulations for some of the more involved metrics are detailed in the following subsections.

CR-F-3: Longevity

The longevity of the candidate solutions is considered as a subset of reliability in terms of anticipated failure rate. There exist many handbooks detailing methods for the theoretical calculation of failure rates for electro-mechanical systems, largely based on empirical studies of components and their failures. Notably, the MIL-217F handbook [168] by the American Department of Defense is a cornerstone often used in these estimations. However, the last release of the handbook was in 1991 (Notice 2), and some studies have examined its obsolescence [169]. Some newer handbooks, such as the NSWC-11 handbook [170] by the American Naval Surface Warface Center, have since been released. Other recent handbooks also exist, such as the Toshiba TDSC Reliability Handbook [171] and the 217 Plus Handbook [172] (a privately developed update to MIL-217F), but these largely focus on the reliability of electrical and semiconductor components. The newest and ongoing attempt to create a reliability standard is FIDES [173], with the latest edition being released in 2022. However, FIDES requires a registration process, and its methodology has been criticized by works such as [174]. The NSWC-11 handbook was found to be the most recent, relevant and reliable source for the failure rate estimation of our electro-mechanical candidate solutions.

In general, the NWSC-11 handbook defines the failure of components by establishing the base failure rate and multiplying this base rate by related factors. These failure rates can be summed up

Table 3.6: Evaluation Criteria for the Intake Module

ID	Criteria	Description	Metric	Goal	RI (t_i)
C-I-1	Exposure to dust	The number of interfaces between moving parts, which are exposed to the regolith.	# of interfaces	Min	0.121
C-I-2	Dust production	The potential for dust production due to agitation, sharp movements, dumping of the regolith, etc.	ordinal scale (Likert 1-5)	Min	0.104
C-I-3	Longevity	The anticipated failure rate of the module based on motor load and transmission [167].	failures per million operating hours	Min	0.133
C-I-4	Reliability	The number of independent points of single failure of the module.	# of points	Min	0.148
C-I-5	Power	The power required to move 30 kg of regolith into the subsystem; calculated as a function of the work required to move the regolith per the anticipated required time according to the delivery flow rate.	Watts	Min	0.103
C-I-6	Mass	The anticipated mass of the module; calculated by taking the estimated surface area of the candidate solution and assuming an average thickness of aluminum of 0.01 m for the structure.	kilograms	Min	0.112
C-I-7	Complexity	The module complexity is measured as the total number of moving parts. This includes motors, gears, vibrating surfaces, and other parts moving in the workspace.	# of moving parts	Min	0.117
C-I-8	Delivery flow rate	The estimated delivery flow rate of the regolith into the filling module.	kilograms per minute	Max	0.0805
C-I-9	Regolith compaction	The anticipated degree of regolith compaction caused by the module, which occurs when inter-particle compression forces are increased.	ordinal scale (Likert 1-5)	Min	0.0809

to approximate the failure rate of a larger system or complex components. Within the context of the design, we consider the reliability of the gears (transmission) and the motor, which is assumed to be a brushless DC motor (common for space applications). The failure rate of gears is defined as

$$\lambda_G = \lambda_{G,B} \cdot C_{GS} \cdot C_{GP} \cdot C_{GA} \cdot C_{GL} \cdot C_{GT} \cdot C_{GF}, \tag{3.34}$$

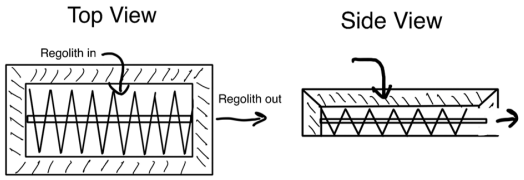
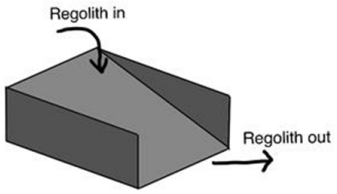
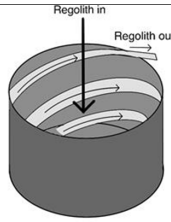
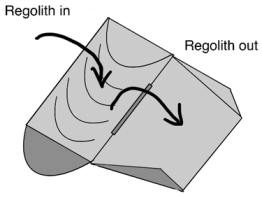
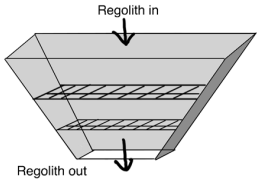
where λ_G is the failure rate of the gear under specific conditions, $\lambda_{G,B}$ is the base failure rate of the gear, and the C terms relate to the operating conditions (see the handbook for full details). All failure rates are considered in terms of failures/million operating hours. The failure rate of a motor is defined as

$$\lambda_M = (\lambda_{M,B} \cdot C_{SF}) + \lambda_{WI} + \lambda_{BS} + \lambda_{ST} + \lambda_{AS} + \lambda_{BE} + \lambda_{GR} + \lambda_C, \tag{3.35}$$

where λ_M is the total failure rate of the motor system, $\lambda_{M,B}$ is the base failure rate of the motor, C_{SF} is the motor load service factor, and the other λ terms relate to the additional components of the motor system (see the handbook for full details). Considering the current stage of design, many factors to perform the full estimation from the handbook are yet to be determined, so we consider a simplified model for the candidate designs

$$\lambda_C = n_{motors} \cdot \lambda_{M,B} \cdot C_{SF} + n_{gears} \cdot \lambda_{G,B}, \tag{3.36}$$

Table 3.7: Intake Mechanism Candidate Solutions

	Concept	Description
Auger and Hopper		A hopper which collects the lunar regolith from IPEx, using gravity to feed downward towards a horizontal auger which forces the regolith into the system.
Vibrating Intake Chute		Leverages the angular vibration of a surface to supplement gravity in feeding the lunar regolith down a slope. The vibration and large surface area of the feeding surface allows for regolith to remain well distributed.
Vibratory Bowl Feeder		Similar to the Vibrating Intake Chute, this concept allows for regolith to be dumped into its centre and leverages vibration to propagate regolith upwards along a feeding path around its cylindrical walls. This method is commonly used for dispensing in terrestrial factory settings.
Front Loader		A traditional loading method, often used in excavation systems, where regolith is dumped into a stationary container, which is then rotated upwards until the regolith falls into the system.
Vibrating Funnel		A funnel which collects regolith from the excavator and propagates it towards a smaller exit leading into the system. This concept also involves a couple sparse grating layers, combined with vibration, which aim to prevent compaction and promote the effective flow of regolith.

where λ_C is the estimated failure rate of the candidate design, n_{motors} is the total number of motors, and n_{gears} is the total number of gears.

All candidate designs for the intake mechanism are considered to have only one brushless DC motor with a base failure rate of 1.75 failures/million operating hours. However, the service factor on these motors varies. The Auger and Hopper System is considered to have a uniform load due to continuous and consistent operation ($C_{SF} = 1.00$). The Front Loader is considered to have a medium impact due to its frequently reversible motor operation and moderate shock during loading and unloading ($C_{SF} = 2.00$). The Vibrating Intake Chute, Vibratory Bowl Feeder, and Vibrating Funnel are considered to have a heavy load due to heavy vibration ($C_{SF} = 3.00$). Only the Auger and Hopper System as well as the Front Loader are estimated to necessitate a transmission due to lower operating speeds. Both of these candidates are estimated to have two gears as a minimum, each with a base failure rate of 1 failure/million operating hours. These formulations allow for the

estimation of the failure rates for the candidates as follows:

$$\lambda_{\text{Auger/Hopper}} = 1 \cdot 1.75 \cdot 1.00 + 2 \cdot 1 = 3.75 \tag{3.37}$$

$$\lambda_{\text{Chute}} = 1 \cdot 1.75 \cdot 3.00 + 0 \cdot 1 = 5.25 \tag{3.38}$$

$$\lambda_{\text{Bowl Feeder}} = 1 \cdot 1.75 \cdot 3.00 + 0 \cdot 1 = 5.25 \tag{3.39}$$

$$\lambda_{\text{Front Loader}} = 1 \cdot 1.75 \cdot 2.00 + 2 \cdot 1 = 5.50 \tag{3.40}$$

$$\lambda_{\text{Funnel}} = 1 \cdot 1.75 \cdot 3.00 + 0 \cdot 1 = 5.25. \tag{3.41}$$

CR-F-5: Power

Although it is challenging to accurately estimate the power consumption of the candidate designs without specific knowledge of the motors, the estimation is done by considering the geometry of the candidate designs and the Work necessary to move one IPEx load (30kg) to the output point. The Work done is then divided by the anticipated time to deliver 30 kg of regolith according to the estimated delivery flow rate (see section 3.3.1). The Work can generally be calculated as

$$W = d \cdot (F_g + F_f), \tag{3.42}$$

where d is the average distance traveled by the lunar regolith, F_g is the force due to gravity, and F_f is the force due to friction. The Power can then be found as

$$P = \frac{W}{(30 \text{ kg}/D) \cdot 60\frac{\text{s}}{\text{min}}}, \tag{3.43}$$

where D is the estimated delivery rate of the candidate in kg/min.

The coefficient of kinetic friction between aluminum (the assumed material of the structure) and the lunar regolith has been empirically found to be 0.8882 [175]. The acceleration due to gravity on the Moon is 1.625 m/s^2. Using these values, the anticipated power of candidate solutions is estimated, starting with the Auger and Hopper System

$$W_{\text{Auger/Hopper}} = 0.30 \text{ m} \cdot (0 + 0.8882 \cdot 1.625\frac{\text{m}}{\text{s}^2} \cdot 30 \text{ kg} \cdot cos(0°)) = 13.00 \text{ J} \tag{3.44}$$

$$P_{\text{Auger/Hopper}} = \frac{13.00 \text{ J}}{(30 \text{ kg}/6.32\frac{\text{kg}}{\text{min}}) \cdot 60\frac{\text{s}}{\text{min}}} = 0.046 \text{ W}. \tag{3.45}$$

For the Vibrating Chute,

$$W_{\text{Chute}} = 0.42 \text{ m} \cdot (0 + 0.8882 \cdot 1.625\frac{\text{m}}{\text{s}^2} \cdot 30 \text{ kg} \cdot cos(16.70°)) = 17.42 \text{ J} \tag{3.46}$$

$$P_{\text{Chute}} = \frac{17.42 \text{ J}}{(30 \text{ kg}/6.99\frac{\text{kg}}{\text{min}}) \cdot 60\frac{\text{s}}{\text{min}}} = 0.068 \text{ W}. \tag{3.47}$$

For the Vibratory Bowl Feeder,

$$W_{\text{Bowl Feeder}} = 10.12 \text{ m} \cdot (1.625\frac{\text{m}}{\text{s}^2} \cdot 30 \text{ kg} + 0.8882 \cdot 1.625\frac{\text{m}}{\text{s}^2} \cdot 30 \text{ kg} \cdot cos(1.36°)) = 931.44 \text{ J} \quad (3.48)$$

$$P_{\text{Bowl Feeder}} = \frac{931.44 \text{ J}}{(30 \text{ kg}/1.68\frac{\text{kg}}{\text{min}}) \cdot 60\frac{\text{s}}{\text{min}}} = 0.87 \text{ W}. \quad (3.49)$$

For the Front Loader,

$$W_{\text{Front Loader}} = 0.47 \text{ m} \cdot (1.625\frac{\text{m}}{\text{s}^2} \cdot 30 \text{ kg} + 0) = 22.91 \text{ J} \quad (3.50)$$

$$P_{\text{Bowl Feeder}} = \frac{22.91 \text{ J}}{(30 \text{ kg}/60\frac{\text{kg}}{\text{min}}) \cdot 60\frac{\text{s}}{\text{min}}} = 0.76 \text{ W}. \quad (3.51)$$

For the Vibrating Funnel,

$$W_{\text{Funnel}} = 0.088 \text{ m} \cdot (0 + 0.8882 \cdot 1.625\frac{\text{m}}{\text{s}^2} \cdot 30 \text{ kg} \cdot cos(55.22°)) = 2.17 \text{ J} \quad (3.52)$$

$$P_{\text{Funnel}} = \frac{2.17 \text{ J}}{(30 \text{ kg}/160\frac{\text{kg}}{\text{min}}) \cdot 60\frac{\text{s}}{\text{min}}} = 0.19 \text{ W}. \quad (3.53)$$

CR-F-6: Mass

The mass of each candidate is estimated by finding the surface area and assuming a generic 0.01m aluminum thickness across the entire surface. The density of aluminum (2710 kg/m^3) is then used to calculate the estimated mass. For each candidate:

$$m_{\text{Auger/Hopper}} = 1.45 \text{ m}^2 \cdot 0.01 \text{ m} \cdot 2710\frac{kg}{m^3} = 43.09 \text{ kg} \quad (3.54)$$

$$m_{\text{Chute}} = 1.44 \text{ m}^2 \cdot 0.01 \text{ m} \cdot 2710\frac{\text{kg}}{\text{m}^3} = 39.08 \text{ kg} \quad (3.55)$$

$$m_{\text{Bowl Feeder}} = 2.67 \text{ m}^2 \cdot 0.01 \text{ m} \cdot 2710\frac{\text{kg}}{\text{m}^3} = 72.36 \text{ kg} \quad (3.56)$$

$$m_{\text{Front Loader}} = 0.63 \text{ m}^2 \cdot 0.01 \text{ m} \cdot 2710\frac{\text{kg}}{\text{m}^3} = 17.07 \text{ kg} \quad (3.57)$$

$$m_{\text{Funnel}} = 0.74 \text{ m}^2 \cdot 0.01 \text{ m} \cdot 2710\frac{\text{kg}}{\text{m}^3} = 23.04 \text{ kg}. \quad (3.58)$$

CR-F-8: Delivery Flow Rate

The delivery flow rate (represented as $\dot{m}$) of the candidates is considered as the flow of regolith at the interface between the intake and the bagging mechanisms. The process for calculating this value varies depending on the nature of the mechanism, e.g., vibration or auger. Starting with the Auger and Hopper System, one study finds that an auger with a 15.24cm radius is able to convey lunar regolith simulant at an average rate of 12.9426 kg/min in vacuum conditions [160]. This result is used to linearly approximate the anticipated output of the Auger and Hopper System, which uses an auger with a 10cm radius. The approximation can be done by considering the surface area of the

augers, since this represents the contact surface with the regolith,

$$\dot{m}_{\text{Auger/Hopper}} = \frac{\pi \cdot r^2_{\text{Auger/Hopper}}}{\pi \cdot r^2_{\text{experiment}}} \cdot \dot{m}_{\text{experiment}} = \frac{(0.1 \text{ m})^2}{(0.1524 \text{ m})^2} \cdot 12.9426 \frac{\text{kg}}{\text{min}} = 8.49 \frac{\text{kg}}{\text{min}}. \quad (3.59)$$

Next, the flow rate for the Vibrating Chute and the Vibratory Bowl Feeder can be linearly approximated using the same study [160], which implements a 91.44cm × 33.02cm × 13.97cm vibratory conveyor for lunar regolith simulant, achieving a mass flow rate of 12.5916 kg/min in vacuum conditions. In this case, the output of the candidates can be approximated by scaling the outlet width

$$\dot{m}_{\text{Chute}} = \frac{w_{\text{Chute}}}{w_{\text{experiment}}} \cdot \dot{m}_{\text{experiment}} = \frac{0.4 \text{ m}}{0.3302 \text{ m}} \cdot 12.5916 \frac{\text{kg}}{\text{min}} = 15.25 \frac{\text{kg}}{\text{min}}. \quad (3.60)$$

However, the Vibrating Chute also needs to account for the additional flow due to gravity caused by the incline. By working out the kinematics, it is found that a factor of $\frac{1}{cos(\alpha)}$, where α is the slope, can account for the additional horizontal flow of a particle down the ramp. Consequently,

$$\dot{m}_{\text{Chute}} = \frac{1}{cos(16.7°)} \cdot 15.25 \frac{\text{kg}}{\text{min}} = 15.92 \frac{\text{kg}}{\text{min}}. \quad (3.61)$$

Similarly, for the Vibratory Bowl Feeder,

$$\dot{m}_{\text{Bowl Feeder}} = \frac{w_{\text{Bowl Feeder}}}{w_{\text{experiment}}} \cdot \dot{m}_{\text{experiment}} = \frac{0.1 \text{ m}}{0.3302 \text{ m}} \cdot 12.5916 \frac{\text{kg}}{\text{min}} = 3.81 \frac{\text{kg}}{\text{min}}. \quad (3.62)$$

In this case, the small upwards slope of 1.36° does not affect the flow rate locally due to the small angle approximation.

The Front Loader is different from the other intake methods, since it is a discrete device. As such, the delivery flow rate for the Front Loader can be approximated to be the entire load at once, giving 60 kg/min.

Finally, the Vibrating Funnel delivery flow rate can be approximated by scaling the outlet surface area of a hopper experiment conducted in the lunar gravity and vacuum conditions [166]

$$\dot{m}_{\text{Funnel}} = \frac{l_{\text{Funnel}} \times w_{\text{Funnel}}}{l_{\text{experiment}} \times w_{\text{experiment}}} \cdot \dot{m}_{\text{experiment}} = \frac{0.3 \text{ m} \times 0.2 \text{ m}}{0.015 \text{ m} \times 0.018 \text{ m}} \cdot 0.72 \frac{\text{kg}}{\text{min}} = 160 \frac{\text{kg}}{\text{min}}. \quad (3.63)$$

Intake Lunar Regolith Module Evaluation

Each intake module candidate solution is designed to be compatible with system requirements, particularly with respect to the interface with the IPEx excavation system. As such, the opening to each candidate solution is capable of capturing regolith from one of the two IPEx bucket drums, which measures 0.625 m × 0.312 m. Further, each candidate solution is designed to intake 60 kg of lunar regolith, which is double the capacity of IPEx. Assuming an average lunar regolith density of 1.58 g/cm^3 [31], this represents a storage volume of 60,000 grams per 1.58 g/cm^3 ≈ 38,000 cm^3 (≈ 0.038 m^3) for each candidate solution. The evaluation of each candidate solution according to the metrics defined by the criteria is shown in Table 3.8.

The quantitative estimations of the criteria are directly used to produce the RP pairwise rankings for all candidates relative to each criterion as detailed in Section 3.2.2. An example of the RP pairwise

Table 3.8: Metric Evaluation Values for Intake Module

ID	Metric	Auger and Hopper	Vibrating Intake Chute	Vibratory Bowl Feeder	Front Loader	Vibrating Funnel
C-I-1	# of interfaces	0	0	0	1	0
C-I-2	ordinal scale (Likert 1-5)	1	2	2	3	2
C-I-3	failures per million operating hours	3.75	5.25	5.25	5.5	5.25
C-I-4	# of points	2	1	1	2	1
C-I-5	Watts	0.046	0.068	0.87	0.76	0.19
C-I-6	kilograms	43.09	39.08	72.36	17.07	23.04
C-I-7	# of moving parts	4	2	2	4	2
C-I-8	kilograms per minute	6.32	6.99	1.68	60	160
C-I-9	ordinal scale (Likert 1-5)	3	1	1	4	2

Table 3.9: RP Matrix for Intake Module for C-I-6 (Mass)

Candidate	Auger and Hopper	Vibrating Intake Chute	Vibratory Bowl Feeder	Front Loader	Vibrating Funnel	RP (f_k^i)
Auger and Hopper	1	1	5	1/2	1/2	0.165
Vibrating Intake Chute	1	1	6	1/2	1	0.200
Vibratory Bowl Feeder	1/5	1/6	1	1/7	1/8	0.0379
Front Loader	2	2	7	1	1	0.324
Vibrating Funnel	2	1	8	1	1	0.278
CR	**0.01** ($\ll 0.1$)					

rankings for C-I-6 (mass) is shown in Table 3.9, performed similarly for each of the intake module criteria.

A summary of the RI of criteria, the RP of the candidate solutions, and the final decision values is shown in Table 3.10, indicating a preference for the Vibrating Funnel, followed by the Auger and Hopper System and the Vibrating Intake Chute.

3.3.2 Deploy Empty RCU

The RCU deployment functional module is also selected for morphological assessment within the RCU Filling subsystem. The evaluation criteria for the module are shown in Table 3.11, along with the normalized RI of criteria, calculated as described in Section 3.2.2. The concept of an automated RCU bagger is studied in [79], with the final design using a series of rollers and hooks to store and open RCUs. Although there has been little further formal research on different methods of RCU storage and deployment, a number of industrial bagging systems can be used as inspiration [176, 177, 178, 179, 180, 181, 182]. Generally, baggers can perform horizontal or vertical filling. In the case of the RCU deployment module, the filling should be done vertically to allow gravity to naturally guide the granular lunar regolith into the bag. Further, the candidates should use a variation of Form-Fill-Seal (FFS) bagging, which is commonly used for granular materials such as feedstock or fertilizer. This would involve the complete opening of the bag mouth prior to filling. However, it is likely that the full "forming" element is unnecessary, since the granular regolith can help force the

Table 3.10: AHP Decision Matrix for Intake Module

		RP (f_k^i)				
ID	**RI** (t_i)	**Auger and Hopper**	**Vibrating Intake Chute**	**Vibratory Bowl Feeder**	**Front Loader**	**Vibrating Funnel**
C-I-1	0.121	0.243	0.243	0.243	0.0270	0.243
C-I-2	0.104	0.379	0.194	0.194	0.0396	0.194
C-I-3	0.133	0.540	0.132	0.132	0.0644	0.132
C-I-4	0.148	0.0354	0.310	0.310	0.0345	0.310
C-I-5	0.103	0.300	0.300	0.0349	0.0732	0.293
C-I-6	0.112	0.165	0.199	0.0338	0.324	0.278
C-I-7	0.117	0.0345	0.310	0.310	0.0345	0.310
C-I-8	0.0805	0.0751	0.0751	0.0676	0.246	0.536
C-I-9	0.0809	0.138	0.283	0.283	0.0351	0.262
Decision Values (D_k)		0.217	0.232	0.185	0.0916	0.275

rest of the non-rigid RCU to open as it is filled.

The RCU deployment functional module can be broken down into two primary tasks, the RCU storage and the RCU deployment. The RCUs could be stored either in a rigid container, or as a part of a roll of RCUs. The RCU deployment task involves the transport mechanism to place the RCU in the filling configuration (open mouth). This could be done using a track (chain-link, wire, or conveyor) or with a series of rollers if the RCU is stored as a roll. Considering these options, the storage and deployment concepts are listed in Table 3.12 and shown in Figure 3.11.

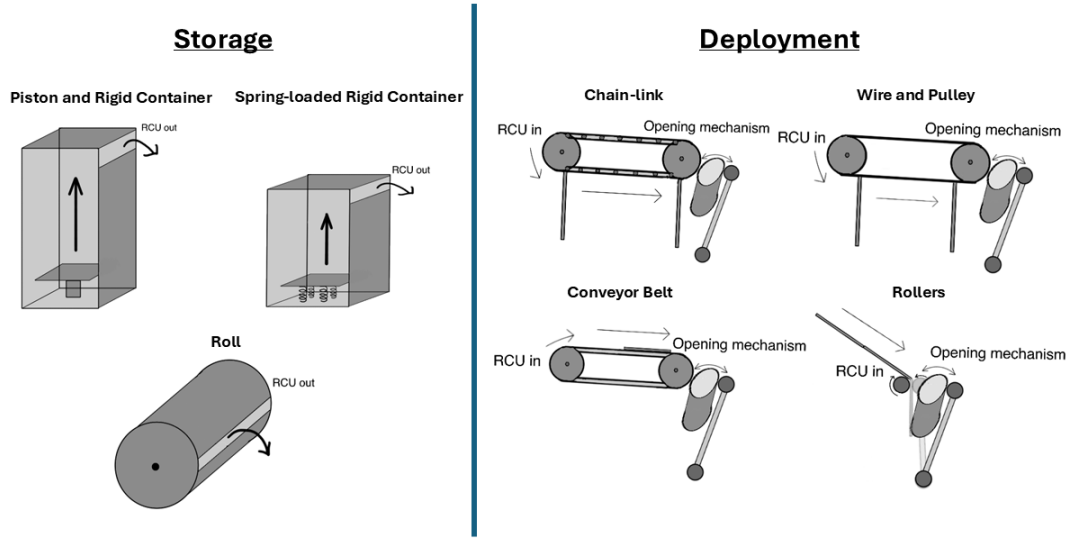


Figure 3.11: Candidate Solution Concepts for RCU Deployment Module

RCU Deployment Module Evaluation

Several key observations can be made from the metric evaluation presented in Table 3.13. Failure rate estimations account for motors and additional mechanical components such as conveyors and springs, referencing data from the NWSC-11 handbook [167]. Among all configurations, the WP and CB designs exhibit the highest estimated failure rates. The SR variants demonstrate the lowest overall mass; however, this is primarily attributed to their reduced RCU volume capacity. In terms of

Table 3.11: Evaluation Criteria for RCU Deployment Module

ID	Criteria	Description	Metric	Goal	RI (t_i)
C-D-1	Exposure to dust	The number of interfaces between moving parts, which are exposed to the regolith.	# of interfaces	Min	0.125
C-D-2	Longevity	The anticipated failure rate of the module based on motor load, transmission, and other mechanism failure rates, such as conveyors belts, chain-link conveyors, and springs [167].	failures per million operating hours	Min	0.159
C-D-3	Reliability	The number of independent points of single failure of the module.	# of points	Min	0.168
C-D-4	Power	The power required to move the RCU along the track to the final deployment location. Approximated by the total number of required motors.	# motors	Min	0.120
C-D-5	Mass	The anticipated mass of the module; calculated by taking the estimated surface area of the candidate solution and assuming an average thickness of aluminum of 0.01 m for the structure.	kilograms	Min	0.118
C-D-6	Complexity	The module complexity is measured as the total number of moving parts. This includes motors, gears, vibrating surfaces, and other parts moving in the workspace.	# of moving parts	Min	0.127
C-D-7	Transport speed	How quickly the RCUs can be moved from the stored to the deployed configuration.	Nominal scale (very slow to very fast)	Max	0.0816
C-D-8	Storage volume	The volume of RCUs that can be stored by the module.	meters cubed	Max	0.101

Table 3.12: Candidate Solutions for RCU Deployment Module

Storage Concept	Deployment Concept	Concept ID*
Piston and Rigid Container	Chain-link	PR-CL
	Wire and Pulley	PR-WP
	Conveyor Belt	PR-CB
Spring-loaded Rigid Container	Chain-link	SR-CL
	Wire and Pulley	SR-WP
	Conveyor Belt	SR-CB
Roll	Rollers	R-R

* IDs used for conciseness due to large number of concepts

deployment speed, PR configurations are generally slower, due to piston loading times, whereas the R-R configuration enables continuous feeding, making it the fastest. The R-R option also provides the highest RCU storage volume, even when constrained to similar external dimensions as rigid container configurations. The quantitative estimations of the criteria are directly used to produce the RP pairwise rankings for all candidates relative to each criterion, as detailed in Section 3.2.2.

A summary of the RI of criteria, the RP of the candidate solutions, and the final decision values is shown in Table 3.14, indicating a strong preference for the R-R deployment configuration, followed by the SR-CL configuration. The R-R configuration suffers from a higher mass requirement, due to the assumption that an aluminum shell protects the RCU bag roll, but performs well in all other criteria.

Table 3.13: Metric Evaluation Values for RCU Deployment Module

ID	Metric	PR-CL	PR-WP	PR-CB	SR-CL	SR-WP	SR-CB	R-R
C-D-1	# of interfaces	3	3	3	3	3	3	3
C-D-2	failures per million operating hours	25.43	140.93	140.93	116	231.5	231.5	9
C-D-3	# of points	4	4	4	3	3	3	3
C-D-4	# motors	3	3	3	2	2	2	2
C-D-5	kilograms	39	39	45	24	24	30	44
C-D-6	# of moving parts	5	5	5	5	5	5	4
C-D-7	Nominal scale (very slow to very fast)	Medium	Medium	Slow	Fast	Fast	Medium	Very Fast
C-D-8	meters cubed	0.0788	0.0788	0.0788	0.0338	0.0338	0.0338	0.126

Table 3.14: AHP Decision Matrix for RCU Deployment Module

		RP (f_k^i)						
ID	**RI** (t_i)	**PR-CL**	**PR-WP**	**PR-CB**	**SR-CL**	**SR-WP**	**SR-CB**	**R-R**
C-D-1	0.125	0.143	0.143	0.143	0.143	0.143	0.143	0.143
C-D-2	0.159	0.263	0.131	0.131	0.140	0.0313	0.0313	0.272
C-D-3	0.168	0.0256	0.0256	0.0256	0.231	0.231	0.231	0.231
C-D-4	0.120	0.0256	0.0256	0.0256	0.231	0.231	0.231	0.231
C-D-5	0.118	0.0913	0.0913	0.0316	0.264	0.264	0.218	0.0401
C-D-6	0.127	0.0667	0.0667	0.0667	0.0667	0.0667	0.0667	0.6
C-D-7	0.0816	0.109	0.109	0.0294	0.210	0.210	0.109	0.223
C-D-8	0.101	0.181	0.181	0.181	0.0368	0.0368	0.0368	0.346
Decision Values (D_k)		0.114	0.0926	0.0790	0.167	0.150	0.136	0.262

3.3.3 Interface with and Secure RCU

The RCU Manipulation subsystem consists of seven functional modules (see Figure 3.8). A primary morphological assessment of the RCU Manipulation subsystem is with regard to the mechanism for interfacing and securing the RCU. The evaluation criteria for the interface module are shown in Table 3.15, along with the normalized RI of the criteria, calculated as described in Section 3.2.2.

Although the RCUs will be non-rigid, the degree of compliance is unknown. However, it is anticipated that due to the high compacting behaviour of the lunar regolith, RCUs will not experience a high degree of deflection (semi-rigid). Some reviews of robotic grippers are conducted in [183], [184] and [185]. A review of automated assembly for non-rigid objects is also presented in [186]. Another domain of interest for lunar surface manipulation is that of compliant grippers (see reviews in [187] and [188]), particularly with regard to mitigating the lunar dust [147]. The definition of the manipulation task as a pick-and-place operation can also be used to constrain the gripping problem, with similar operations being done in industrial settings [189]. The concepts for the interface module, derived from common gripper configurations, are illustrated in Table 3.16, including some specific examples from literature and industry.

RCU Interface Module Evaluation

Some key observations can be drawn from the evaluation process, summarized in Table 3.17. Only the Compliant and Magnetic grippers are immune to jamming caused by dust exposure at moving interfaces. The Flatbed configuration is particularly prone to generating dust, due to the scooping

Table 3.15: Evaluation Criteria for the RCU Interface Module

ID	Criteria	Description	Metric	Goal	RI (t_i)
C-S-1	Exposure to dust	The number of interfaces between moving parts, which are exposed to the regolith.	# of interfaces	Min	0.104
C-S-2	Dust production	The potential for dust production due to agitation, sharp movements, etc.	ordinal scale (Likert 1-5)	Min	0.0894
C-S-3	Longevity	The anticipated failure rate of the module based on motor load [167].	failures per million operating hours	Min	0.117
C-S-4	Reliability	The number of independent points of single failure of the module.	# of points	Min	0.127
C-S-5	Power (Active/Passive)	Whether the interface mechanism requires active power to grip the RCU.	Yes/No	No	0.0917
C-S-6	Mass	The anticipated mass of the module based on reference designs.	kilograms	Min	0.0873
C-S-7	Complexity	The module complexity is measured as the total number of moving parts. This includes motors, gears, vibrating surfaces, and other parts moving in the workspace.	# of moving parts	Min	0.100
C-S-8	Attachment force	The force that can be enacted by the interface on the RCU.	Newtons	Max	0.0855
C-S-9	Rigid attachment	Whether the interface rigidly attaches to the RCU. Rigid attachment allows for easier manipulation of the RCU.	Yes/No	Yes	0.0850
C-S-10	Attachment points	How many individual points of attachment the interface has on the RCU.	# of attachment points	Max	0.0594
C-S-11	Attachment speed	How quickly the interface module can move into a secured configuration.	Nominal scale (very slow to very fast)	Max	0.0533

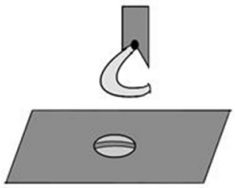
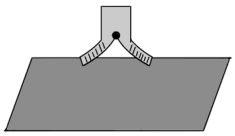
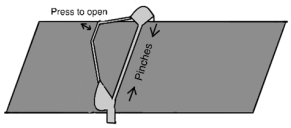
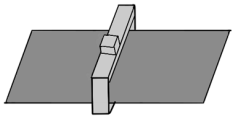
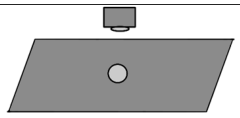
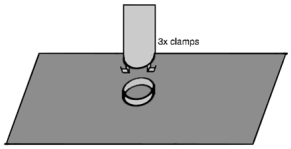
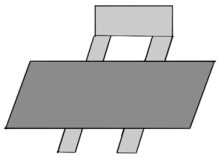
motion it employs, which disturbs the surrounding regolith. The longevity estimates only consider the operational lifespan of motors, and the Flatbed is the only design that relies on a non-powered mechanical interface. The attachment force calculations depend on the friction coefficient between Kevlar, used as an RCU material analogue, and aluminum (structural frame material), estimated to be 0.18 [212]. Amongst all options, the Gecko gripper demonstrates the lowest attachment force, reflecting its status as a relatively new and less mature technology. Only the Magnetic and Coupling configurations provide rigid or quasi-rigid attachment interfaces. The Gecko, Compliant and Rigid grippers have effectively infinite attachment points, as they can interface with the RCU at nearly any location; this was cut off at a value of 10 for comparison. In terms of attachment speed, the Hook and Gecko designs are the slowest, due to their deliberate attachment process, whereas the Magnetic gripper enables instantaneous engagement. The quantitative estimations of the criteria are directly used to produce the RP pairwise rankings for all candidates relative to each criterion, as detailed in Section 3.2.2.

A summary of the RI of criteria, the RP of the candidate solutions, and the final decision values is shown in Table 3.18, indicating a preference for the Magnetic interface, followed by the Flatbed. The preference for these can largely be attributed to the simplicity of both designs, with the fewest moving parts and no motors.

3.3.4 Transport RCU to Desired Pose

Another primary morphological assessment of the RCU Manipulation subsystem is with regard to the functional module for transporting the RCUs to the desired pose. The evaluation criteria for

Table 3.16: RCU Interface Mechanism Candidate Solutions

	Concept	Examples
Hook		[190, 191, 192, 193]
Gecko		[194, 195, 196]
Compliant Gripper		[197, 147, 198, 199]
Rigid Gripper		[200, 201, 202]
Magnetic		[203, 204, 205]
Coupling		[206, 207, 191]
Flatbed		[208, 209, 210, 211]

the manipulation of RCUs are shown in Table 3.19, along with the normalized RI of the criteria, calculated as described in Section 3.2.2. It is important to note that the selected manipulation morphology will be paired with a mobile base, which will also expand its effective workspace volume.

There is a number of manipulation mechanisms that could perform the required functions for the RCU transport module (see Figure 3.12), which can be divided into planetary space and Earth applications. One planetary robotic manipulator is the Light Exploration Robotic Arm (LERA) by MDA, an articulated 4-Degree-of-Freedom (DoF) manipulator being developed for the lunar surface [213], similar to NASA's COLDArm [127]. Another higher DoF manipulator is the Perseverance robotic arm, which is an articulated 5-DoF manipulator, primarily for scientific purposes, that has

Table 3.17: Metric Evaluation Values for RCU Interface Module

ID	Metric	Hook	Gecko	Compliant Gripper	Rigid Gripper	Magnetic	Coupling	Flatbed
C-S-1	# of interfaces	1	1	0	2	0	3	0
C-S-2	ordinal scale (Likert 1-5)	2	1	3	3	1	2	5
C-S-3	failures per million operating hours	3.5	3.5	3.5	3.5	0	3.5	0
C-S-4	# of points	2	2	2	2	1	1	1
C-S-5	Yes/No	Yes	Yes	Yes	Yes	Yes	Yes	No
C-S-6	kilograms	0.045	0.05	5	8	1.9	1	3
C-S-7	# of moving parts	2	2	2	3	0	4	1
C-S-8	Newtons	216	6.5	100	1000	300	1000	216
C-S-9	Yes/No	No	No	No	No	Yes	Yes	No
C-S-10	# of attachment points	2	∞	∞	∞	2	2	4
C-S-11	Nominal scale (very slow to very fast)	Very Slow	Very Slow	Fast	Fast	Very Fast	Slow	Slow

been deployed to Mars [214, 215, 216]. A different approach to the RCU transport task can be taken by NASA's LSMS lunar cranes (see Section 2.4.3), specifically by the L35 crane that has payload and reach capabilities within the reasonable range for RCU manipulation [119]. There are a few terrestrial alternatives that can provide additional morphologies, particularly when considering the task as a pick-and-place problem. One manipulator configuration optimized for such tasks in industrial settings is the Selective Compliance Assembly Robot Arm (SCARA). The Yamaha YK1200 series [217] is a family of terrestrial SCARA manipulators with payload specifications matching the requirements. A final alternative morphology is an automated forklift system, similar to those produced by Rockwell Automation, Agilox, Hyster, or Toyota [208, 209, 210, 211, 218].

Table 3.18: AHP Decision Matrix for Interface with and Secure RCU Module

		RP (f_k^i)						
ID	**RI** (t_i)	**Hook**	**Gecko**	**Compliant Gripper**	**Rigid Gripper**	**Magnetic**	**Coupling**	**Flatbed**
C-S-1	0.104	0.172	0.172	0.182	0.0891	0.182	0.0232	0.182
C-S-2	0.0894	0.161	0.202	0.125	0.125	0.202	0.161	0.0231
C-S-3	0.117	0.0435	0.0435	0.0435	0.0435	0.391	0.0435	0.391
C-S-4	0.127	0.0323	0.0323	0.0323	0.0323	0.290	0.290	0.290
C-S-5	0.0917	0.0667	0.0667	0.0667	0.0667	0.0667	0.0667	0.6
C-S-6	0.0873	0.182	0.182	0.0891	0.0230	0.175	0.179	0.172
C-S-7	0.100	0.151	0.151	0.151	0.0805	0.261	0.0284	0.177
C-S-8	0.0855	0.103	0.0330	0.0567	0.301	0.103	0.301	0.103
C-S-9	0.0850	0.0435	0.0435	0.0435	0.0435	0.391	0.391	0.0435
C-S-10	0.0594	0.0303	0.273	0.273	0.273	0.0303	0.0303	0.0909
C-S-11	0.0533	0.0326	0.0326	0.227	0.227	0.275	0.102	0.102
Decision Values (D_k)		0.0945	0.107	0.105	0.104	0.226	0.150	0.215

Table 3.19: Evaluation Criteria for the RCU Transport Module

ID	Criteria	Description	Metric	Goal	RI (t_i)
C-T-1	Exposure to dust	The number of interfaces between moving parts, which are exposed to the regolith.	# of interfaces	Min	0.0982
C-T-2	Dust production	The potential for dust production due to agitation, sharp movements, etc.	ordinal scale (Likert 1-5)	Min	0.0775
C-T-3	Longevity	The anticipated failure rate of the module based on motor load [167].	failures per million operating hours	Min	0.105
C-T-4	Reliability	A combination of the mechanical and task reliability. Mechanical reliability refers to the likelihood of single-point failure and task reliability considers the risk of task failure when faced with imperfect setups, such as non-optimally positioned RCUs.	ordinal scale (Likert 1-5)	Max	0.113
C-T-5	Power	The relative power consumption as proxied by the number of gravity-loaded joints.	# of gravity-loaded joints	Min	0.0815
C-T-6	Mass	The anticipated mass of the module; estimated by reference designs.	kilograms	Min	0.0759
C-T-7	Complexity	The module complexity is measured as the total number of moving parts. This includes motors, gears, vibrating surfaces, and other parts moving in the workspace.	# of moving parts	Min	0.0859
C-T-8	Workspace - module volume ratio	A measure of the reachable workspace given a specific module configuration.	unitless	Max	0.0822
C-T-9	Manipulation accuracy	A measure of the error between the desired RCU pose and the actual achieved RCU pose.	ordinal scale (Likert 1-5)	Min	0.0786
C-T-10	Manipulation resolution	The smallest measurable distance or increment that a robot can physically produce.	meters	Min	0.0660
C-T-11	Dexterity	A measure of the degrees of freedom, the degree of articulation, and the ability of the module to enter a variety of configurations within its workspace.	ordinal scale (Likert 1-5)	Max	0.0680
C-T-12	Structural rigidity	The resistance to deflections caused by external forces or torques.	Nominal scale (very stiff to very flexible)	Min	0.0690

RCU Transport Module Evaluation

Several notable insights emerge from the evaluation process, summarized in Table 3.20. The Forklift design is the most susceptible to dust production, due to direct ground contact during RCU pickup and drop-off operations. For longevity, only motor components were considered, naturally favouring configurations with fewer motors. The SCARA manipulator, despite having additional degrees of freedom, features only one gravity-loaded joint, making its power consumption equivalent to that of the Forklift. In terms of workspace efficiency, the 4-DoF manipulator and the Crane perform best relative to module volume, whereas the Forklift performs worst, due to its constrained single-axis movement. The Forklift also scores lowest in manipulation accuracy, primarily relying on coarse x–y positioning from its mobile base. The Crane similarly suffers from reduced accuracy, due to potential sway in its cable system. Manipulation precision was considered equal for all candidates, as accurate estimation would require detailed specifications of actuation and sensing hardware. Finally, the Crane is deemed the most flexible, due to its non-rigid cable architecture, followed by the 4-DoF and 5-DoF manipulators whose slender limbs enable some structural deflection. The SCARA and Forklift designs are comparatively more rigid. The quantitative estimations of the criteria are

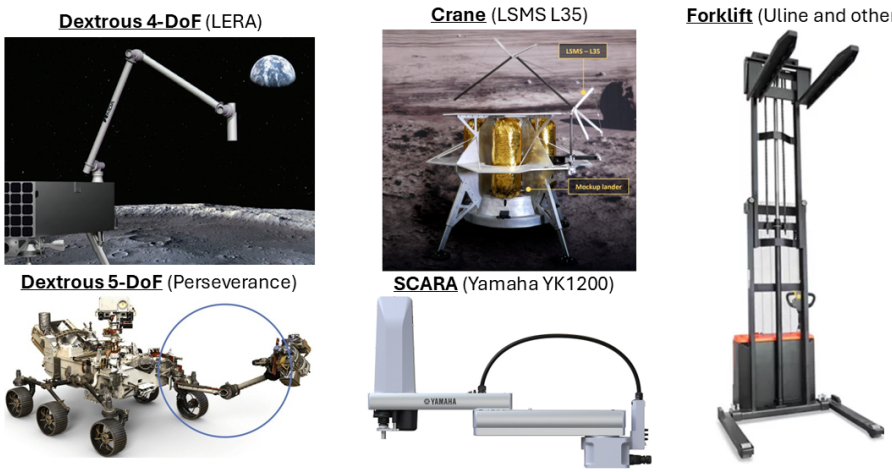


Figure 3.12: RCU Transport Module Candidate Solution Concepts

directly used to produce the RP pairwise rankings for all candidates relative to each criteria, as detailed in Section 3.2.2.

Table 3.20: Metric Evaluation Values for RCU Transport Module

ID	Metric	4-DoF	5-DoF	SCARA	Forklift	Crane
C-T-1	# of interfaces	4	5	3	2	4
C-T-2	ordinal scale (Likert 1-5)	2	2	2	4	3
C-T-3	failures per million operating hours	14	17.5	14	7	14
C-T-4	ordinal scale (Likert 1-5)	5	4	3	2	1
C-T-5	# of gravity-loaded joints	3	3	1	1	2
C-T-6	kilograms	40	76	96	20	11
C-T-7	# of moving parts	4	5	4	3	4
C-T-8	unitless	900	294	41	5	900
C-T-9	ordinal scale (Likert 1-5)	2	2	1	5	3
C-T-10	meters	1	1	1	1	1
C-T-11	ordinal scale (Likert 1-5)	4	5	3	1	1
C-T-12	Nominal scale (very stiff to very flexible)	Medium	Medium	Stiff	Stiff	Very Flexible

A summary of the RI of criteria, the RP of the candidate solutions, and the final decision values is shown in Table 3.21, indicating a preference for the 4-DoF morphology, closely followed by the SCARA morphology.

Table 3.21: AHP Decision Matrix for Transport RCU to Desired Pose Module

		RP (f_k^i)				
ID	**RI** (t_i)	**4-DoF**	**5-DoF**	**SCARA**	**Forklift**	**Crane**
C-T-1	0.0982	0.161	0.0419	0.305	0.331	0.161
C-T-2	0.0775	0.276	0.276	0.276	0.0301	0.141
C-T-3	0.105	0.189	0.0462	0.189	0.387	0.189
C-T-4	0.113	0.353	0.270	0.219	0.119	0.0399
C-T-5	0.0815	0.0397	0.0397	0.365	0.365	0.190
C-T-6	0.0759	0.258	0.109	0.0372	0.294	0.302
C-T-7	0.0859	0.194	0.0396	0.194	0.379	0.194
C-T-8	0.0822	0.383	0.143	0.0496	0.0424	0.383
C-T-9	0.0786	0.239	0.239	0.291	0.0345	0.196
C-T-10	0.0660	0.200	0.200	0.200	0.200	0.200
C-T-11	0.0680	0.382	0.302	0.230	0.0433	0.0433
C-T-12	0.0690	0.232	0.232	0.252	0.252	0.0331
Decision Values (D_k)		0.236	0.161	0.218	0.213	0.171

3.3.5 Final LRCS Morphology Selection

Now that the candidate solutions for the key functional modules have been evaluated, it is possible to assemble a high-level morphology for the LRCS. It is worth noting that the best morphology may not necessarily be the combination of the best performing candidates during evaluation, since not all candidates can be physically combined across the various modules, e.g., the flatbed interface and the crane transport module. The decision value $(D_k)_i$ for each functional module i is scaled to account for the spread in the decision value:

$$(D'_k)_i = \frac{N_i}{N}(D_k)_i \quad ; \quad i = 1, \ldots, F \quad ; \quad k = 1, \ldots, N_i \ , \tag{3.64}$$

where N_i is the total number of candidate solutions for module i, F is the total number of functional modules evaluated ($F = 4$ in this case), and N is the total number of candidate solutions across all modules. This effectively transforms $(D_k)_i$ values into a common scale $(D'_k)_i$, such that they can be summed up without giving additional preference to modules with fewer candidate solutions. The combination of all functional modules will then form a set of $\lambda = \prod_{i=1}^{F} N_i$ solutions. The decision values of such solutions are $D' = \{D'_1, D'_2, \ldots, D'_\lambda\}$, where

$$D'_j = \sum((D'_{k_j})_1, (D'_{k_j})_2, \ldots, (D'_{k_j})_F) \quad ; \quad j = 1, \ldots, \lambda. \tag{3.65}$$

For the LRCS evaluation studied in this thesis, $F = 4$ and $\lambda = 5 \times 7 \times 7 \times 5 = 1225$ module combinations, resulting in the top 10 compatible morphologies listed in Table 3.22. The clear favourites for the Intake Lunar Regolith and Deploy Empty RCU functional modules are the Vibrating Funnel and the R-R morphologies, respectively. The top candidates for the Interface with and Secure RCU functional module are the Magnetic and Flatbed morphologies, with a preference for Magnetic. Top preferences for the Transport RCU to Desired Pose functional module include the 4-DoF and SCARA morphologies. Moving forward, the design process will continue with the top-most combined morphology, including the Vibrating Funnel, R-R deployment, Magnetic interface, and 4-DoF manipulator.

Table 3.22: Morphological Table for Four Primary Functional Modules

Intake Lunar Regolith	Deploy Empty RCU	Interface with and Secure RCU	Transport RCU to Desired Pose	D'_j
Vibrating Funnel	R-R	Magnetic	4-DoF	0.249
Vibrating Funnel	R-R	Flatbed	4-DoF	0.246
Vibrating Funnel	R-R	Magnetic	SCARA	0.245
Vibrating Funnel	R-R	Magnetic	Forklift	0.244
Vibrating Funnel	R-R	Flatbed	SCARA	0.242
Vibrating Funnel	R-R	Flatbed	Forklift	0.241
Vibrating Chute	R-R	Magnetic	4-DoF	0.240
Hopper and Auger	R-R	Magnetic	4-DoF	0.237
Vibrating Chute	R-R	Flatbed	4-DoF	0.237
Vibrating Chute	R-R	Magnetic	SCARA	0.236

3.4 Selection of Surrounding Architecture

Since the morphology of the primary functional modules has been established, it is now possible to select technologies for the surrounding architecture. This is done methodically by considering the functional modules listed in Figure 3.8. These functional modules, the selected candidates, the placement, and the reasonings are broken down in Table 3.23. Since the RCU bagging and manipulation tasks are decoupled (see Section 3.1.1), the placement locations are broken down into the RCU Bagger (RB) and Mobile Manipulator (MM).

Table 3.23: LRCS Architecture Breakdown

	Function	Solution	Placement	Reasoning
Power Management	Generate power	Solar power	RB	Solar arrays gain a mass advantage over nuclear reactors at polar sites illuminated for over 75% of a lunar cycle. [219]
	Store power	Li-ion batteries	RB + MM	Higher TRL than alternatives like Regenerative Fuel Cells (RFCs), and mass advantage for short discharge durations under 10–18 hours [220, 221, 222, 223].
	Distribute power	Power bus	RB + MM	A custom Power Management and Distribution (PMAD) architecture will need to be designed according to the system's needs, which can follow standards, such as the AIAA S-122 [224] or the ECSS-E-ST-20-20C [225].
Onboard Computing	Collect and monitor system health information; Process and communicate telemetry data; Memory management; Parse and process incoming commands; Command and Control	CPUs # 1 and # 2, and Communications Bus	RB + MM	Two Central Processing Units (CPUs) will manage all of the Onboard Computing subsystem functionalities on the RB (CPU # 1) and MM (CPU #2). Additionally, an internal communications architecture will need to be implemented based on the hardware, such as MIL-STD-1553B or SpaceWire [226].
RCU Filling	Intake lunar regolith; Fill RCU with lunar regolith	Vibrating Funnel	RB	Previously selected (Section 3.3.5)
	Beneficiate lunar regolith	Grizzly bars	RB	Has been found to be effective for removing larger rock sizes when used with RASSOR (IPEx predecessor) [39].
	Deploy Empty RCU	R-R morphology	RB	Previously selected (Section 3.3.5)
	Seal RCU	Thermal sealing	RB	Thermal sealing is a common strategy in industrial bagging processes, and allows for less moving parts as compared to other methods like sewing or folding. The sealing bar can be integrated into the opening mechanism and rollers of the Deploy Empty RCU R-R configuration [227, 228, 229].

Table 3.23 (continued)

	Function	Solution	Placement	Reasoning
RCU Transfer	Receive RCU; Internally transfer RCU; Prepare RCU for manipulation	Pallet conveyor	RB	Allows for the accumulation of multiple RCUs and easy low-power dispensing to the MM [107, 230].
RCU Manipulation	Interface with and secure RCU; Release RCU	Magnetic interface	MM	Previously selected (Section 3.3.5)
	Identify desired RCU pose; Calculate trajectory; Identify obstacles; Verify Pose	Sensors and CPU # 2	MM	The MM CPU will need to process and fuse sensor data to construct a map of its surroundings for path planning and localization. This is essential for both the manipulation and MM trajectory generation. The specific sensors necessary for this task require further investigation.
	Transport RCU to desired pose	4-DoF	MM	Previously selected (Section 3.3.5)
Communications	Process signals	Transceiver	RB + MM	Common signal processing component
	Send and receive communications signals	K/Ka antenna	RB + MM	It is anticipated that lunar communications terminals, such as the RB, would leverage the K/Ka band when communicating with Earth, as well as with lunar surface rovers, such as the MM [231].
Thermal Management	Monitor thermal levels	Thermistors	RB + MM	Common thermal sensor
	Thermal actuation (passive or active); Radiate excess heat; Intake heat	Thermal surfaces	RB + MM	Further thermal analysis is required to determine if or where active thermal actuation is needed, as well as the placement of passive coatings and thermal conducting structures (e.g., thermal straps).
Mobility	Localize system within environment; Obstacle avoidance	Sensors, and CPUs # 1 and # 2	RB + MM	Same functionality as previously described for the sensors and CPU # 2. The RB will also need to perform sensor fusion for simple localization and self-inspection.
	Navigate and move within environment; Deploy from launch vehicle	Differential 4-Wheel Drive and 6-Wheel Rocker-Bogie	RB + MM	The RB will require limited mobility, such as for deploying from the launch vehicle, which can be achieved by a simple differential 4-wheel drive mobility system. The 6-wheel rocker-bogie suspension system has been implemented on the Mars Science Laboratory, Mars Exploration Rovers, Pathfinder, and Perseverance. The latest iteration includes 4 individually steerable front and back wheels, allowing the vehicle to swerve, make arcing turns, and turn 360 degrees in place. This flexibility in steering would allow the MM the freedom to enter any necessary configurations for manipulation [214, 216].
Lunar Dust Mitigation	Lunar Dust Mitigation	EDS, Surface Coatings, and Seals	RB + MM	The Electrodynamic Dust Shield (EDS) is an active dust mitigation tool with high TRL, which leverages the electrostatic charging of lunar regolith to keep surfaces, such as solar panels, clear from dust [232, 233, 234, 235, 236, 237, 238]. Passive dust mitigation techniques can also be implemented through surface coatings that minimize the dust attraction, such as the lotus leaf coating [239, 49, 48] and topology modification [240, 241, 48], as well as seals to prevent dust ingress to critical interfaces, such as the spring-loaded Teflon seal [242, 243, 48].

Chapter 4

Systems Engineering

4.1 LRCS Mission Scenario

The LRCS mission scenario is illustrated in Figure 4.1. As an initial step, the system will need to be delivered to the surface by a lunar lander. Numerous landers are being developed by private companies and space agencies, summarized in Table 4.1. It is important to note that while landers have distinct mass capacities, they lack specific volumetric constraints, likely due to the fact that the payload bays of the landers are customized for the payloads they carry. The landers have been categorized into light (< 300 kg), medium (300–900 kg), and heavy (> 900 kg) classes according to their payload capacity. The LRCS has an anticipated total mass of 542.61 kg (see Section 4.6.2), requiring a medium-class lander.

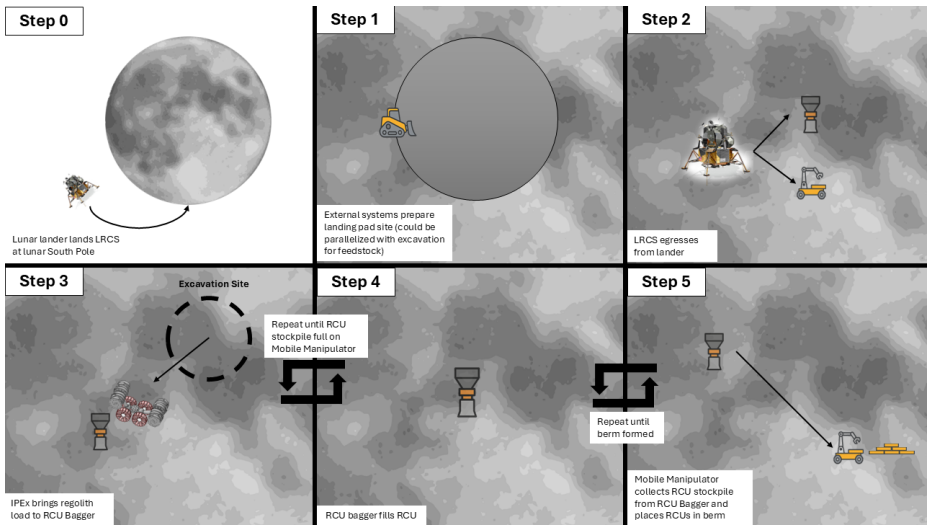


Figure 4.1: LRCS Mission Scenario

The landing pad area will need to be prepared as a preliminary step by other robotic systems. This can involve bulldozing and grading the surface, as well as creating the landing pad, although the latter is not required for the LRCS to operate. It is also possible that the site preparation be conducted in parallel with the LRCS operation, since excavation systems may also act as bulldozers

Table 4.1: Payload Capacities of Selected Lunar Landers

Lander	Company	Payload Capacity	Class	Reference
Griffin	Astrobotic	625 kg	Medium	[244]
Peregrine	Astrobotic	120 kg	Light	[244]
Blue Moon Mark 1	Blue Origin	3000 kg	Heavy	[245]
Blue Ghost	Firefly Space	240 kg	Light	[246]
Nova-C	Intuitive Machines	130 kg	Light	[247]
Nova-D	Intuitive Machines	1500–2500 kg	Heavy	[248]
APEX 1.0	iSpace	300–500 kg	Medium	[249]
Argonaut	Thales Alenia Space	1500 kg	Heavy	[250]

and graders, feeding the cleared regolith to the LRCS. Once the landing pad site is prepared, the LRCS operation can begin. As determined by the design conceptualization (Chapter 3), the LRCS is composed of two separate machines, an RCU Bagger (RB) and a Mobile Manipulator (MM). Considering that a lander has successfully transported the LRCS to the lunar surface near the landing pad location, both machines will need to be offloaded from the lander. Once offloaded, the LRCS will need to initialize its systems, including the deployment of the solar arrays on the RB and MM machines. The LRCS will then traverse to the center of the landing pad location, where the RB will remain for the bagging task. The bagger will then begin producing RCUs, which will be loaded into the MM up to its capacity. This stage relies on an external lunar regolith excavating system to bring regolith feedstock to the LRCS, namely NASA's ISRU Pilot Excavator (IPEx) [17]. Some of the IPEx properties are listed in Table 3.1. The excavator will continuously bring regolith to the RB, with a load-dump cycle time of 13.5 min, within which 1 RCU should be created. The IPEx maximum load is 30 kg, and the mass of one RCU is 22 kg. It is approximated that one IPEx load can create one RCU with a 27% margin. Once the MM has filled its stockpile of RCUs, it will traverse to the desired berm location, and stack the RCUs using an articulated 4-Degree-of-Freedom (DoF) manipulator. The MM then returns to the RB to collect more RCUs, and the process is repeated until the berm is formed.

Another important mission-level consideration is the anticipated location on the lunar surface. Most of the proposed habitat locations on the Moon's surface involve the lunar South Pole. For the Artemis Base Camp, NASA has identified 13 candidate landing regions within 6 degrees of latitude of the lunar South Pole [2, 18]. NASA cites the presence of water ice and other in-situ resources in Permanently Shadowed Regions (PSRs), and long periods of continuous sunlight, as major considerations for these selections. Similarly, the ESA's Moon Village project intends to target the Shackleton crater at the lunar South Pole [3, 4]. For the ILRS, the CNSA has identified the lunar South Pole as a target habitat location, with researchers paying particular attention to the Amundsen and Malapert craters [19]. The maximum bolometric temperature for the lunar South Pole is shown in Figure 2.2, illustrating the temperature variability between shadowed regions (craters) and illuminated regions. This indicates the importance of illumination conditions on the operating environment, affecting factors such as temperature and solar power availability. Therefore, the lunar day-night cycle is of particular importance to the LRCS mission scenario. Assuming a mission launch in 2030, the solar illumination conditions for the Shackleton crater, at -89.67° latitude and 129.78° longitude [251], are shown in Figure 4.2, using ephemeris data from NASA's SPICE information system for planetary bodies [252]. Note that a 90° solar incidence angle indicates that

the Sun is directly on the horizon, showing that a best-case solar day of approximately 6 months (≈ 184 days) can be expected.

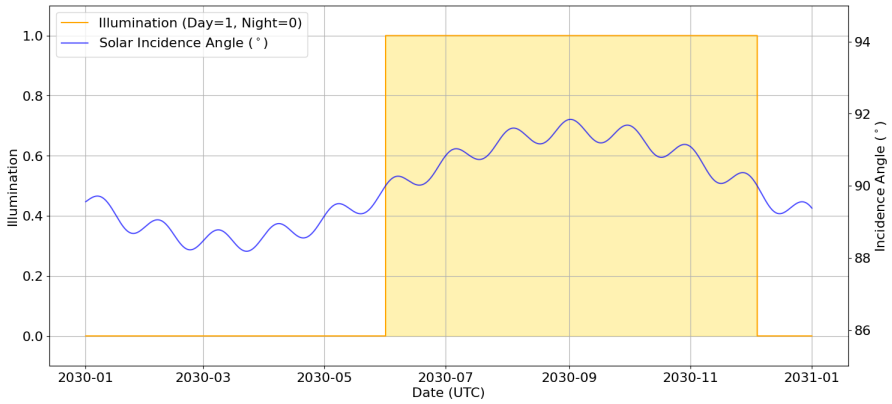


Figure 4.2: Lunar South Pole Illumination Conditions

As determined in Sections 4.3 and 4.4, the RB has a bagging speed of 44 s/RCU, or approximately 6 min per 8 RCUs, and a maximum traversal speed of 0.2 m/s. The MM has an average RCU placement speed of 84 s/RCU, and a maximum traversal speed of 0.24 m/s, placing 8 RCUs per trip. Various RCU stacking patterns can be considered for creating the berm. A limiting factor for the RCU stacking patterns is the shadow caused by the berm itself. From Figure 4.2, it is evident that the solar incidence angle is never more than 3° above the horizon, which matches the angle that the fully constructed berm would create with the center of the landing pad. This limits the feasible stacking patterns, since simply constructing the berm around the RB or between the RB and the Sun would require the implementation of a solar array tower at least 2.6 m tall to ensure consistent power generation over the mission duration. Four feasible patterns are considered in Table 4.2, contrasting estimations for the average RCU placement time, the total duration, and the total power consumption. Some stacking patterns involve moving the RB to follow the MM. These patterns build the berm in quarters to minimize the amount of movement operations for the RB, during which RCUs cannot be made, and to avoid shadowing the RB's solar panels. Pattern 2 outperforms all the other patterns in every metric. Despite the fact that moving the RB requires some additional energy and bagging downtime, the reduced distance that the MM needs to travel is substantial, resulting in a lower overall duration and power consumption. This stacking pattern results in a stacking rate of approximately 8 RCUs every 25 minutes.

All stacking patterns assume continuous regolith intake from IPEx. Since one IPEx load roughly translates to one RCU created, the duration of the mission is also constrained by the IPEx load-dump time of 13.5 min. Table 4.3 shows an estimate of the mission duration, using the previously calculated average bagging and placement time per batch of 8 RCUs. The bottleneck with IPEx can be alleviated by using multiple IPEx systems. However, the LRCS placement speed becomes the limiting factor once 6 IPEx systems are deployed simultaneously, since 8 RCUs can be produced before the MM is able to collect them. Considering the importance of completing the LRCS berm construction mission within one lunar day, at least 6 IPEx systems would be required for the year 2030 (see Figure 4.2). This clearly identifies the excavation rate as the limiting factor for the berm

Table 4.2: Feasible RCU Placement Patterns

Pattern	Description	Placement Time (s/RCU)	Duration (days)	Power (W)
1	The RB remains stationary in the center of the landing pad. The MM builds the berm, but leaves an opening where the Sun can shine onto the RB's solar panels. This opening is closed at the end.	220	208	643
2	The RB follows the MM along a circular path within the landing pad (10 m from berm), building the berm in quarters, starting with the side furthest from the Sun to avoid being shadowed.	190	180	556
3	The RB remains stationary (10 m) outside the landing pad, on the side closest to the Sun. The MM builds the berm, leaving an opening to enter and exit the landing pad area.	242	229	686
4	The RB follows the MM along a circular path outside the landing pad area (10 m from berm), building the berm in quarters to avoid shadowing itself on the side furthest from the Sun.	193	183	564

Note: Time estimates based on average distance to berm and speed. Power estimates based on average case power draw from Section 4.6.1.

building concept. The primary contributors to the mission duration are the excavation rate, the total number of RCUs, and the MM RCU stockpile capacity. The small size of the IPEx system makes it suitable for swarm operations. Also, note that the above estimates are based on the current landing pad diameter of 100 m, which is expected to reduce to a diameter as small as 25 m with the advancement of current technologies.

Table 4.3: Estimated Mission Duration vs. Number of IPEx Systems

# of IPEx Systems	Load-Dump Time per IPEx (min)	Estimated Mission Duration (days)
1	13.50	808
2	6.75	425
3	4.50	297
4	3.38	234
5	2.70	195
6	2.25	177
7	1.92	177
8	1.69	177

4.2 LRCS Design Overview

The conceptual design of the LRCS is performed in Chapter 3, resulting in the morphological selection of the four primary functional modules. The conceptualization process selected a vibrating funnel for regolith intake, a series of rollers for RCU deployment, a magnet for RCU interfacing, and an articulated 4-DoF manipulator for RCU transport. These morphologies constitute the bases for the subsystem design in Sections 4.3 and 4.4.

Since the LRCS will need to manipulate the lunar regolith at scale, it should be considered a dusty system. This is important given the propensity for the lunar regolith to jam and scratch mechanical components, as well as damage electronic components [31]. The LRCS conceptualization finds that reliability, longevity, exposure to dust, and complexity are primary design factors. The system should also minimize power consumption due to limited power availability on the lunar surface.

Some initial material selection should also be considered. As a starting point, Aluminum 6061-T6 is selected for the majority of the structure, due to its relatively high strength-weight ratio [253] and its common usage in many space systems, thus frequently tabled in space system handbooks [254, 255]. Further, Aluminum 6061-T6 has been shown to be able to withstand a very wide range of temperatures, performing particularly well at very low temperatures [256], which can be expected on the lunar surface. A thorough analysis of the material properties for Aluminum 6061-T6 can be found in the the Metallic Materials Properties Development and Standardization (MMPDS-01) handbook [257], including a density of 2713 kg/m^3. The handbook further suggests a standard thickness range of 0.002-0.005 m for aluminum sheets and tubes. General guidelines for structural design can be taken from NASA's STD-5001B standard: Structural Design and Test Factors of Safety for Spaceflight Hardware [254]. Specifically, the standard suggests a factor of safety of 1.4 for metallic structures. Combining this with the above, a wall thickness of 0.002 m $\times$ 1.4 $\approx$ 0.003 m is considered for the load-bearing elements of the chassis and frame, and 0.002 m for other non-load-bearing structures and plates. Standard Aluminum 6061 plate and tube properties can be found through McMaster-Carr, indicating that 0.002m thick plates and 0.003m thick 0.025 m $\times$ 0.025 m tubes are reasonable selections [258, 259]. After initial loading and thermal analysis, material selection and thickness should be revisited to reinforce high-load areas and consider heat distribution. The design of the RCUs is not included in this thesis. Kevlar is used as a material analogue wherever the RCU material strength or friction coefficients are considered [14].

Based on these factors, some design guidelines are established for the first iteration of the LRCS, as follows:

Design Considerations for LRCS

- The number of interfaces susceptible to dust ingress, particularly in direct contact with the lunar regolith, should be minimized.
- All rotary shaft interfaces should be protected by Spring-loaded Teflon seals [48].
- All lead-screw mechanisms should have their own housings to prevent jamming along the threads, and interfaces should be protected by Zephyr fiberglass brush seals, clearing excess dust [260].
- The minimal amount of motors should be used to actuate all mechanisms.
- All motors should be protected inside sealed housings.
- Aluminum 6061-T6 is used as the primary structural material, with a thickness of 0.003 m for load-bearing structures and 0.002 m for non-load-bearing structures.
- RCUs are made of Kevlar as an analogue material.

4.3 LRCS System Engineering: RCU Bagger

Since the RCU filling and transport tasks are decoupled, the LRCS can be considered as two separate machines: an RCU Bagger (RB) and a Mobile Manipulator (MM). This section focuses on the primary subsystems required for the RB, namely RCU Filling, RCU Transfer, and Mobility. The RCU filling and transfer subsystems are inspired by an automated RCU bagger described in [79], using a series of rollers and hooks to store and open RCUs. Although there exists little further formal research on different methods of RCU bagging, a number of terrestrial industrial bagging systems are also considered in the design [176, 177, 178, 179, 180, 181, 182]. Generally, baggers perform either horizontal or vertical filling. The RB performs vertical filling for Form-Fill-Seal (FFS) bagging, allowing gravity to naturally guide the granular lunar regolith into the RCU once its mouth is opened.

4.3.1 RCU Filling

The RCU Filling subsystem consists of two mechanical assemblies, i.e., the Vibrating Funnel for regolith intake and the RCU bagging mechanism, working together to produce an RCU for manipulation. A 3D-model of the Vibrating Funnel is shown in Figure 4.3. The dimensions of the funnel opening are constrained by the dimensions of the IPEx bucket wheels (0.625m $\times$ 0.312m), while the funnel exit is constrained by the dimensions of the RCU (0.43m $\times$ 0.1m). The funnel exit dimensions assume side filling to minimize the required vertical vertical space, which is important in order to accommodate IPEx's limited dump height (discussed in Section 4.3.2). Further, the funnel is designed to intake 60 kg of lunar regolith, which is double the capacity of IPEx. Assuming an average lunar regolith density of 1.58 g/cm^3 [31], this represents a storage volume of 60000 grams per 1.58 g/cm^3 $\approx$ 38000 cm^3 ($\approx$ 0.038 m^3). Considering the entry and exit dimensions, the desired volume, and the geometry of the funnel (trapezoidal prism), a height of 0.185 m is required.

Grizzly bars are used for the coarse beneficiation of the lunar regolith, ensuring that rocks larger than 0.04 m diameter, less than the 0.05 m grating spacing, do not enter the system. These rocks are cleared by vibrating the Grizzly bars at an angle toward a third-party Regolith Analysis and

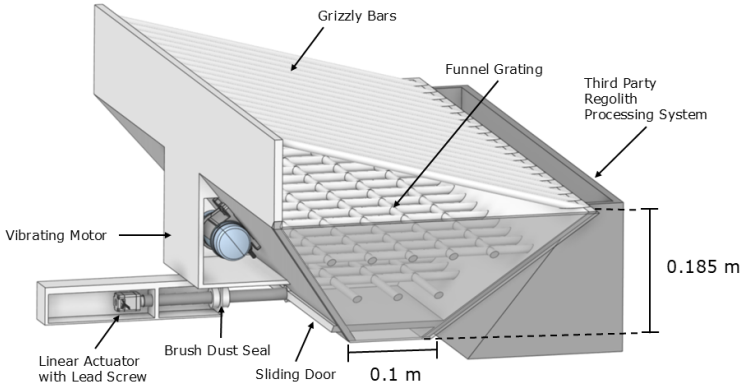


Figure 4.3: 3D Model of Vibrating Funnel

Processing subsystem, supporting science objectives and minimizing wasted materials. A brushless Direct Current (DC) Eccentric Rotating Mass (ERM) vibratory motor is mounted to the wall of the funnel, initially intended to operate at 500 Hz [261]. The purpose of the vibration is to fluidize the lunar regolith, promoting the effective flow through the hopper, as well as preventing its compaction and formation of arches. The vibration also serves the secondary purpose of clearing the Grizzly bars of excess materials. The necessary vibration amplitude to clear the Grizzly bars can be related to the acceleration a_v due to vibration

$$a_v = A \times (2\pi f)^2 \tag{4.1}$$

where A and f are the vibration amplitude and frequency, respectively. Clearing can be achieved once

$$a_v \geq g_{\text{Moon}} \times (\sin\theta + \mu_s \cos\theta) \tag{4.2}$$

where g_{Moon} is the lunar gravitational acceleration (1.625 m/s^2), $\theta = 14.04°$ is the angle of the bars, and μ_s is the coefficient of static friction ($\mu_s = 1.211$ between lunar regolith and aluminum [175]). Combining equations (4.1) and (4.2), the vibration amplitude required for displacing excess materials is determined as

$$A \geq \frac{g_{\text{Moon}}(\sin\theta + \mu_s \cos\theta)}{(2\pi f)^2} = 0.233\,\mu\text{m} \tag{4.3}$$

for $f = 500$ Hz. The frequency and amplitude for the intake funnel is further investigated using the AGX Dynamics physics-based simulation software for the simulation of the granular lunar regolith [262]. The software is used to produce a granular physics model for representing the lunar regolith's physical and dynamic properties. A simulation of the flow of the granular regolith through the intake funnel is conducted, shown in Figure 4.4. The objectives of the simulation are to analyze the impact of vibration frequency and amplitude on bulk flow rate and RCU packing density. The simulation considers 22 kg of regolith, composed of 3mm-diameter particles and 5cm-diameter rocks, falling in lunar gravity. In Table 4.4, a grid of frequencies and amplitudes is considered, populated using equation 4.3. The simulation finds that a vibration frequency of 500 Hz with an amplitude of 23.3 μm best promotes the bulk flow of regolith. Configurations at 500 Hz appeared to perform

best in terms of flow rate, while amplitude had little consistent effect. None of the configurations significantly affected the RCU packing density.

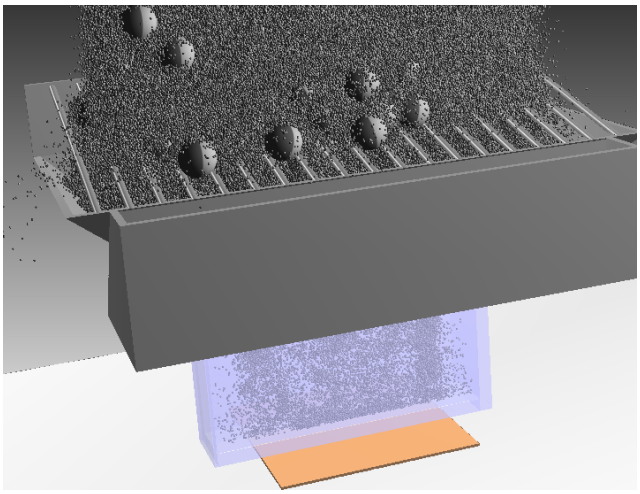


Figure 4.4: AGX Simulation of Regolith Intake Funnel

Table 4.4: Payload Capacities of Selected Lunar Landers

Frequency (Hz)	**Amplitude (μm)**	**Bulk Flow Rate (kg/s)**	**Packing Density (kg/m^3)**
5	0.233	0.533	0.732
5	23.3	0.626	0.754
5	2330	0.522	0.728
50	0.233	0.521	0.740
50	23.3	0.533	0.731
50	2330	0.537	0.725
500	0.233	0.653	0.756
500	23.3	0.786	0.774
500	2330	0.537	0.726

Considering an ERM of mass m_{ERM} at radius r_{ERM}, then the required torque is

$$\tau = (a_v \times m_{\text{ERM}}) \times r_{\text{ERM}} = 0.000115\,\text{Nm} \tag{4.4}$$

corresponding to a power of

$$P = \tau \times \omega = 0.000115\,\text{Nm} \times 2\pi(500\,\text{Hz}) \approx 0.36\,\text{W}\,. \tag{4.5}$$

In addition, the funnel exit incorporates a sliding door to enable the start and stop of regolith into the RCU. The door is operated by a lead-screw mechanism, powered by a linear actuator that serves to open the sliding door when the RCU is being filled, and close it once the filling is completed or between operations. The motor and lead screw are intentionally kept outside of the path of the lunar regolith to help prevent dust contamination in the mechanism. To achieve this, the screw is concentrically passed directly along a threaded path within the sliding door, with the outer contact point of the funnel sealed with the Zephyr fiberglass brush seal [48]. The sliding door never opens past the inner funnel wall, keeping the dust and regolith from spreading to the rest of the RB. The

open and close conditions are detected by a torque sensor incorporated within the motors of the rollers. This same condition determines whether the vibratory motor should be turned on, to avoid the compaction of lunar regolith when the sliding door is closed. The lead-screw and sliding door mechanisms will need to withstand a maximum load of 60 kg $\times$ 1.625 m/s^2 = 97.5 N in the case that the funnel is completely full with regolith. The sliding door weighs 0.000504 m^3 $\times$ 2713 kg/m^3 = 1.37 kg, requiring a force of 1.37 kg $\times$ 1.625 m/s^2 $\times$ 1.05 = 2.34 N to move, where $\mu = 1.05$ [263] is the aluminum-aluminum coefficient of static friction. The linear actuator will need to produce a torque of [253]

$$\tau = \frac{F \times d_m}{2}\left(\frac{l + \pi\mu \times d_m}{\pi d_m - \mu \times l}\right) \tag{4.6}$$

where F is the required force , d_m is the diameter of the screw, μ is the coefficient of friction, and l is the pitch or lead of the screw ($l = 0.00125$ m). Using common handbook values for a 1cm-diameter lead screw ($d_m = 0.01$ m), the actuator is required to produce a torque magnitude of 0.014 Nm to move the sliding door.

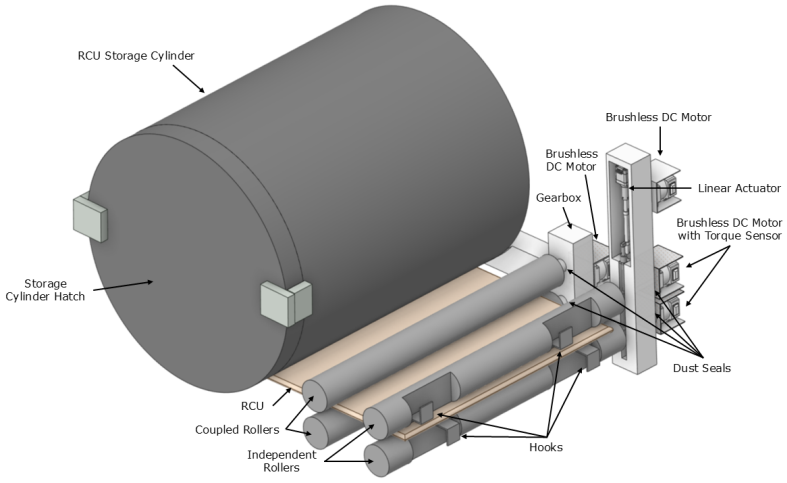


Figure 4.5: 3D Model of Bagging Mechanism

The RCU bagging assembly, shown in Figure 4.5, consists of a series of rollers that extract a pre-formed RCU from a roll, moves it to an open-mouth configuration below the intake funnel, and then seals the RCU before releasing it to the RCU Transfer subsystem (see Section 4.3.2). Since the RCUs are side-filled in this configuration, the RCU storage cylinder should measure at least 0.43 m in height. The storage capacity of the cylinder will depend on the properties of the RCUs, such as thickness of the material, which is not considered in this thesis, but a cylinder diameter of 0.4 m is nominally decided to match the dimensions of the RB. This outer diameter can be increased to improve the storage capacity. Given a certain RCU thickness t, the capacity of the RCU storage cylinder is

$$N_{RCU} = \frac{D^2_{cylinder} - D^2_{spindle}}{4t^2}. \tag{4.7}$$

Although the number of RCUs cannot be determined without the thickness, the mass of the roll of RCUs can be estimated using the density of Kevlar (analogue material), giving a mass of $\pi \times (0.2\,\text{m})^2 \times 0.43\,\text{m} \times 1440\,\text{kg/m}^3 \approx 77.8\,\text{kg}$ [264]. The central spindle should be able to wrap at least

one RCU, giving

$$D_{spindle} \geq \frac{L_{RCU}}{\pi} = \frac{0.25\,\mathrm{m}}{\pi} = 0.08\,\mathrm{m} \tag{4.8}$$

as the resulting spindle diameter. Further, the storage cylinder has a set of clamps at one end to enable the replacement of the RCU spindle, so that more RCUs may be loaded into the LRCS. The storage cylinder spindle also has a rotary encoder incorporated to determine how many RCUs have been dispensed.

Next, there exist two pairs of rollers. The first pair is actuated by a motor that powers a gearbox resulting in opposite rotations. The diameter of the rollers is set to 5 cm nominally, but should be revisited to account for the RCU material and thickness, ensuring that a sufficient amount of friction is applied. It is also recommended that these rollers be textured to increase the friction between the rollers and the RCU. The second pair of rollers are for opening the RCU using a series of hooks mounted on the rollers, and they are attached to a rotating arm that manipulates the RCU from the storage cylinder to the funnel.

The RCU filling process is shown in Figure 4.6, and the RCU filling subsystem, with masses and dimensions, is shown in Figure 4.7. The process begins with the two rollers attached to the storage cylinder pulling out approximately half of the RCU (Step 1, , Figure 4.6). The exact length to be extracted is dependent on the RCU material compliance, but there should be no droop in the material exposed to allow for the second set of rollers to interface with the RCU. Next, the rotating arm brings the two bag-opening rollers to the exposed RCU, and a linearly actuated clamp mechanism brings the rollers in contact with the RCU (Step 2, , Figure 4.6). Then, the rollers rotate so that their mounted hooks grapple the RCU. The gripping action is to engage the hooks with the slotted band mounted on the RCU, as shown in Step 3 of Figure 4.6. These rollers are independently actuated by two motors, allowing for redundancy if the hooks are not perfectly aligned. During this interfacing, the first set of rollers is locked, allowing the second set of rollers to create a torque during the hooking operation. It is important that the hooks between the rollers be offset to prevent their collision during close operation. At this stage, both sets of rollers are in contact with the RCU. The arm rotates the RCU up to the filling position in two stages (Step 4, , Figure 4.6). First, it pulls the rest of the RCU through the first passive set of rollers. Then, the first set of rollers locks, applying a holding torque as the arm rips the RCU away from the roll. This tear-away procedure minimizes the need for additional moving parts (e.g., a blade), but will need to carefully consider the design of the tear-away strip to resist the internal friction of the RCU roll.

Now that the RCU is free from the roll, it can be moved by the rotating arm to the filling configuration, directly below the funnel. Once in this position, the clamp mechanism is opened, thereby opening the mouth of the bag (Step 5, , Figure 4.6). At this point, the bag can be filled by opening the sliding door on the funnel and activating the vibration. The stop condition for the filling is detected by weight, using the torque sensors in the rollers. Considering that each roller supports half the weight of the RCU, then

$$\tau_{roller} = \frac{1}{2} r_{roller} W_{RCU} \tag{4.9}$$

and

$$W_{RCU} = m_{RCU} \times g_{Moon}\ . \tag{4.10}$$

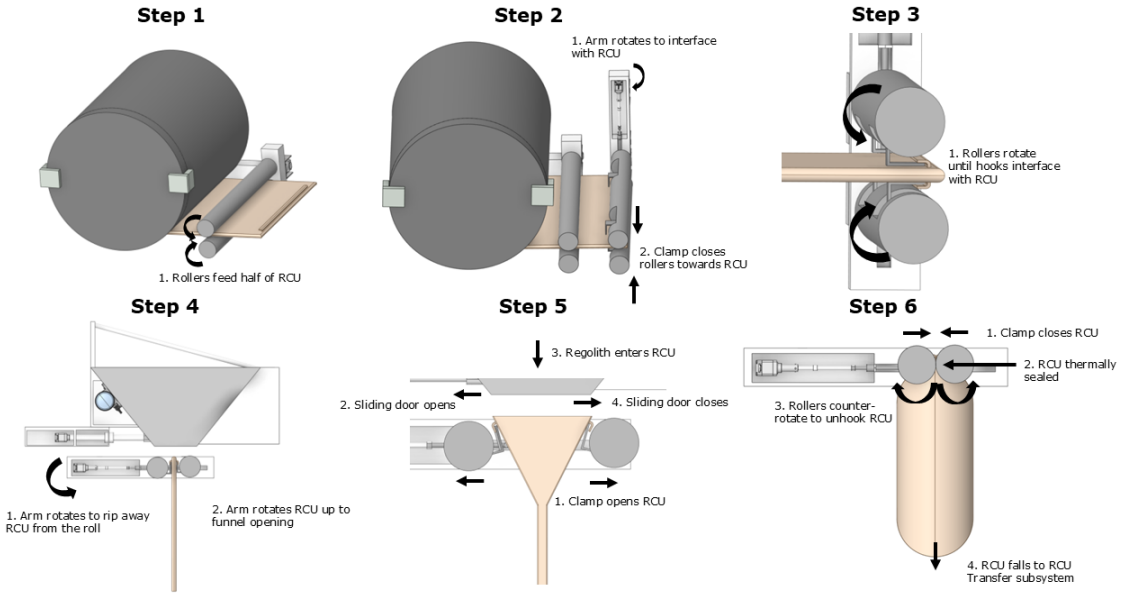


Figure 4.6: RCU Filling Process

By combining the two equations, a stop condition can be formed such that filling occurs when

$$m_{RCU} = \frac{2\tau_{measured}}{r_{roller} \times g_{Moon}} \leq 22 \text{ kg} \tag{4.11}$$

where the 22kg condition is as Section 3.1.3 for the case of RCU with the largest possible regolith density at $\rho = 1.92$ g/cm^3 [41]. This mass also acts as the loading requirement for the hooks and rollers, meaning that they should be able to withstand as much as 22 kg $\times 1.625$ m/s^2 = 35.75 N downward force. This force is split between the two bag-filling rollers. Consequently, the required holding torque for one roller is (35.75 N / 2) $\times$ 0.025 m = 0.45 Nm. Since this torque is relatively low, the limiting factor for these rollers is likely the resistance to the tear-away force when ripping away the RCU from the roll. More accurate estimates can be determined depending on the specifics of the RCU design. The motor in the arm will also need to hold the RCU, the arm, and the rollers. The force required to move both rollers, each with a mass of 0.44 kg, is 0.88 kg $\times$ 1.625 m/s^2 = 1.43 N. This load is used to quantify the torque required by the linear actuator when the arm is down (Step 2, Figure 4.6), using equation 4.6 with the same parameters and finding a torque value of 0.0081 Nm. The force required to move the arm is 0.36 kg $\times$ 1.625 m/s^2 = 0.59 N. Combining the moments for these forces, the torque required by the motor should be no less than (35.75 N $\times$ 0.15 m) + (0.44 kg $\times$ 1.625 m/s^2 $\times$ 0.1 m) + (0.44 kg $\times$ 1.625 m/s^2 $\times$ 0.2 m) + (0.59 N $\times$ 0.1 m) = 5.64 Nm when the arm is horizontal. Further, the linear actuator in the arm will need to withstand a 35.75 N + 1.43 N = 37.18 N downwards force on the lead screw when the arm is horizontal.

Once the filling is completed, the clamp mechanism closes and an element inside the roller is electrically heated, thermally sealing the RCU (Step 6, Figure 4.6). The specifics of the thermal sealing process, such as melting temperature, are dependent on the RCU material and whether a thermal heating strip is sown into the RCU. However, the sealing could be achieved by a Nichrome

wire element, the power for which can be found using the resistance and voltage,

$$P = \frac{V^2}{R} = \frac{V^2}{\rho L/A} \quad , \tag{4.12}$$

where ρ is the resistivity of Nichrome, L is the length of the strip, and A is the cross-sectional area of the strip. Considering some strip of thickness 0.0001 m, width 0.01 m and length 0.5 m, with a resistivity of 1.09×10^{-6} Ωm [265], operated at a voltage of 15 V, an estimated power of 413 W is obtained. Now that the RCU is sealed, the rollers rotate in the opposite direction, unhooking the RCU and allowing it to fall onto the RCU Transfer subsystem, further detailed in Section 4.3.2. The time required for the bagging process in Figure 4.6 is estimated in Table 4.5. Since the anticipated torque rating of the actuators far exceeds the required torques (see Section 4.3.4), and actuator RB motor operations do not carry significant loads, average speeds of 40°/s and 0.1 m/s are considered for the rotary and linear actuators, respectively. Further, an estimate of funnel flow rate can be formulated by scaling the outlet surface area of a funnel experiment conducted in lunar gravity and vacuum conditions [166]:

$$\dot{m}_{\text{Funnel}} = \frac{l_{\text{Funnel}} \times w_{\text{Funnel}}}{l_{\text{experiment}} \times w_{\text{experiment}}} \cdot \dot{m}_{\text{experiment}} = \frac{0.43 \text{ m} \times 0.1 \text{ m}}{0.015 \text{ m} \times 0.018 \text{ m}} \cdot 0.72 \frac{\text{kg}}{\text{min}} = 114.7 \frac{\text{kg}}{\text{min}}. \tag{4.13}$$

Table 4.5: LRCS Bagging Process Time Estimation (see Figure 4.6)

	Steps	Time (s)
Step 1		
	Step 1.1 - Rollers feed half an RCU	5
Step 2		
	Step 2.1 - Arm rotates to interface with RCU	2.25
	Step 2.2 - Clamp closes rollers towards RCU	0.5
Step 3		
	Step 3.1 - Rollers rotate until hooks interface with RCU	6.75
Step 4		
	Step 4.1 - Arm rotates to rip away RCU from roll	0.25
	Step 4.2 - Arm rotates RCU up to funnel opening	2
Step 5		
	Step 5.1 - Clamp opens RCU	0.5
	Step 5.2 - Sliding door opens	1
	Step 5.3 - Regolith enters RCU	16
	Step 5.4 - Sliding door closes	1
Step 6		
	Step 6.1 - Clamp closes RCU	0.5
	Step 6.2 - RCU thermally sealed	3
	Step 6.3 - Rollers counter-rotate to unhook RCU	4.5
	Step 6.4 - RCU falls to RCU Transfer subsystem	1
	Total Time	**44 s**

4.3.2 RCU Transfer

The RCU Transfer subsystem, shown in Figure 4.8, is closely related to the RCU Filling subsystem, interfacing with it to collect the sealed RCU and move it out of the system to a position where it can be manipulated. Considering the design of the RCU Filling subsystem (see Section 4.3.1),

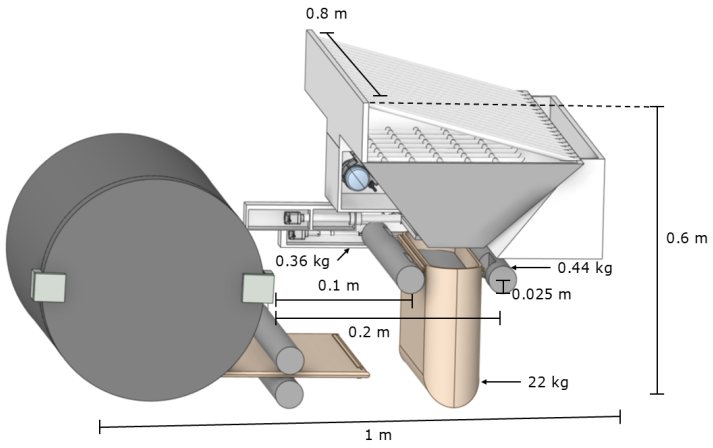


Figure 4.7: 3D Model of the LRCS RCU Filling Subsystem

a mechanism is required to move the RCU from underneath the Vibrating Funnel. In traditional industrial bagging, this is typically achieved by a conveyor. Although a non-powered conveyor that leverages gravity would be ideal; the LRCS is constrained in the vertical dimension by the height of IPEx (see Figure 3.6). In fact, the maximum dump height of IPEx can be estimated by considering the angle required for the clearance of the 31.2cm-diameter bucket drums over the wheels of equal dimension. Considering that only half a bucket drum would need clearance from the wheel axle height (due to the dumping pattern), it can be found that IPEx has a maximum dump height of (Figure 3.6)

$$\underbrace{(0.682\,\mathrm{m} - 0.156\,\mathrm{m})^2}_{\text{bucket drum actuator length}} - \underbrace{\left(\frac{0.312\,\mathrm{m}}{2}\right)^2}_{\text{wheel radius}} - \underbrace{\frac{0.312\,\mathrm{m}}{2}}_{\text{bucket drum radius}} = 0.346\,\mathrm{m}\,. \tag{4.14}$$

Comparing the maximum dump height with the total height of the RCU Filling subsystem in Figure 4.7, it is evident that IPEx will require additional dumping height to successfully dump its load into the LRCS. This motivates the addition of a ramp to the RB, the maximum angle of which is 30° according to Table 3.1. The final height for this ramp will be determined according to the combined height of the RCU Filling and RCU Transfer subsystems. The limited vertical space necessitates the implementation of a powered conveyor for the RCU Transfer subsystem.

Conveyors have been proposed for the movement of lunar regolith, with analogies being drawn to dusty ore mining environments [48, 266]. However, the materials used in traditional conveyors, e.g., rubber, are not suitable to the lunar environment, due to temperature variations. Alternative conveyors, such as chain-link, suffer from jamming under dust exposure, and require maintenance frequently. Another option is the implementation of a lead-screw mechanism, but this requires too long a screw to push along the length of the bag, and the threads would be under constant dust exposure, leading to jamming. Therefore, a simple lever arm is implemented to push the RCU out of the system along a chute, using a motor with an Aeroflex dust-tolerant seal. The required force and subsequent torque can be calculated by considering the force required to overcome the static friction of the chute. Using a coefficient of static friction of 0.18 [212], assuming an aluminum chute

and Kevlar as the RCU material, a force of at least 22 kg $\times$ 1.625 m/s^2 $\times$ 0.18 = 6.44 N must be enacted on the RCU. This corresponds to a torque of 6.44 N $\times$ 0.25 m = 1.61 Nm. Additionally, the chute has sloped walls to align the RCU as it falls from the rollers. The chute is designed to accommodate one extra RCU, sticking out of the RB by one RCU length for the MM to collect.

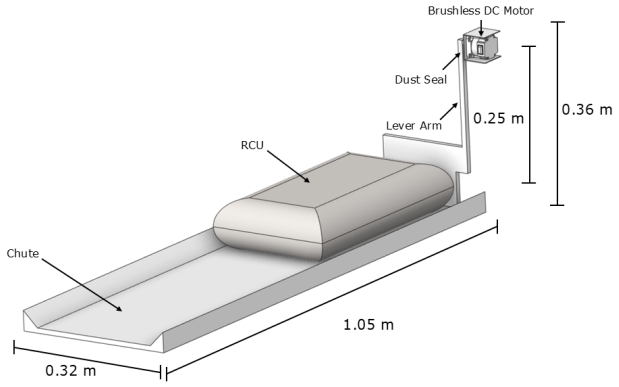


Figure 4.8: 3D Model of RCU Transfer Subsystem

4.3.3 RCU Bagger Mobility

The RCU Bagger requires a Mobility subsystem that will deploy it from the launch vehicle, and allow it to displace itself in the environment. The RB can be displaced by a simple set of differentially steered wheels under the primary supports. Since the bagger does not require a high displacement speed, field-tested wheels designed for the lunar surface can be implemented, such as the wire mesh wheels developed by NASA and General Motors [267]. Further, these wheels will not be required to traverse complex terrain, since the RB will primarily operate within the landing pad area. Therefore, it is reasonable to assume a maximum obstacle traversal height of 0.07 m, similar to IPEx (see Table 3.1). Continuing to take IPEx as a reference design, wheels of approximately 0.3 m diameter should be able to achieve this obstacle clearance. The mass of each wheel can be scaled from the original Lunar Roving Vehicle (LRV), giving a mass of 2.02 kg per wheel. The wheels are to be individually actuated by 4 motors. To quantify the torque required by the motors, we can find the force required to move the system of mass $m = 197.98$ kg forward at the speed of $v = 0.2$ m/s (similar to MM) on terrain with maximum grade of $\theta = 30°$ (similar to IPEx) and for a conservative rolling resistance of $\mu_r = 0.02$ [31], reaching the top speed in $t = 5$ s

$$F_{total} = mg\mu_r \cos(\theta) + mg \sin(\theta) + m\frac{v_{desired}}{t_{acc}} \approx 174\,\text{N}.$$

Consequently, the required torque per motor would be ($r_{wheel} = 0.15$ m)

$$\tau = \frac{F_{total}}{4} \times r_{wheel} = 6.5\,\text{Nm}\,. \tag{4.15}$$

Note that the mass of the system is estimated based on a combined mass of 78 kg for the solar panels and batteries prior to sizing (see Section 4.6.1).

The RB must also consider some sensors to enable localization and traversing in the workspace. However, since the RB does not require detailed environmental mapping, this is achieved using a classical stereo camera apparatus with two optical cameras. These are mounted on top of the RB using a mast, allowing for a clear view of the surroundings. Further, the mast has two degrees of freedom, with a 360° planar view of the environment and a 180° vertical range, allowing it to both view the environment as well as inspect the RB for jamming, damage or faults. This setup is similar to that of Perseverance's Mastcam-Z stereo cameras, which are used as a reference design [268]. The RB also implements an Inertial Measurement Unit as a standard input for most dead-reckoning and localization algorithms, which will be kept close to the other Onboard Computing architecture (see Section 4.5.1). Figure 4.10 shows the RB Mobility subsystem along with the chassis.

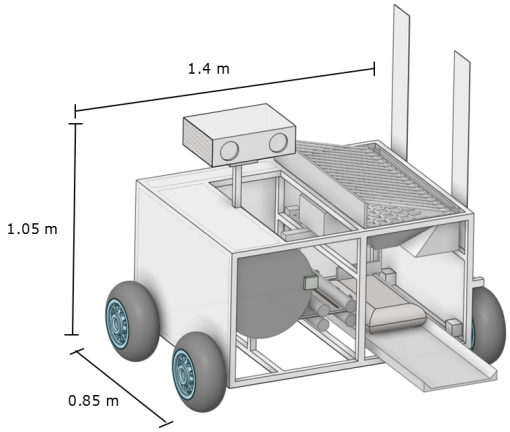


Figure 4.9: 3D Model of RCU Bagger

4.3.4 RCU Bagger Design

The final design for the RCU Bagger, including the primary subsystems and plating, is shown in Figure 4.9. The figure does not include the Power, Onboard Computing, or Communications subsystems, which are further detailed in Section 4.5. Note that the RB structure does not seal the system, to allow the dust to escape, but individual components are sealed against the dust. Now that the primary subsystems of the RB have been determined, it is possible to find the height of the ramp for IPEx, as mentioned in Section 4.3.2. The height from the ground to the funnel opening is 0.7 m. Considering that IPEx's dump height is approximately 0.35 m (see Section 4.3.2), with an additional 0.05 m margin, and assuming a ramp that is angled at IPEx's maximum traversal slope of 30° (see Table 3.1), a ramp length of 0.9 m is determined. This ramp is actuated so that it can be stowed during traversal and deployed during intake. The ramp is actuated by a single motor, and must withstand a 35 kg $\times$ 1.625 m/s^2 $\times$ cos(30) = 49.3 N normal force from IPEx. Assuming a 0.03 m (load-bearing) thick Aluminum 6061-T6 ramp of density 2713 kg/m^3, the mass of the ramp is estimated to be approximately 1.96 kg. Consequently, the ramp motor must be able to produce 1.96 kg $\times$ 1.625 m/s^2 $\times$ 0.4 m = 1.3 Nm of torque. To verify the stability of the ramp under the

IPEx load, we need to ensure that

$$\sigma_{bending} = \frac{Mc}{I} \leq \frac{\sigma_{yield}}{\text{Safety Factor}} \tag{4.16}$$

where $M = FL/4$ being the bending moment for a for a force $F = 49.3$ N applied to a simply supported beam of length $L = 0.9$ m, $c = 0.0015$ m being the distance to the neutral axis (half the plate thickness), and $I = wt^3/12$ being the moment of inertia for a rectangular plate of width $w = 0.14$ m and thickness $t = 0.003$ m. Taking $\sigma_{yield} = 241.3\,\text{MPa}$ [259] and safety factor of 1.4 [254],

$$\sigma_{bending} = 13.2\,\text{MPa} \leq \frac{241.3\,\text{MPa}}{1.4} = 172.4\,\text{MPa}\,, \tag{4.17}$$

which is well below the yield stress for the ramp.

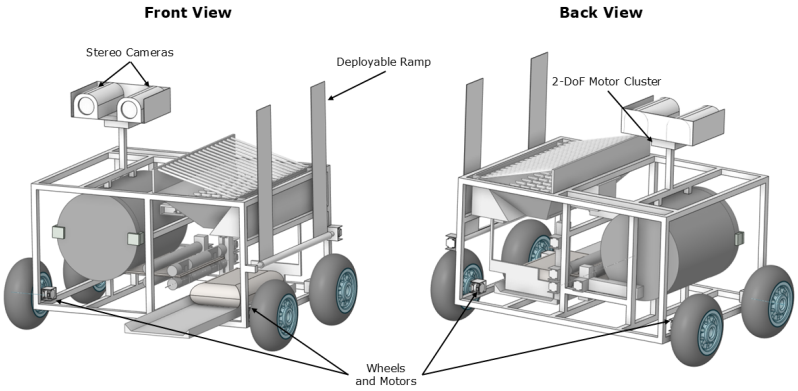


Figure 4.10: 3D Model of RCU Bagger Chassis, Mobility Subsystem, and Deployable Ramp

Only a few actuators have been built that could survive the lunar environment, such as those used in COLDArm [127], which is a lunar manipulator designed to operate in temperatures ranging from 93 K to 373 K with no additional survival heaters. Therefore, the COLDArm motors are considered as the baseline for the RB actuators. The COLDArm brushless DC motors can function on the lunar surface, and are rated for a torque of $\tau_{\max} = 98$ Nm at a mass of 1.4 kg per actuator, exceeding the necessary capabilities of all actuators even considering conservative factors of safety. The COLDArm actuators operate on a 28V power rail, and can draw up to 2 A of current, implying a peak power draw of $P_{\max} = 56$ W. Since torque and current are directly proportional, this can be scaled linearly to estimate the average power draws of the actuators in the system for the Power budget in Section 4.6.1, so that

$$P(\tau) \approx P_{max}\left(\frac{\tau}{\tau_{max}}\right). \tag{4.18}$$

The RCU Bagger chassis is shown in Figure 4.10, including the Mobility subsystem and deployable ramp. The chassis is designed to support the load bearing elements of the RB, such as the RCU storage cylinder, the funnel, and the RCU Transfer subsystem chute. All motors have attachment points on the chassis for stability. Further, a basic structural analysis of the chassis is performed using the Finite Element Analysis (FEA) in ANSYS. The result of the FEA analysis is shown in Fig-

ure 4.11, indicating structural stability under lunar gravity. The FEA demonstrates that no stress values approach the yield strength of 241.3 MPa for Al 6061-T6. The highest loads are experienced directly under the RCU storage cylinder, as expected for the heaviest element in the system.

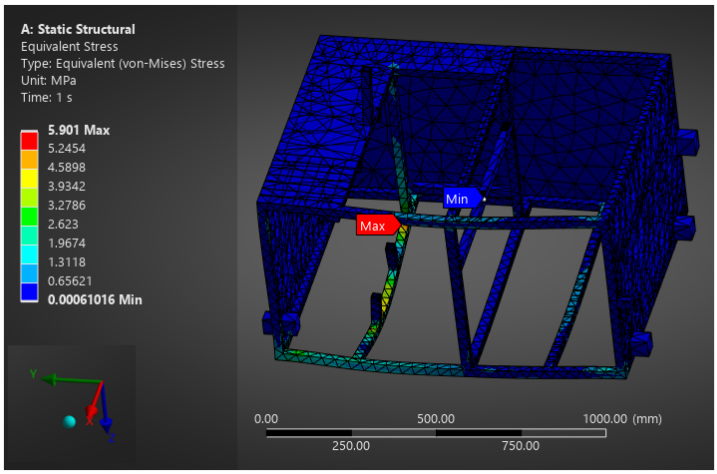


Figure 4.11: Stress Model of RB in Lunar Gravity

4.4 LRCS System Engineering: Mobile Manipulator

The second machine of the LRCS is the Mobile Manipulator, shown in Figure 4.12, which uses an articulated 4-Degree-of-Freedom (DoF) manipulator to collect RCUs from the RCU Bagger, transports them to the berm location, and then places them into the berm. The MM takes inspiration from similar systems, namely GITAI's Lunar Rover [135, 132] and MDA's Skymaker. These two solutions combine well, since GITAI's Lunar Rover is a mobility system created for the lunar environment and is already designed to operate with manipulators of similar size.

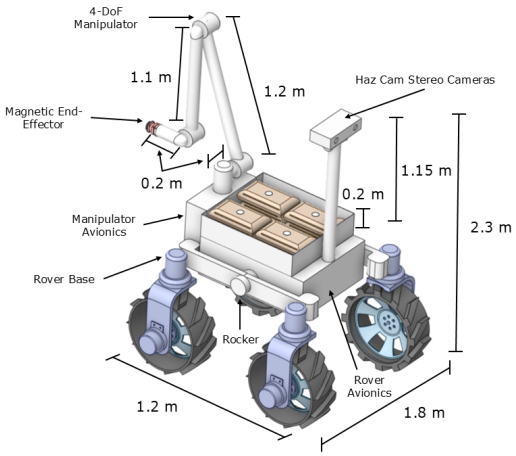


Figure 4.12: 3D Model Mobile Manipulator

4.4.1 Mobility

The MM Mobility subsystem is designed according to GITAI's Lunar Rover [135]. This eliminates the need to further design the mobility base, and ensures stability when operated with the 4-DoF manipulator, i.e., does not tip over. Further, it fits within the volumetric constraints, and offers a simpler solution than alternatives such as ATHLETE [126, 124, 125]. The rover implements a 4-wheel dual-rocker suspension, reaching top speeds of 0.24 m/s. The wheels measure 0.6 m in diameter, and are individually actuated by 4 brushless DC motors. All actuators implement dust mitigation technologies. The rover also implements two Haz Cam stereo cameras [269] for navigation, mounted on a 1.15m-tall mast. The processing is done by a Jetson Xavier onboard computer. The exact payload capacity of the rover is not specified; however, it can be seen from lunar analogue demonstrations [270] that the rover is capable of supporting at least 2 additional robotic systems and their payloads. The first of these robots is the Inchworm manipulator [134], with a mass of 50 kg and a payload capacity of 5 kg. The second robot is the S2 dual-arm manipulator, with a mass of at least 100 kg and payload of at least 10 kg (estimated as two Inchworm manipulators). Further, the base is capable of carrying an additional 20-40 kg of payload. This results in a total payload capacity of at least 200 kg for the rover mobility base. Overall, the system components have reached Technological Readiness Level (TRL) of 7 as of 2023.

The LRCS Mobility subsystem will need to stockpile RCUs, in order to be able to place more than one RCU at a time in the blast berm. Considering the payload capacity of 200 kg, at least 8 RCUs, leaving some margin, could be transported per trip from the RB to the berm. Several RCU arrangement configurations can be considered for the stockpile, and three basic ones are shown in Figure 4.13. Configuration 1 is selected in this design since it allows for a higher manipulation error margin (more space between stacks) and all RCUs are in the same orientation, which simplifies the manipulation task. The stack of RCUs requires walls of 0.2 m to keep the RCUs secure, as shown in Figure 4.12.

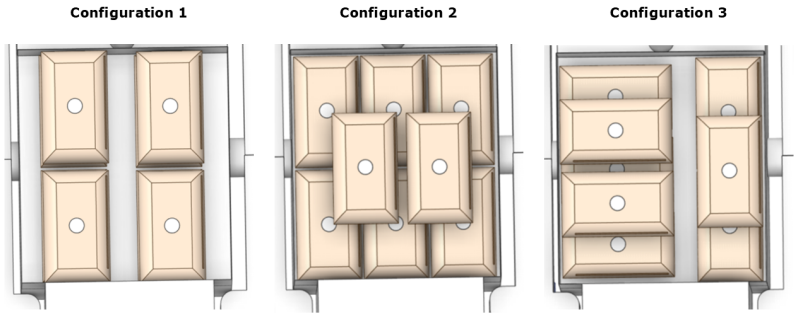


Figure 4.13: RCU Stockpiling Configurations

4.4.2 RCU Manipulation

The RCU Manipulation subsystem consists of two primary physical pieces: the RCU manipulation interface and the articulated 4-DoF manipulator. The RCU manipulation interface refers to the magnetic end-effector on the 4-DoF manipulator, as well as the receiver on the RCU itself. The 4-DoF manipulator is mounted on the MM (see Section 4.4.1), and takes inspiration from COLDArm [127].

RCU Manipulation Interface

The LRCS Manipulator implements a magnetic end-effector to grasp the RCUs. The required attachment force for the end-effector per RCU is 22 kg $\times$ 1.625 m/s^2 = 35.75 N, which turns into 71.5 N with a safety factor of 2, in order to account for inefficiencies, e.g., dust on interface. Generally, to select an appropriate electromagnet and contact area, one can consider the magnetic force formula

$$F_{magnetic} = \frac{B^2 \times A}{2\mu_0} \geq 71.5\ N, \tag{4.19}$$

where B is the magnetic flux, A is the contact surface area, and μ_0 is the permeability of free space. In practice, electromagnetic gripper manufacturers quantify these attachment forces for their electromagnet products. For example, the HGE-SQ-052-FR-EP-P-I electromagnetic gripper from Goudsmit Magnetics would match the requirements, with a working payload of 140 N, contact area of 0.0057 m diameter, and mass of 1 kg [271]. This already includes a safety factor of 3, enhancing redundancy. Further, Goudsmit Magnetics has also quantified the thickness of low carbon steel required to produce these attachment forces with varying air gaps, requiring 0.003m-thick steel to produce 100 N of magnetic force at a 0.0001 m air gap [272]. These low-carbon steel interfaces could be woven into the fabric of the RCUs, and the distribution of the attachment force across the RCU surface should be investigated to prevent tearing. Additionally, the surface area of the steel could be expanded beyond the 0.0057 m diameter to add tolerance during interfacing. For the lack of space-grade electromagnetic grippers, an industrial gripper can be selected in the design, but it would need to be retrofitted for the lunar environment.

4-DoF Manipulator

According to the requirements defined in Table 3.2, the 4-DoF Manipulator will need to reach a workspace volume of 2m (depth) $\times$ 2.6m (height) $\times$ 2m (width), which aligns with the dimensions of the rover bed, discussed in Section 4.4.1. The two candidate articulated 4-DoF lunar manipulators are Skymaker [**lera**], with a maximum reach of 2.5 m, and COLDArm [127], with a maximum reach of 2 m. Considering that the Manipulator should be mounted on a rover bed of approximately 1 m high (see Section 4.4.1), both manipulators could reach the desired workspace. The end-effector payload capacities for Skymaker and COLDArm are 57 N and 40 N, respectively. Given the payload of 35.75 N per RCU (see Section 4.3.1), both Skymaker and COLDArm can effectively handle the payload. However, Skymaker is selected for the MM, due to the larger workspace reach and payload capacity. Further, similar to COLDArm, Skymaker is designed for the lunar environment, including the mitigation of lunar dust, temperature variations, and other environmental factors. An illustration of the reachable workspace for Skymaker, relative to the desired workspace, is shown in Figure 4.14, illustrating that it can comfortably cover the desired workspace. Table 4.6 lists

Skymaker's operation time estimation. Given a 4°/s tip speed within the workspace, the average RCU placement speed from within the rover bed to a target 2 m away is 83.75 s.

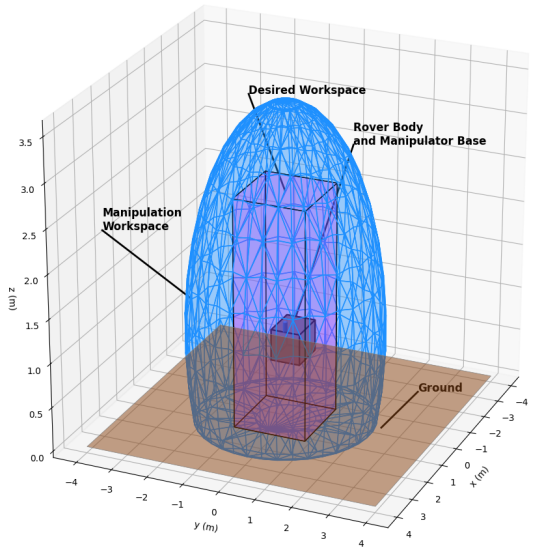


Figure 4.14: Skymaker Workspace versus Desired Workspace

Table 4.6: LRCS RCU Placement Time Estimation

Step	Operation	Time (s)
Step 1	Shoulder (z-axis) joint rotates 90°	22.5
Step 2	Shoulder/Elbow/Wrist (y-axis) joint rotates 45°	11.25
Step 3	Magnet engages RCU	1
Step 4	Shoulder/Elbow/Wrist (y-axis) joint rotates 45°	11.25
Step 5	Shoulder (z-axis) joint rotates 90°	22.5
Step 6	Shoulder/Elbow/Wrist joint (y-axis) rotates 45°	11.25
Step 7	Fine placement (y and z axes)	3
Step 8	Magnet disengages RCU	1
	Total Time	**83.75 s**

4.5 LRCS System Engineering: Common Subsystems

The RCU Bagger and the Mobile Manipulator also share some common subsystems, shown in Figure 4.15, namely, Onboard Computing (Section 4.5.1), Power Management (Section 4.5.2), Communications (Section 4.5.3), Thermal Management (Section 4.5.4), and Lunar Dust Mitigation (Section 4.5.5). Since the MM is largely based on the Skymaker and GITAI Lunar Rover reference designs, the majority of the subsystem development is focused on the RB.

4.5.1 Onboard Computing

The LRCS onboard computer is generally required to manage command and control of the system, including performing global and local task planning, processing and fusing sensor data, driving the

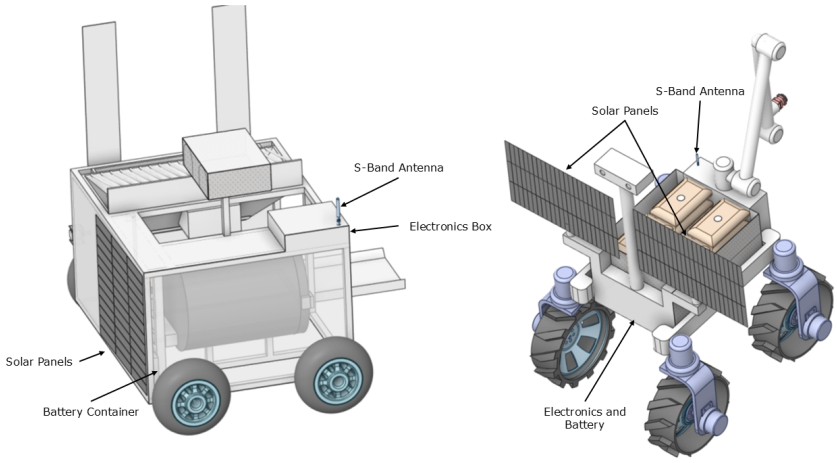


Figure 4.15: Common Subsystems for RB (left) and MM (right)

actuators, processing incoming and outgoing communications, and organizing memory resources. These functions, which are needed for both the RB and the MM, require a highly capable processor. In this design, the Jetson Xavier computer [273] is selected for both machines, inspired by GITAI's lunar rover [135]. This computing platform enables the implementation of advanced real-time vision systems, as well as high processing speeds to manage the variety of sensors and actuators simultaneously. The Jetson Xavier development kit has the complete electronics on a 1.68kg printed circuit board, including 32 GB of storage, in a 0.105m $\times$ 0.105m $\times$ 0.065m form factor [274].

The motor controller for the RB actuators is the same as the one for the COLDArm , which is a Dual-Axis Controller for Extreme Environments (DACEE), capable of controlling two motors at a time [127]. The controller design is based on the Bravo dual-axis motor controller from Motiv Space Systems [275]. Considering the bagging process in Figure 4.6 and the RB's differential steering (see Section 4.4.1), it is expected that four motors be active at any one time, requiring two DACEEs. Further, the onboard computer will process thermal input data from thermistors and output commands to the thermal actuators. A high-level overview of the RB data flow is shown in Figure 4.16.

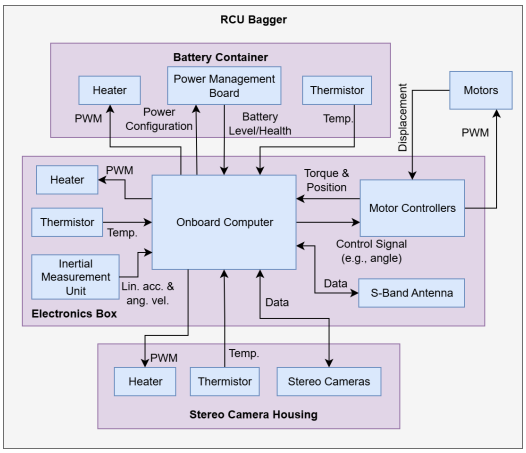


Figure 4.16: Data Flow for the RB Onboard Computer

4.5.2 Power Management

For power generation, solar power concepts seek to leverage near-constant illumination conditions of the lunar South Pole. These concepts find that solar arrays gain a mass advantage over nuclear reactors at polar sites illuminated for over 75% of a lunar cycle [219]. The state-of-the-art solar cells have energy densities up to 350 W/kg [276], as compared to 5 W/kg provided by the latest space-rated nuclear reactors, such as NASA's KRUSTY kilopower reactor [277]. Therefore, solar power is considered for the LRCS power generation. Both the RB and the MM will require solar panels; however, the MM will also require an actuator to rotate the panels towards the Sun.

The LRCS also needs to store power throughout the duration of the mission scenario, accounting for variability in the solar power input and enabling the distribution of peak load power. Lithium-ion (Li-ion) batteries and Regenerative Fuel Cells (RFCs) are the primary technologies considered for long-duration energy storage on the lunar surface [222, 219, 223]. Although Li-ion batteries have the higher TRL, their specific energy level is typically around 200 Wh/kg [220], lower than the 300 to 1000 Wh/kg provided by RFCs [222, 278]. It has been found that Li-ion batteries perform better for short discharge durations, while RFCs require lower mass for discharge durations over 10-18 hours, such as the lunar night [223, 221]. Considering that the LRCS mission scenario should be completed within a single lunar day, thus requiring shorter discharge durations, it is reasonable to select Li-ion batteries for power storage.

The power storage and generation architecture consists of some combination of Li-ion batteries and solar power. To evaluate the specific configuration and size of these components, a system-level power budget is developed in Section 4.6.1. A high-level power distribution architecture is presented for the RB in Figure 4.17 and for the MM in Figure **??**. For both the RB and the MM, the solar panels act as the primary power source, supporting average system loads for the entirety of the mission scenario. The batteries act as a buffer for the system, giving a stable power source for the system to draw upon and allowing for up to 1 minute of peak loading. The primary power rails for the RB include one high-voltage 28 DC voltage rail for the motors and one low-voltage 15 DC voltage rail for other electronics, such as the onboard computer. The power distribution for the MM is abstracted in Figure **??**, since the power systems for the GITAI lunar rover and Skymaker manipulator are managed independently.

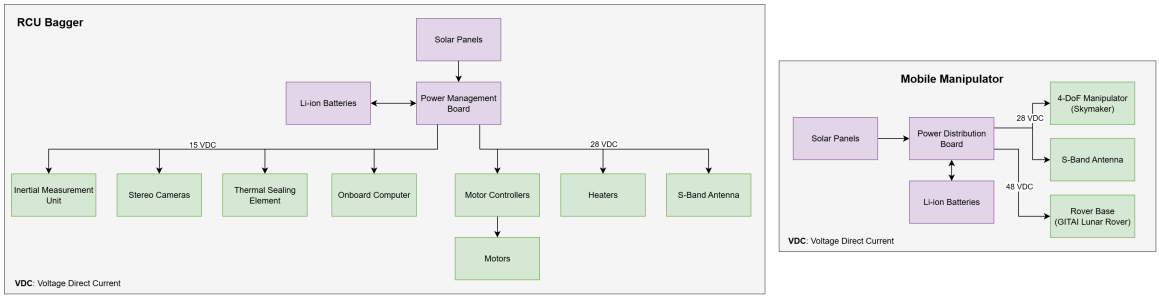


Figure 4.17: RB Power Distribution Architecture

4.5.3 Communications

According to a NASA study, lunar communications terminals, such as the RB, can leverage the K/Ka band when communicating with Earth, as well as with lunar surface rovers, such as the MM [231]. The study anticipates that unmanned surface systems will require an estimated 10-25 Mbps throughput bandwidth. However, no existing systems currently meet this requirement. Current systems, such as India's Chandrayaan 3 and ESA's Pathfinder, communicate with Earth using the X-band [279, 280]. Realistic mission scenarios will likely need to include relay satellites, such as ESA's Moonlight program [281]. In the absence of established lunar relay satellites, current mission scenarios typically leverage their lunar landers as communication relays [282], mitigating the need for a large communications array. One NASA study anticipates minimum X-band data rates of 17.1 Mbps for uplink and 1.9 Mbps for downlink from lunar relay satellites, as well as 2.17 Mbps for uplink and 2.31 Mbps for downlink from the lunar surface [283]. The Carnegie Mellon day-night lunar rover, a similar system to the LRCS, implements a 1kg S-band antenna with 1 W of input power, achieving a 6 Mbps inter-rover communication data rate at a distance of up to 2 km [284]. This is used as the closest estimate for the LRCS Communications subsystem, assuming some lander (or orbital) relay, without fully designing a custom antenna and dish based on anticipated downlink/uplink data rates. The communications data flow for the LRCS is shown in Figure 4.18.

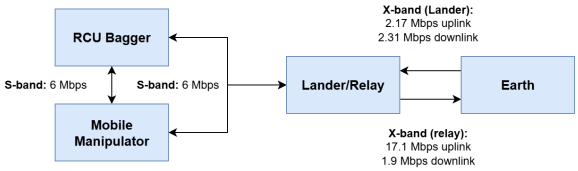


Figure 4.18: LRCS Communications Flow

4.5.4 Thermal Management

The lunar South Pole experiences a wide range of surface temperatures. Diviner data collected over 10 years from the Lunar Reconnaissance Orbiter are used to make a comprehensive mapping of lunar surface temperatures [33], indicating that the temperature range and distribution for the lunar South Pole vary slightly throughout seasons, owing to the Moon's lack of weather and low axial tilt. However, there are slight increases in the size of cold spots during the winter. Generally, temperatures vary from over 300 K in the hottest regions to below 60 K in the coldest regions, the latter typically coinciding with PSRs in the craters.

The large temperature variation in the lunar South Pole poses challenges for surface systems such as the LRCS, for both material selection and electronics. For the LRCS thermal management, it is imperative to identify the components that require tight thermal control. The usage of the COLDArm motors and motor controllers, which are designed for the lunar temperature range between 93 K to 373 K, dramatically reduces the complexity of the thermal management task [127]. Since the GITAI mobility base and Skymaker are both designed for the lunar surface, it is assumed that the thermal management for the MM is already considered for these systems. For the RB, the thermal management task can be further simplified by keeping the most sensitive electronic components grouped in a few locations. As such, only three locations within the RB may require

active thermal management if passive thermal management is not sufficient. The first location is the battery container, which houses the power management board and the Li-ion batteries. The second location is the electronics box, which contains the onboard computer (see Section 4.5.1) and the S-band antenna (see Section 4.5.3), with related electronics, e.g., transceivers. The third location is the stereo camera housing. These three locations can be monitored using thermistors and thermally actuated using electronic heaters (see Figure 4.16). Small Kapton film heaters can be used to heat relevant components [285, 286]. Lastly, it is assumed that the electronics of the system can be thermally coupled to the chassis for heat dissipation, eliminating the need for radiators. Further, some components, such as the onboard computer, already have heat dissipation built-in. However, further analysis is required in the follow-up stages of design to determine whether heaters, insulation, and radiators are necessary to maintain operable temperature ranges.

4.5.5 Lunar Dust Mitigation

The lunar dust has been documented as being highly abrasive and adhesive, as well as electrostatically charged [31]. This poses a challenging environmental factor for machinery, such as the LRCS, necessitating the implementation of mitigation technologies. To protect the solar panels, the Electrodynamic Dust Shield (EDS) can be employed, which is a high-TRL active dust mitigation technology, leveraging the electrostatic charging of lunar regolith to keep surfaces clear from dust [232, 233, 234, 235, 236, 237, 238]. Active methods, such as the EDS, will need to be combined with passive dust mitigation techniques. These techniques include surface coatings that minimize the dust attraction, such as the lotus leaf coating [239, 49, 48] and topology modification [240, 241, 48]. It is recommended that every dust-exposed surface of the LRCS be textured using a combination of these technique, with the specific implementation depending on the final design and material selection. Further, it is imperative that dust ingress to critical rotary interfaces be minimized, which can be achieved by implementing rotary dust seals, such as the spring-loaded Teflon seal [242, 243, 48]. Mature dust mitigation technologies for other interfaces are yet to be developed. One such challenging interface is the sealing of dust from prismatic movement interfaces, such as for lead-screw mechanisms. This could be achieved by modifying dust brushing mechanisms, such as the Zephyr fiberglass brush, to seal and clear dust from the interface [260].

4.5.6 Regolith Containment Unit

The detailed design of the RCUs is not performed in this work. The RCUs can be formed using Beta Cloth from Bron Aerotech, a material used in other space applications [287]. One trade study finds that Beta Cloth performs best as a barrier material for lunar regolith [48]. Some properties of Beta Cloth are summarized in Table 4.7. Of particular importance is the melting point of the material at 934 K. Given that Nichrome heating elements can operate up to 1673 K [265], Beta Cloth RCUs can be feasibly heat sealed without the addition of a melting strip. Further, the density of Beta Cloth is 1300 kg/m^3, lower than that of Kevlar (1440 kg/m^3) that is used as an analogue in this work. Consequently, using Beta Cloth would reduce the estimated total system mass.

Table 4.7: Beta Cloth Properties (Courtesy: Bron Aerotech)

Substrate	Beta Fiberglass
Coating	PTFE
Weight	1300 kg/m^3
Thickness	0.0002 m
Breaking Strength (Warp)	2679 kg/m
Breaking Strength (Fill)	3036 kg/m
Melting Point	923 K

4.6 System Budgeting

4.6.1 Power Budgeting

To perform power budgeting for the LRCS, it is necessary to define the modes of operation of the system, which are broken down as follows:

LRCS Modes of Operation

- **Filling** - The RB intakes lunar regolith, deploys the empty RCU, fills the RCU, and seals the RCU.
- **Transfer** - The RB pushes the RCU out of the machine to be collected by the MM.
- **Manipulation** - The MM manipulates the RCU from the initial pose to the desired pose in the environment.
- **Idle** - The RB and/or MM remains in a live but inactive state; communications are still active.
- **Traversal** - The RB or MM traverses between various lunar surface locations.

Considering these modes of operation, a system-level power budget is presented in Table 4.8, adopting a set of power margins defined by NASA Ames [288]. Accordingly, a 30% power margin is taken for system-level design. A key assumption is made in formulating the power budget with regard to the mass of the solar panels and Li-ion batteries. Considering the constraint of 810 kg for the total mass of the LRCS (see Table 3.2), the power consumption for the RB drive motors is initially estimated based on a combined mass of 78 kg for its solar panels and batteries, prior to budgeting. Since the power consumption of the MM drive motors is already specified by the GITAI Lunar Rover, no initial estimate of the mass of the MM solar panels and batteries is needed.

Since the entire mission scenario is expected to take place during the lunar day, solar power will be the primary source for the LRCS. The battery acts as a buffer to distribute the power to the system, enabling some power storage for temporary periods of shadow and supporting peak loads. Figure 4.19 shows the load of the system (P_{RB} and P_{MM}) for one round-trip placement cycle to deploy 8 RCUs. Therefore, the cycle should be repeated 81760 RCUs / 8 RCUs = 10220 times to complete the berm (see Section 4.1). The amount of power generated by solar panels can be quantified as [222, 289]

$$P_{\text{solar}} = (1 - \chi_d) \times f_{\text{sc}} \times \eta_{\text{sc}} \times A \times \phi sin(\alpha + \beta) \tag{4.20}$$

where χ_d is the percentage of dust coverage, f_{sc} is the solar array fill factor, η_{sc} is the solar cell efficiency, A is the solar array area, α and β are the Sun and array inclination angles, respectively, and ϕ is the solar flux. For the LRCS operation, we can assume $\chi_d = 0$, $f_{\text{sc}} = 0.89$ [289], $\eta_{\text{sc}} = 0.32$ [276],

Table 4.8: Power Budget for LRCS

Component	Power (W, Avg/Peak)	Idle	Fill	Transfer	Traverse	Manipulate	Source
RCU Bagger (RB)							
Power Distribution Unit	4 / 7	100%	100%	100%	100%	100%	MarsFast
Onboard Computer	10 / 30	100%	100%	100%	100%	100%	Jetson Xavier
K/Ka Band Antenna	1 / 2	20%	20%	20%	20%	20%	Carnegie Mellon
Vibratory Motor	0.36 / 56	0%	36%	0%	0%	0%	COLDArm
Linear Actuator (Funnel)	0.008 / 56	0%	5%	0%	0%	0%	COLDArm
Linear Actuator (Arm)	0.005 / 56	0%	3%	0%	0%	0%	COLDArm
Rotary Motor (Arm)	3.2 / 56	0%	89%	0%	0%	0%	COLDArm
Roller Motors (Arm ×2)	0.6 / 112	0%	80%	0%	0%	0%	COLDArm
Thermal Sealing Element	413 / 600	0%	7%	0%	0%	0%	Estimate
Roller Motor (Storage)	1 / 56	0%	11%	50%	0%	0%	COLDArm
Rotary Motor (Lever Arm)	1 / 56	0%	0%	100%	0%	0%	COLDArm
Rotary Motor (Ramp)	0.8 / 56	0%	0%	0%	1%	0%	COLDArm
Stereo Cameras (×2)	2 / 4	0%	20%	20%	100%	0%	Mastcam-Z
Inertial Measurement Unit	0.4 / 0.8	0%	0%	0%	100%	0%	MarsFast
Mast Motors (×2)	0.26 / 112	0%	20%	20%	100%	0%	COLDArm
Drive Motors (×4)	26 / 224	0%	0%	0%	100%	0%	COLDArm
Motor Controllers (×2)	5 / 10	0%	100%	100%	100%	0%	DACEE (COLDArm)
Heater (Electronics Box)	5 / 10	100%	100%	100%	100%	100%	OMEGA
Heater (Stereo Cameras)	5 / 10	100%	100%	100%	100%	100%	OMEGA
Heater (Battery)	20 / 40	100%	100%	100%	100%	100%	OMEGA
RB Total Load (W)	–	44 / 97	82 / 343	51 / 215	78 / 449	44 / 97	–
RB Total Load with Margin (W)	–	57 / 127	107 / 446	66 / 279	101 / 583	57 / 127	–
Mobile Manipulator (MM)							
K/Ka Band Antenna	1 / 2	20%	20%	20%	20%	20%	Carnegie Mellon
Rover Base (incl. avionics)	750 / 1500	10%	10%	10%	100%	10%	GITAI
4-DoF Manipulator (incl. avionics)	220 / 310	10%	10%	10%	10%	100%	Skymaker
Electromagnet	4.8 / 120	0%	0%	0%	0%	100%	Goudsmit Magnetics
MM Total Load (W)	–	97 / 181	97 / 181	97 / 181	772 / 1531	300 / 580	–
MM Total Load with Margin (W)	–	126 / 236	126 / 236	126 / 236	1004 / 1991	390 / 755	–
LRCS Total Load (W)	–	141 / 279	180 / 524	148 / 396	850 / 1980	344 / 678	–
LRCS Total Load with Margin (W)	–	184 / 362	235 / 682	193 / 515	1105 / 2574	447 / 881	–

and $\alpha+\beta = 90.5°$ (see Figure 4.2). The solar flux on the lunar surface varies from $\phi = 1308.9\ \mathrm{W/m^2}$ to $\phi = 1425.7\ \mathrm{W/m^2}$ [146], and the lowest value is considered for a conservative power budgeting. To power the system entirely using solar energy, $(P_{\mathrm{RB}})_{\mathrm{solar}} = 108\,W$ and $(P_{\mathrm{MM}})_{\mathrm{solar}} = 1004\,W$. Solving equation 4.20 for A using these values, results in solar panels of area $A = 0.26\,\mathrm{m^2}$ for the RB and $A = 2.4\,\mathrm{m^2}$ for the MM. This corresponds to one 0.5m × 0.5m panel for the RB and two 1m × 0.6m panels for the MM. Considering a solar cell mass density of 0.32 $\mathrm{kg/m^2}$ [276], the total mass of the solar arrays will then become 0.08 kg for the RB and 0.8 kg for the MM. The size of these panels can be reduced by optimizing the power system balance with the battery, relying on stored power for the most power-hungry modes of operation, and recharging the battery during periods of lower load.

The battery properties can be determined by considering the peak loads of the system from Table 4.6.1. To support one minute of peak load for the RB and the MM, where $(P_{\mathrm{RB}})_{\mathrm{peak}} = 583\,\mathrm{W}$ and $(P_{\mathrm{MM}})_{\mathrm{peak}} = 1991\,\mathrm{W}$, the system batteries need to contain

$$E_{\mathrm{battery\,RB}} = P_{\mathrm{peak\,RB}} \times t_{\mathrm{peak}} = 583\,\mathrm{W} \times 60\,\mathrm{s} = 34980\,\mathrm{J} \tag{4.21}$$

$$E_{\mathrm{battery\,MM}} = P_{\mathrm{peak\,MM}} \times t_{\mathrm{peak}} = 1991\,\mathrm{W} \times 60\,\mathrm{s} = 119460\,\mathrm{J} \tag{4.22}$$

of energy capacity. Accounting for a battery charging efficiency of $\eta_{\mathrm{charging}} = 0.99$ [222] and an

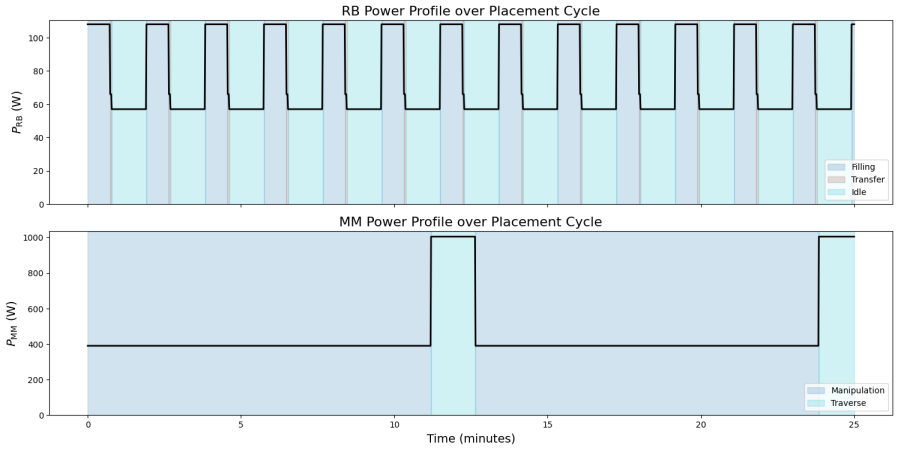


Figure 4.19: LRCS Power Consumption Over One Placement Cycle

allowable depth of discharge (DoD) of 40% [288], the desired battery capacity (C) can be found as

$$C_{RB} = \frac{E_{\text{battery RB}}}{\text{DoD} \times \eta_{\text{charging}}} = \frac{34980\,\text{J}}{0.4 \times 0.99} = 88333\,\text{J} = 24.5\,\text{Wh} \tag{4.23}$$

$$C_{MM} = \frac{E_{\text{battery MM}}}{\text{DoD} \times \eta_{\text{charging}}} = \frac{119460\,\text{J}}{0.4 \times 0.99} = 301667\,\text{J} = 83.8\,\text{Wh}\,. \tag{4.24}$$

Using a Li-ion energy density of $\rho_{\text{li-ion}} = 200$ Wh/kg or 720000 J/kg, the total battery mass for the RB and MM can be calculated as

$$m_{\text{battery RB}} = \frac{C_{RB}}{\rho_{\text{li-ion}}} = \frac{88333\,\text{J}}{720000\,\text{J/kg}} = 0.12\,\text{kg} \tag{4.25}$$

$$m_{\text{battery MM}} = \frac{C_{MM}}{\rho_{\text{li-ion}}} = \frac{301667\,\text{J}}{720000\,\text{J/kg}} = 0.42\,\text{kg}\,. \tag{4.26}$$

4.6.2 Mass Budgeting

The mass budgeting of the LRCS is detailed in Table 4.9. The American Institute of Aeronautics and Astronautics (AIAA) recommends a set of mass margins and mass growth allowances (MGA) [290], which are adopted in this thesis. Component masses are taken from the aforementioned reference designs, such as COLDArm for the actuators or Skymaker for the 4-DoF Manipulator. However, additional reference designs include MarsFast [291] for some common system components, such as the power distribution unit, and Carnegie Mellon's lunar rover [284] for the S-band antenna.

4.6.3 Cost Budgeting

A preliminary cost budgeting for the LRCS is performed by applying the Advanced Mission Cost Model (AMCM) [292]. The AMCM considers the system mass, launch year (2030), difficulty, and maturity [293]. The maturity of the system is based on the Block Number, representing the level of design inheritance, e.g., two or more for a second-generation design onward. These factors vary between the RB and the MM; therefore, they are considered separately in Table 4.10. The AMCM finds an anticipated cost of 631 million USD for RB and 709 million USD for the MM. The estimated total cost, including a 30% margin would end up to 1742 million USD. The estimated costs are

particularly influenced by the programmatic and technical difficulty that are foreseen for the system and the task it is expected to perform. Since both the RB and the MM machines are new complex systems, the level of difficulty adopted for the system is selected as "very high."

Table 4.9: System-Level Mass Budget with Subsystem Breakdown

Subsystem	Component	RB/MM	Maturity	MGA (%)	Mass (kg)	Mass + MGA (kg)	Source
Structural					**31.68**	**34.85**	
	Chassis	RB	E2	10%	14.40	15.84	Model
	Plating	RB	E2	10%	10.27	11.30	Model
	Ramp	RB	E2	10%	1.96	2.16	Model
	Motor (Ramp)	RB	E2	10%	1.40	1.54	COLDArm
	RCU Stockpile Storage	MM	E2	10%	3.65	4.02	Model
Thermal					**0.27**	**0.31**	
	3 × Heater	RB	E2	15%	0.02	0.02	OMEGA
	Insulation	RB	E2	15%	0.25	0.29	MarsFast
Power Management					**18.42**	**22.06**	
	Solar Panels	RB	E2	10%	0.08	0.09	Section 4.6.1
	Solar Panels	MM	E2	10%	0.80	0.88	Section 4.6.1
	Li-Ion Batteries	RB	E2	10%	0.12	0.13	Section 4.6.1
	Li-Ion Batteries	MM	E2	10%	0.42	0.46	Section 4.6.1
	Power Distribution Unit	RB	E2	15%	7.00	8.05	MarsFast
	Cabling	RB	E2	20%	10.00	12.00	Carnegie Mellon
Communications					**2.00**	**2.30**	
	S-Band Antenna	RB	E2	15%	1.00	1.15	Carnegie Mellon
	S-Band Antenna	MM	E2	15%	1.00	1.15	Carnegie Mellon
Onboard Computing					**2.05**	**2.37**	
	CPU and Memory	RB	E2	15%	0.84	0.97	Jetson Xavier
	CPU and Memory	MM	E2	15%	0.84	0.97	Jetson Xavier
	2 × Motor Controller	RB	E2	15%	0.37	0.43	DACEE (COLDArm)
Filling					**104.51**	**114.09**	
	4 × Rollers	RB	E2	10%	1.76	1.94	Model
	Storage Cylinder	RB	E2	10%	4.78	5.26	Model
	RCU Roll	RB	E2	10%	77.80	85.58	Model
	1-DoF Arm	RB	E2	10%	0.36	0.40	Model
	Vibrating Funnel	RB	E2	10%	8.10	8.91	Model
	Sliding Door	RB	E2	10%	1.54	1.69	Model
	Vibratory Motor	RB	E2	15%	1.40	1.61	COLDArm
	2 × Linear Actuator	RB	E2	15%	2.80	3.22	COLDArm
	4 × Motor	RB	E2	15%	5.60	6.44	COLDArm
	Gearbox	RB	E2	10%	0.37	0.41	Model
Transfer					**6.08**	**6.76**	
	Chute	RB	E2	10%	4.32	4.75	Model
	Lever Arm	RB	E2	10%	0.36	0.40	Model
	DC Brushless Motor	RB	E2	15%	1.40	1.61	COLDArm
Manipulation					**41.00**	**43.15**	
	4-DoF Manipulator	MM	C3	5%	40.00	42.00	Skymaker
	Electromagnet	MM	E2	15%	1.00	1.15	Goudsmit Magnetics
Mobility					**299.54**	**325.50**	
	2 × Stereo Camera	RB	E2	15%	2.76	3.17	Mastcam-Z
	Inertial Measurement Unit	RB	E2	15%	0.30	0.35	MarsFast
	2 × Motor (Mast)	RB	E2	15%	2.80	3.22	COLDArm
	4 × Wheels	RB	E2	10%	8.08	8.89	LRV Wheels
	4 × Motor	RB	E2	10%	5.60	6.16	COLDArm
	4-Wheel Rover	MM	C3	5%	280.00	294.00	GITAI
RCU Bagger					**177.84**	**197.98**	
Mobile Manipulator					**327.71**	**344.63**	
Total					**505.55**	**542.61**	

Table 4.10: Illustrative AMCM Parameter Breakdown for LRCS Elements

System	Mass [kg]	Difficulty	Block No.	Cost Estimate [Million USD]
RB	197.98	Very High	1	631
MM	344.63	Very High	2	709
LRCS Total				**1340**
LRCS Total with Margin*				**1742**

* Total with 30% margin applied

Chapter 5

Conclusion

This thesis has presented the Pre-Phase A design of a Lunar Robotic Construction System (LRCS) intended to enable the use of raw lunar regolith for infrastructure construction through the bagging and manipulation of Regolith Containment Units (RCUs). The work was motivated by the need for scalable construction systems capable of building protective berms, as well as other infrastructure, where reliance on in-situ resources will be critical to sustainable surface operations. The RCUs require minimal regolith processing, making them ideal candidates to build simple structures, as well as to provide surface structures physical, radiation, and thermal protection.

This thesis began with a review of lunar construction, regolith geotechnics, and robotic construction system architectures, with particular focus on confinement formation and regolith bagging. The problem was then formalized by defining the system requirements and constraints, as well as performing a functional analysis that identified four primary functional modules: intake lunar regolith, deploy empty RCU, interface with and secure RCU, and transport RCU to desired pose. Candidate solutions were surveyed and their merits were systematically evaluated using the Analytic Hierarchy Process (AHP). Experts were surveyed to rank design criteria priorities, and the rankings were aggregated using the ZM-II algorithm. This process minimized the cognitive load that pairwise comparisons would have had on experts, reconciling simple rankings with the AHP process. Further, the pairwise comparisons between candidate solutions was evaluated using various metrics, essentially automating the AHP. The selected system design, consisting of a vibrating funnel for regolith intake, series of rollers for RCU deployment, magnetic RCU interface, and articulated 4-DoF manipulator for RCU transport, was then designed to the Pre-Phase A level. Subsystem designs were developed for the two LRCS machines, the RCU Bagger and the Mobile Manipulator. Power, mass, and cost budgeting was performed, as well as basic structural analysis.

Future studies should extend the work through further subsystem engineering, physical prototyping, and laboratory experiments with high-fidelity regolith simulants, particularly in order to validate intake, filling, and stacking mechanisms under lunar-representative conditions. Simulation studies should also be extended to creating more realistic lunar regolith and RCU modeling. The integration of autonomy, perception, and dust-mitigation technologies will also be critical to achieving scalable and reliable operations. Expanding the scope of construction tasks beyond berm formation to include landing pad foundations, radiation shielding, and modular habitat walls would further demonstrate the validity and versatility of the LRCS.

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